

Same-well imaging and transcriptomics in a 50,000-well one-bead-one-compound microchip recover compound-resolved phenotypes

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Abstract

Phenotypic screens usually buy scale by giving up biological depth, or buy depth by giving up compound throughput. High-content imaging can profile very large libraries, but the readout is morphological. Single-cell transcriptomics gives a deeper view of cell state, but at a higher cost per treatment. ActiveSeq was built to test a narrower and more practical question: can the same Z-Screen micro-well provide a useful paired image and RNA readout from the same 5 to 10 cell colony?

We analyzed 20,222 matched wells from two HEK293 pilot experiments, ACT010 and ACT011. The dataset contains 11,511 wells from a 35-compound control panel and 8,711 ZEL026 dummy-bead wells used as a negative-control QC stress test. The image representation used for the main analyses is an internally calibrated 32-dimensional latent learned from a frozen per-well image embedding on this same control-panel domain. It should therefore be read as a calibrated assay representation, not as evidence that the uncalibrated vision embedding generalizes to new chemistry or cell types.

At the compound level, both readouts were reproducible. Split-half top-1 retrieval reached 98.8% for RNA and 94.2% for the calibrated image latent in ACT010, and 80.1% / 86.8% in ACT011, against 2.9% chance. ACT010 centroids retrieved the same ACT011 compounds at 71.4% top-1 for RNA and 85.7% top-1 for image. Image and RNA also preserved related, non-identical compound maps, with cross-modal distance correlations of 0.718 in ACT010 and 0.607 in ACT011.

The strongest operational result was the held-out classification gain from combining image and RNA. A linear discriminant trained on merged image-plus-RNA features reached 55.5% balanced accuracy, compared with 41.4% for RNA alone and 29.2% for image alone; the same ordering held within both runs. A same-compound shuffled-fusion control matched true same-well fusion on this task, so the supervised gain should be read as multimodal compound-level additivity. Analyses that ask the stricter same-well question found positive but modest within-compound structure and should be treated as exploratory support.

The RNA branch remained interpretable at low pass depth. Centered pseudobulk gene signatures retrieved the correct compound across runs at 48.6% top-1 and 68.6% top-3, with compound-specific gene programs suitable for triage and follow-up hypothesis generation. Relative to scGeneScope, the

closest public treatment-matched Cell Painting and scRNA-seq reference, ActiveSeq is much smaller and uses a simpler imaging modality. After normalizing for label-space difficulty, its multimodal classification effect falls in the same range, which provides magnitude context rather than a direct benchmark victory.

Significance

Treatment-matched Cell Painting and Perturb-seq experiments describe separate cell populations exposed to the same perturbation. ActiveSeq instead reads image and RNA from the same multicellular colony in the same micro-well, on a chip designed for 50,000 wells in parallel. In this pilot, both readouts recover compound identity, their maps are aligned but not redundant, and their merged representation improves held-out classification across the 35-compound panel. For early discovery, the useful claim is not that same-well pairing is already fully proven in every sense; it is that a paired well can carry enough orthogonal image and RNA information to make triage more informative than either readout alone.

Introduction

Early-stage discovery needs enough throughput to explore chemical space and enough biological resolution to decide whether a hit is worth pursuing. Existing platforms usually emphasize one side. Cell Painting and imaging-first ML systems can run at very large scale, but they infer biology from morphology [1]. Perturb-seq and related transcriptomic resources provide deeper cell-state measurements, but at lower compound concurrency and higher cost per profile [3]. Public multimodal datasets bridge the two by matching treatments across replicate cell populations; scGeneScope is the strongest current public example, pairing Cell Painting and scRNA-seq across treatments in U2-OS cells [4].

Z-Screen asks whether the pairing can be made physically tighter. In ActiveSeq, a one-bead-one-compound micro-well is imaged and then sequenced, so each well contributes one image vector and one low-pass RNA vector from the same 5 to 10 cell colony. That design does not remove all biological heterogeneity, and this pilot does not yet prove that exact same-well pairing is superior to every treatment-level fusion strategy. It does let us ask whether a single well carries two reproducible, aligned, and operationally additive descriptions of compound response.

The Results follow the decision sequence a skeptical screen designer would use: 1. Did the matched wells and negative controls behave well enough to analyze, and what exactly is the calibrated image representation? 2. Does each modality recover compound identity reproducibly? 3. Are the image and RNA compound maps aligned without being redundant? 4. Does combining the two readouts improve a held-out operational task? 5. Is there evidence for additional same-well structure beyond compound labels, and how large is it? 6. Does low-pass RNA remain interpretable enough for biological triage? 7. How large are these effects relative to scGeneScope, without treating that comparison as a direct win/loss benchmark?

Results

Matched wells pass QC, and image calibration defines the representation tested here

Before interpreting multimodal signal, we first asked whether the matched wells and assay controls behaved as expected. The dataset contained 20,222 matched wells from ACT010 and ACT011: 11,511 wells assigned to the 35-compound control panel and 8,711 ZEL026 dummy-bead wells. ZEL026 dummy wells contain no compound and were retained as a negative-control assay-chemistry stress test, not as a biological discovery condition. They showed the expected collapse in RNA yield: median UMIs dropped from 5,167 to 342 in ACT010 and from 4,750 to 293 in ACT011; detected genes dropped from 2,826 to 301 and from 2,463 to 248. This failure mode is visible in the readout itself and is reported as Supplementary Figure 1.

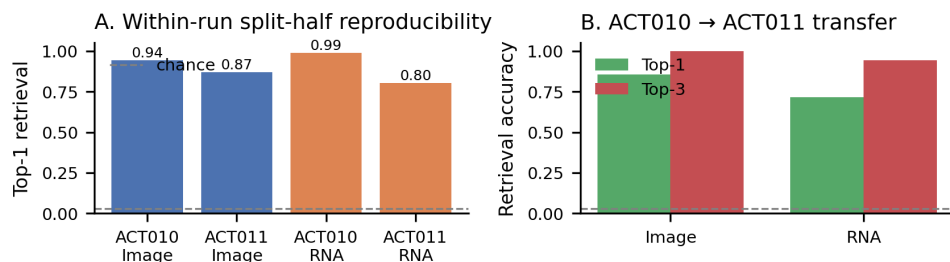
The next question was whether the image branch had a representation suitable for compound-level analysis. The original per-well image embedding came from a frozen pretrained vision model. In raw form, such an embedding can be dominated by experiment, position, or microscopy effects rather than compound response. We therefore used a 32-dimensional domain-adversarial latent calibrated on the matched control-panel domain. The calibration objective rewards compound discrimination while penalizing experiment-of-origin discrimination. The calibrated image latent is consequently an internal assay representation: it can reveal compound-resolved image signal in this pilot, but it is not yet evidence of off-panel image generalization.

With QC and representation scope defined, the first biological question was whether each modality reproducibly encoded the 35-compound panel.

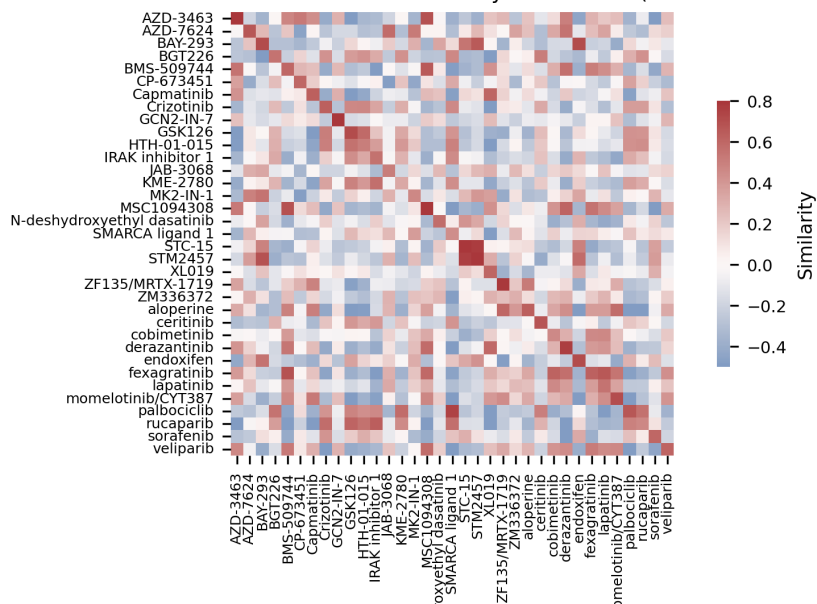
Each modality reproducibly encodes compound identity

Wells for each control compound were split into two pseudo-replicates, and one half was queried for retrieval of the other among the 35 compounds. RNA latent space was highly stable: split-half top-1 retrieval was 98.8% in ACT010 and 80.1% in ACT011. The calibrated image latent was similar at 94.2% in ACT010 and 86.8% in ACT011. Top-3 retrieval reached 100% for RNA in ACT010 and 99.6% for the calibrated image latent.

Cross-run transfer is the stricter test of a control-panel measurement. Compound centroids learned in ACT010 retrieved the same compounds in ACT011 at 71.4% top-1 (94.3% top-3) in RNA and 85.7% top-1 (100% top-3) in the calibrated image latent, against a 2.9% top-1 chance level. These results show compound-level reproducibility and transferability. They do not yet show that exact same-well pairing carries extra information, because split-half and centroid retrieval are driven by compound labels aggregated across many wells. That stricter pairing question is addressed later.



C. RNA centroid similarity across runs (ACT010 rows × ACT011 cols)



D. Image centroid similarity across runs

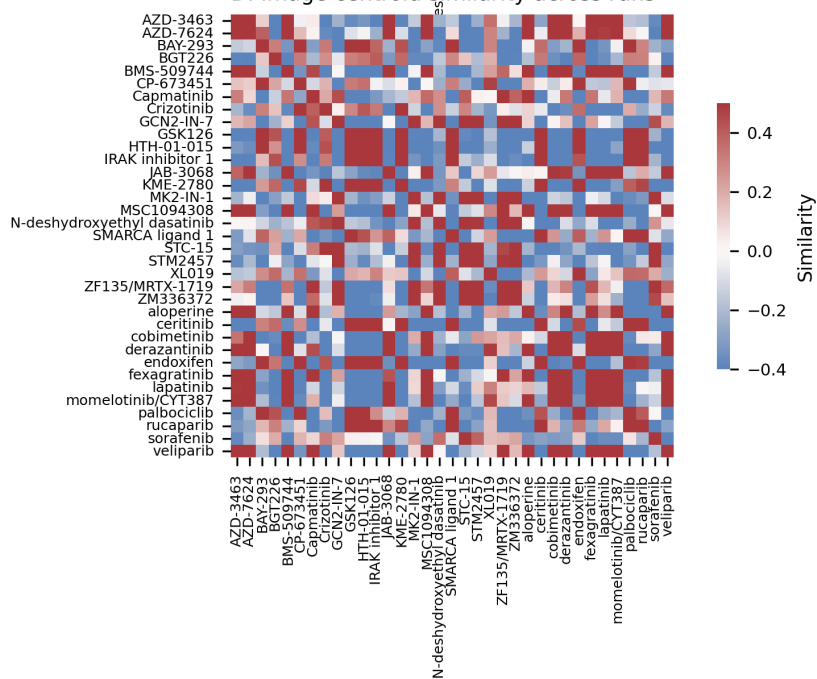


Figure 1. Compound-level reproducibility and cross-run centroid transfer in the 35-compound control panel. Both modalities are reproducible within run, and both transfer well above chance from ACT010 to ACT011. These metrics establish compound-level signal, not the incremental value of exact same-well pairing.

Image and RNA share a compound map but are not redundant

If both modalities recover compound identity, the next question is whether they organize compounds in the same way. Exact label recovery alone could hide two unrelated representations that happen to separate the 35 classes. We therefore compared pairwise compound distances in image and RNA space within each run.

The two maps were aligned. Spearman correlations between pairwise compound-distance matrices were 0.718 in ACT010 and 0.607 in ACT011. The largest perturbations also recurred across experiments: cross-run effect-magnitude rankings correlated at Spearman 0.799 in image space and 0.662 in RNA space. Several compounds, including GCN2-IN-7, KME-2780, BGT226, ZF135 / MRTX-1719, and CP-673451, appeared among the stronger responses in both modalities in at least one run.

The caveat is important for triage: a shared map is not an identical map. The image latent and RNA latent agree enough that image-derived neighborhoods are biologically meaningful, but they preserve different aspects of response. That nonredundancy motivates a direct test of whether combining them improves an operational task.

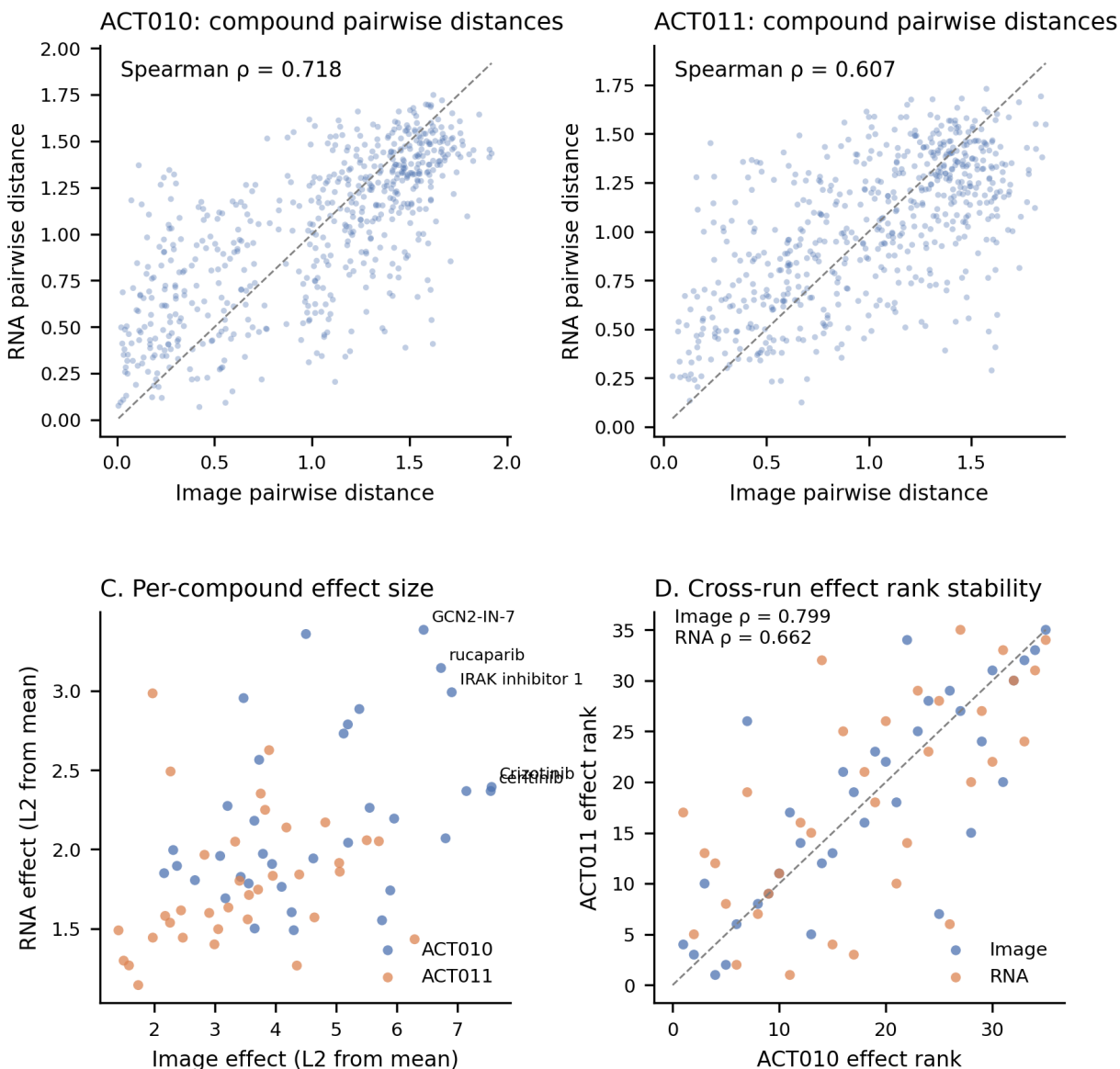


Figure 2. Image and RNA preserve a shared but non-identical compound geometry within run, and effect-size rankings are stable across runs. Pairwise compound-distance correlations are 0.718 in ACT010 and 0.607 in ACT011; cross-run per-compound effect magnitudes correlate at Spearman 0.799 for image and 0.662 for RNA.

Combining same-well image and RNA improves held-out compound classification

The practical question is whether the two same-well readouts help a decision model more than either readout alone. Each matched well was reduced to one calibrated image vector and one RNA vector. A linear discriminant model was trained on image alone, RNA alone, or the raw concatenation of image plus RNA, then evaluated on held-out wells.

Merged image-plus-RNA features outperformed RNA alone in every cut: 55.5% balanced accuracy

versus 41.4% pooled, 60.8% versus 47.3% within ACT010, and 45.1% versus 30.5% within ACT011. Image alone reached 29.2% balanced accuracy pooled. Overall accuracy followed the same pattern, rising from 42.5% to 54.9% pooled, from 48.3% to 60.2% in ACT010, and from 31.6% to 45.5% in ACT011.

The gain was broad rather than concentrated in a few easy classes. Merged features improved per-compound recall for 35 of 35 compounds in the pooled analysis, 33 of 35 in ACT010, and 30 of 35 in ACT011. ACT010 confusion matrices show the same pattern: RNA performs well on its own, and adding the calibrated image latent moves probability mass onto correct diagonal entries and resolves several near-confusions.

This is the strongest evidence in the paper that the paired readout is useful for compound classification. It should still be interpreted precisely. A same-compound unpaired-fusion control, in which image and RNA vectors were shuffled among wells of the same compound and run before fitting the same classifier, matched the true same-well merged result almost exactly. Across 100 shuffles, pooled balanced accuracy averaged 55.4% (95% interval 54.7% to 56.1%), compared with 55.5% for true same-well fusion. ACT010 and ACT011 showed the same pattern. The supervised gain therefore reflects multimodal compound-level additivity in the ActiveSeq readout, not an isolated contribution of exact well-to-well pairing on this label task.

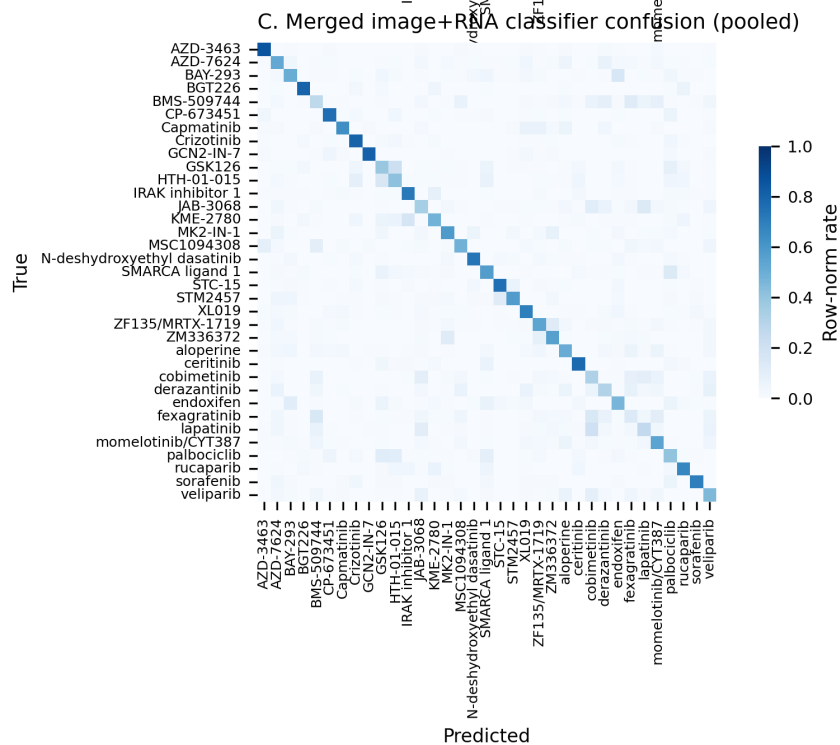
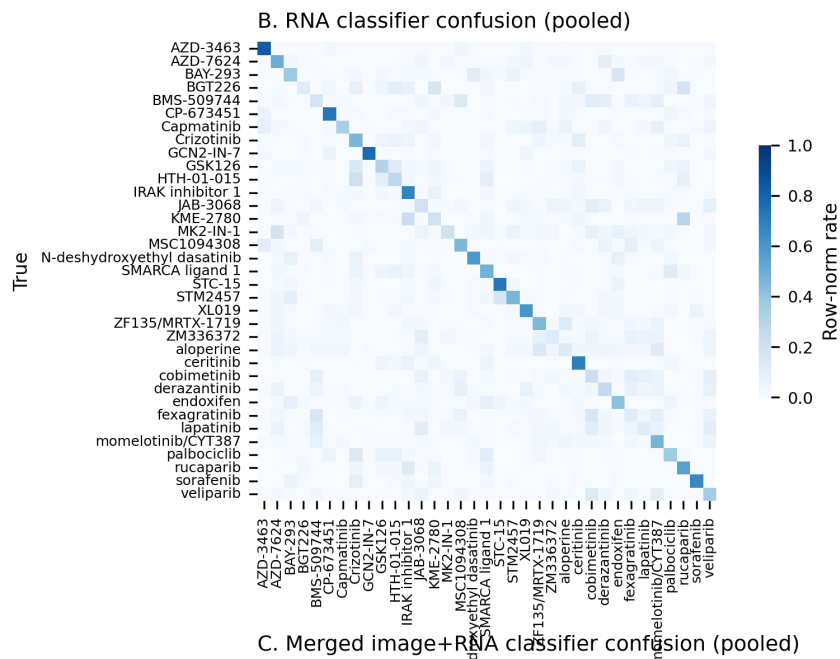
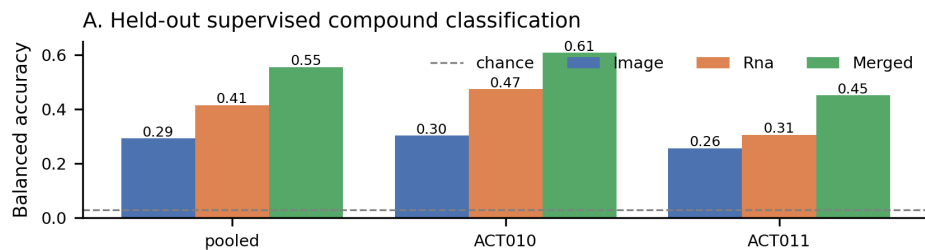


Figure 3. Same-well multimodal classification on held-out wells. Merged image-plus-RNA features outperform RNA alone in pooled and per-run analyses, and per-compound confusion matrices show the gain is from reduced class confusion across most compounds rather than from a few easy classes.

Within-compound same-well structure is positive but modest

The remaining question is the stricter same-well one: after compound identity is held constant, do image and RNA preserve matching well-to-well structure? This matters because exact same-well pairing would be most valuable if it captured response-state differences among wells exposed to the same compound, not only the average compound effect.

In the harder held-out centroid-retrieval task, the calibrated image latent reached 25.1% / 21.7% top-1 across the two runs, RNA reached 44.8% / 24.3%, and a naive image-plus-RNA concatenation reached 49.1% / 33.7%, exceeding RNA alone in both runs. A lightweight CLIP-style same-well contrastive objective on top of the calibrated latent did not improve retrieval further, indicating that calibration had already exposed most of the recoverable compound-identity information.

The exact same-well signal was above random but small. Top-10 exact image-to-RNA retrieval after contrastive training was 2.1% in ACT010 and 2.5% in ACT011, compared with random expectations of 0.4% and 1.0%. Within each compound, image and RNA distance structure was positive in 34 of 35 compounds in both runs, but the median cross-modal distance correlations were only 0.087 in ACT010 and 0.055 in ACT011. Mean kNN overlap gains were 0.006 and 0.020. These effects are compatible with real within-compound response-state structure, but they are not large enough to carry the main claim. The shuffled-fusion control above reinforces that boundary: exact same-well pairing is visible in exploratory within-compound summaries, but the current supervised classification gain is driven by compound-level multimodal information.

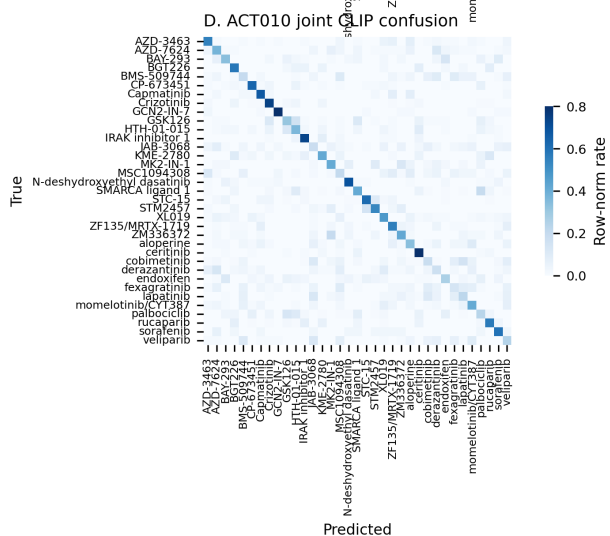
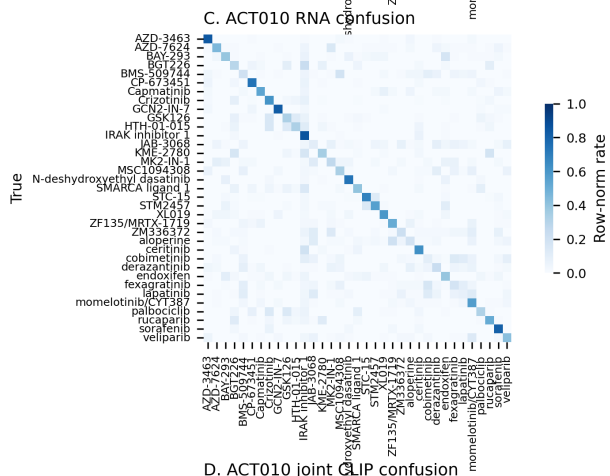
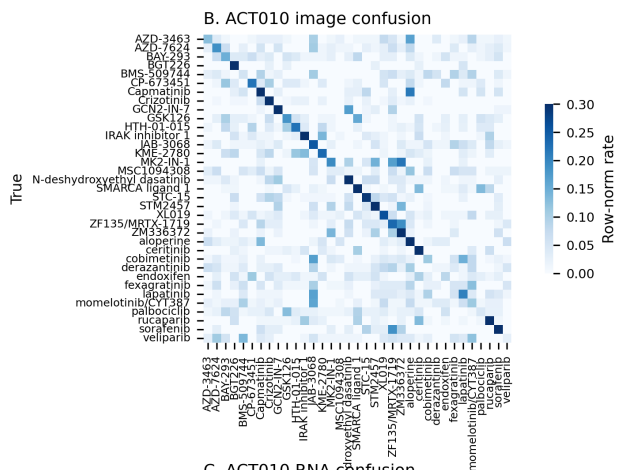
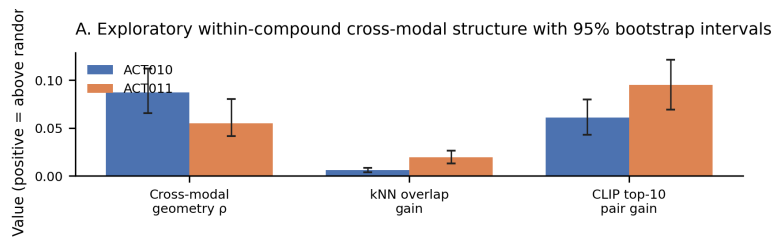


Figure 4. Exploratory same-well structure beyond supervised classification. Within-compound cross-modal agreement is usually above random but modest in magnitude. These data support the possibility of same-well response-state information, while leaving the incremental value of exact pairing to a dedicated control analysis.

Low-pass RNA remains interpretable enough for triage

The classification results say that RNA helps prediction. Drug hunters also need to know whether the RNA readout remains interpretable enough to guide follow-up. We therefore aggregated raw counts from matched control wells into per-compound pseudobulk signatures, converted them to log-CPM, and centered each run by subtracting its experiment-wide mean. This deliberately treats the compound-by-run pseudobulk as the robust unit, rather than overfitting per-well differential expression at low RNA depth.

Centered signatures retrieved the correct compound across runs at 48.6% top-1 and 68.6% top-3. Representative compounds carried coherent gene programs, including palbociclib (ARC, RASD1, CDKN1C, EGR1), BAY-293 (MT1X, MT1F, MT2A, SLC30A1), rucaparib (ITPRIP, SLC37A4, NACC2, COL4A1), crizotinib (SORCS2, ZNF34, AC253572.2), and GSK126 (DUSP10, SPATA6L, CCN2). These examples should be read as interpretable rank signatures for triage, not as validated mechanisms of action in HEK293. The useful result is that low-pass ActiveSeq RNA preserves enough compound-specific gene structure to cluster hits, flag stress or metal-response programs, and nominate follow-up biology.

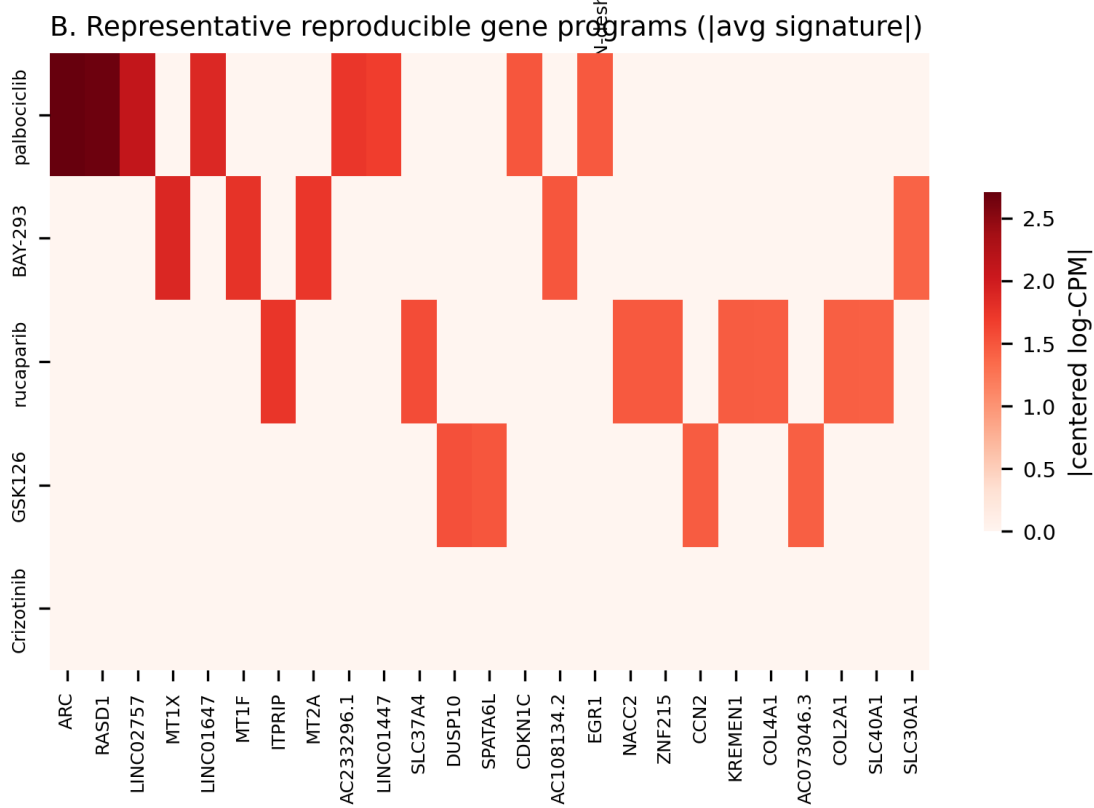
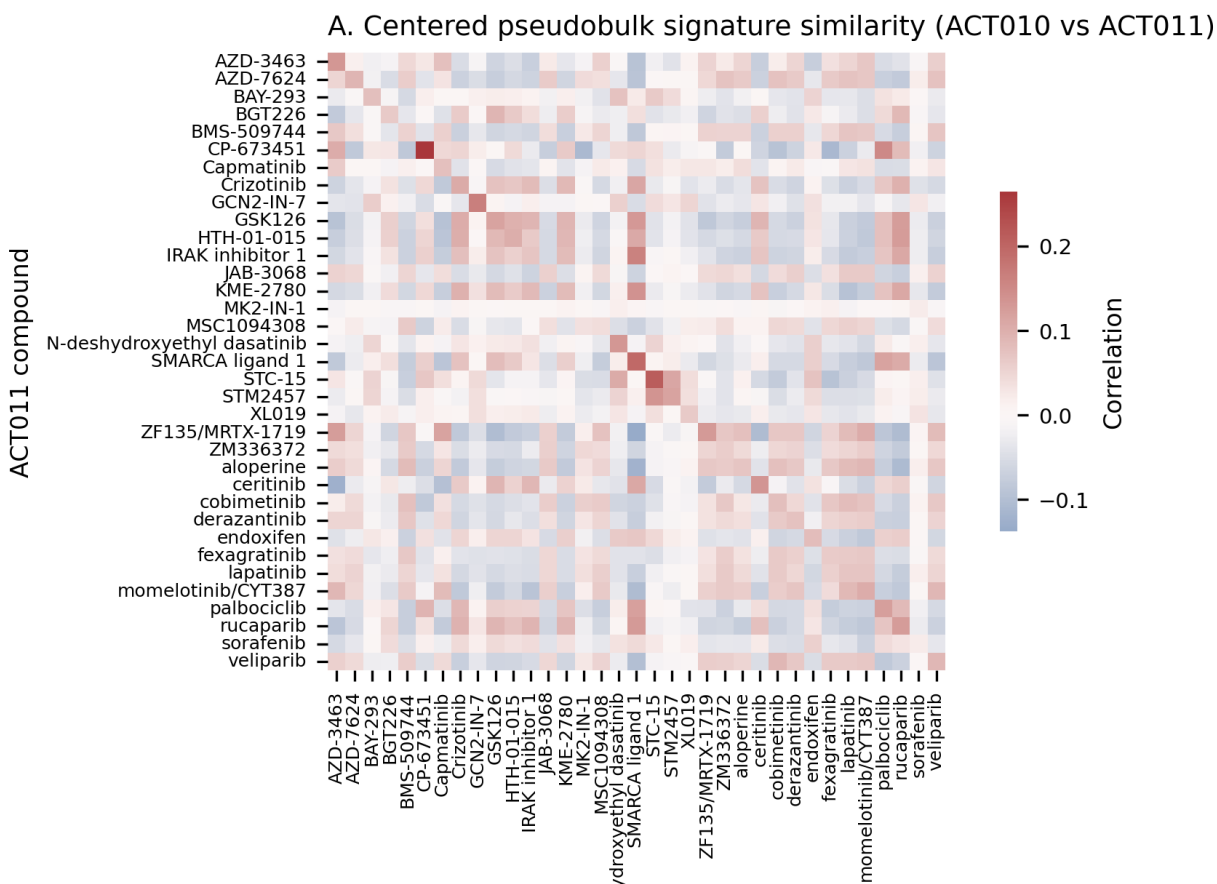


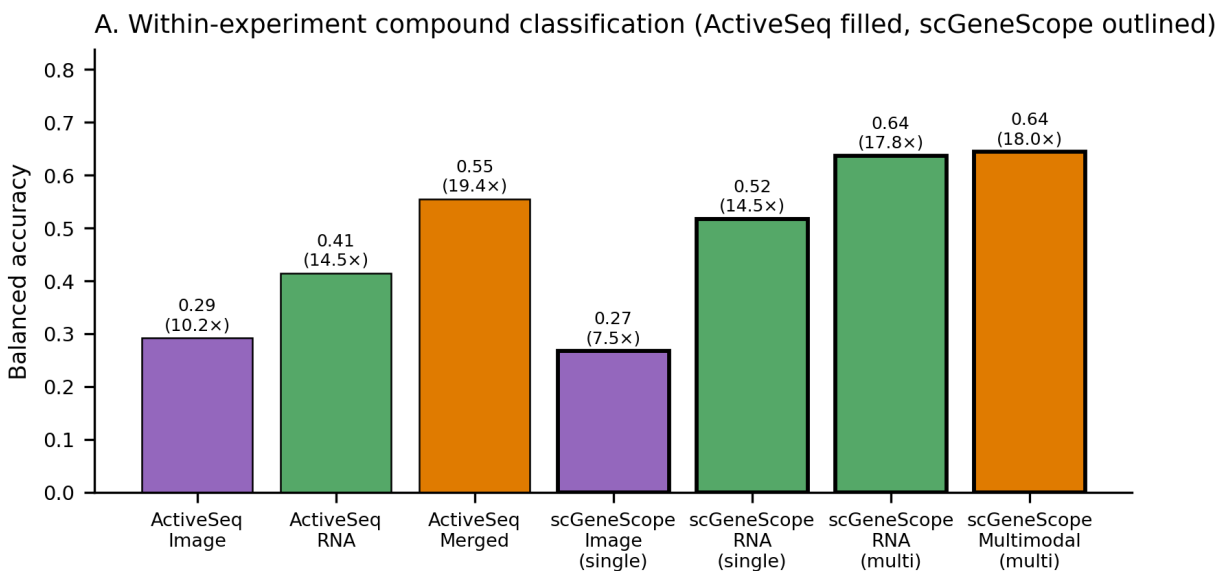
Figure 5. Centered pseudobulk gene signatures preserve compound-specific transcriptional programs across runs and recover the correct compound at 48.6% top-1 and 68.6% top-3 retrieval. The gene lists support RNA interpretability and triage rather than standalone mechanism calls.

scGeneScope provides external magnitude context

The final question is not whether ActiveSeq “beats” a public benchmark. The designs differ too much for that framing. scGeneScope profiles approximately 716,767 single-cell images and 627,704 single-cell RNA profiles across 28 treatments in U2-OS cells, with treatment-level matching across replicate populations [4]. This ActiveSeq pilot contains 11,511 control-panel matched wells across two HEK293 experiments, each read as a 5 to 10 cell pseudobulk, with a simpler imaging modality and a calibrated internal image latent.

The comparison is still useful as an order-of-magnitude check. ActiveSeq trails the strongest scGeneScope multimodal reference on raw within-experiment balanced accuracy, consistent with the scale, cell-line, and imaging-richness differences. Because the label spaces differ, fold over random-chance performance gives a rough normalization: pooled ActiveSeq RNA reaches 14.5 times chance, matching the scGeneScope single-profile RNA value; pooled ActiveSeq image reaches 10.2 times chance, compared with 7.5 times for the scGeneScope single-profile image value; pooled ActiveSeq merged features reach 19.4 times chance, compared with 18.0 times for the scGeneScope multimodal multiprofile reference.

The correct conclusion is restrained: a small same-well pilot reaches the same broad performance range after chance normalization, while measuring a different experimental unit. That supports continued scale-up. It does not establish superiority over scGeneScope or any imaging-first production system.



B. Context for interpreting the gap

Aspect	ActiveSeq (Z-Screen)	scGeneScope
Single-cell scale	~60-110k cells (5-10 × 11.5k wells)	~717k images, ~628k RNA
Label space	35 compounds	28 treatments
Pairing	Same matched micro-well	Treatment-matched replicate
Imaging	Brightfield + calibrated latent	5-ch Cell Painting + ViT
Pooled image	29.2% (10.2× chance)	26.7% single (7.5×)
Pooled RNA	41.4% (14.5× chance)	51.7% single (14.5×)
Pooled multimodal	55.5% (19.4×)	64.4% multi (18.0×)
Best ActiveSeq run	ACT010 merged 60.8% (21.3×)	n/a
Takeaway	Smaller pilot; same range after chance noise	Larger public benchmark on raw accuracy

Figure 6. Within-experiment compound classification for ActiveSeq versus the published scGeneScope reference, annotated with fold over random-chance performance, plus a context table summarizing the design differences that make the two benchmarks complementary rather than directly interchangeable.

Discussion

This pilot answers a foundational question, not the whole platform question. In 11,511 control-panel matched wells, both RNA and a calibrated image latent recover compound identity within and across runs. The two modalities share a compound map without collapsing into the same representation. When combined in a simple linear model, they improve held-out compound classification across essentially the whole panel. A same-compound shuffled-fusion control shows that this supervised gain is compound-level multimodal additivity rather than a measurable advantage of exact well-to-well pairing on the classification task.

The result is strongest at the compound level. Split-half reproducibility, cross-run centroid transfer, and supervised classification all depend on compound labels aggregated across many wells. They establish that ActiveSeq can produce a reproducible multimodal compound phenotype. They should not be described as proof that exact same-well pairing is the sole source of the gain. The within-compound analyses ask the stricter same-well question and are encouraging but modest: exact pair retrieval and distance-structure effects sit above random, but at small absolute magnitude. Larger replicate depth, stronger perturbational diversity, and prospective designs that enrich within-compound response heterogeneity are needed to determine when exact same-well pairing becomes operationally decisive.

The image result should be read with the same discipline. Calibration made compound-resolved image structure accessible, but the calibrated latent was learned on the same control-panel domain analyzed here. That is a useful internal representation for this pilot and a rational path for assay development. It is not yet a claim that the frozen image embedding or the calibration procedure will generalize unchanged to new libraries, cell types, or imaging runs.

For drug discovery, the most important downstream feature is that the RNA branch remains interpretable. The centered pseudobulk signatures do not provide definitive mechanisms of action, but they do preserve compound-specific gene programs well enough for triage: clustering hits, flagging broad stress programs, and nominating follow-up assays. This makes the multimodal gain more useful than a black-box classification result.

The competitive context is therefore directional. scGeneScope shows what is possible in a much larger treatment-matched public dataset with richer imaging and single-cell readouts. ActiveSeq reaches a similar broad range after chance normalization despite being smaller and pseudobulked, while measuring image and RNA from the same micro-well. The next experiments are clear: larger control panels, explicit unpaired-fusion controls, broader cell-line coverage, prospective chemistry beyond the calibration domain, and deeper analysis of whether same-well response-state structure becomes stronger with scale.

Methods

Dataset

The analysis uses the canonical Z-Screen ActiveSeq workspace under `data/ZScreen_Canonical_Dataset/`. Each ActiveSeq well is annotated with an experiment id (ACT010 or ACT011), a global well id, a compound assignment, an image embedding, and an RNA latent. Gene-level pseudobulk counts come from the matched H5AD object that backs the same-well RNA pipeline, repaired as documented in `DATA_REPAIR_REPORT.md`. The scGeneScope comparison uses published benchmark values transcribed from OpenReview Table 1 into `paper1a/tables/scgenescope_reference.csv`.

Per-well image embedding calibration

The original per-well image embedding was produced by a frozen pretrained vision model on raw imaging tiles. That representation captures generic visual structure but does not necessarily align compound identity with the principal directions of variance: experiment-of-origin, plate position, and microscopy artifacts can dominate compound signal in the raw geometry. Direct held-out compound classification on that embedding therefore tests the representation as much as the biology.

A 32-dimensional control-panel-calibrated latent was learned from the original embedding by a lightweight domain-adversarial network trained on the matched control panel. The objective rewards compound discrimination and penalizes experiment-of-origin discrimination, which pushes compound information onto the principal directions of variance without retraining a vision model from scratch. Calibration was learned on the same control-panel domain studied here, so the calibrated-latent results should be read as an internal representation refinement; broader-library and additional-cell-line generalization is the next test.

Matched-well preprocessing

Imaging detections were aggregated by `experiment_id` and `gwid` so each matched well contributed one image vector. Wells with an assigned control name were treated as the 35-compound panel. Wells without an assigned control name were the ZEL026 dummy-bead wells used for QC and were excluded from the compound-response analyses.

Reproducibility and transfer analyses

Within-run reproducibility was measured by repeated split-half pseudo-replication within each compound (cosine similarity between pseudo-replicate centroids after standardization within each matrix). Cross-run transfer used compound centroids learned in ACT010 and queried in ACT011.

Cross-modal geometry

For each run, pairwise distances among compound centroids were computed separately in image and RNA space. Agreement between distance matrices was summarized with Spearman correlation over upper-triangular entries, which does not assume the two scales are matched. Cross-run effect-size stability used Spearman correlation over per-compound effect magnitudes.

Supervised multimodal classification

Held-out classification used a 50:50 compound-stratified split, either pooled across runs or fit within each run. Linear discriminant analysis (eigen solver, automatic shrinkage) was trained on calibrated image features alone, RNA features alone, or their raw concatenation. LDA was used in preference to a deep classifier so that any reported gain is traceable to a linear function of the input features. Performance was summarized with overall accuracy, balanced accuracy, and per-compound recall. To test whether the supervised gain depended on exact image/RNA well pairing, a same-compound unpaired-fusion control shuffled RNA vectors among wells sharing the same compound and experiment before fitting the same merged classifier. The control used 100 shuffle iterations with the same train/test split.

Same-well representation learning

A lightweight CLIP-style dual encoder with a symmetric contrastive objective was trained on same-well pairs from the training split, evaluated by nearest-centroid compound retrieval and by exact held-out pair retrieval. Within-compound analyses compared image and RNA distance structure among wells assigned to the same compound and summarized distance-correlation, kNN-overlap, and same-compound pair-retrieval gains over random baselines. These analyses were treated as exploratory because the absolute effects were small.

Gene-level pseudobulk analysis

Raw counts were summed per experiment and control compound, converted to log-CPM, and centered by subtracting the experiment-wide mean signature. Cross-run retrieval used correlation similarity between centered signatures. Per-cell differential expression was not performed because each well averages 5 to 10 cells; pseudobulk plus centering is the unit at which the signal is robust at present depth.

External benchmark context

Within-run supervised metrics from `supervised_multimodal_metrics.csv` were compared to scGeneScope within-experiment values transcribed from OpenReview Table 1 into `scgenescope_reference.csv`. Because the label spaces differ (35 vs 28), the comparison reports both raw balanced accuracy and fold over random-chance performance.

Limitations

The pilot is intentionally narrow: one cell line (HEK293), 35 control compounds, and two matched experiments. The calibrated image latent was learned on the same control panel studied here; broader libraries and additional cell lines are needed to evaluate calibration generalization. The supervised fusion result shows additive value for combined image and RNA readouts, but the same-compound shuffled-fusion control indicates that exact well-to-well pairing does not add measurable classification value on this compound-label task. Exact same-well cross-modal retrieval and within-

compound structure remain modest in absolute terms. Gene-level signatures are summarized through pseudobulk aggregation rather than full differential-expression modeling with biological replicates. These caveats define the next analyses rather than negate the central conclusion: ActiveSeq supports reproducible compound-resolved multimodal profiling, with interpretable RNA and additive image-plus-RNA classification value.

Data availability

All data tables, derived latent representations, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper1a/` and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper1a/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

References

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Combinatorial chemistry yields a learnable transcriptomic design map in Z-Screen

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Abstract

Most phenotypic screens end as a list of active wells. Z-Screen was designed to leave behind something more reusable: a map from combinatorial chemistry provenance to measured RNA state. The platform profiles one-bead-one-compound libraries in 50,000-well microchips, with each microwell containing 5 to 10 cells and carrying both compound identity and a low-pass transcriptomic readout. Across 615,793 repaired RNA profiles spanning 12 combinatorial libraries and 4 cell lines (615,721 with valid scVI latent coordinates), the present pilot asks whether that RNA-first chemistry record is already structured enough to support prediction, candidate triage, and reuse.

The assay floor is high across the main systems. Median split-half cosine for control compounds is 0.993 in A549, 0.991 in H1650, 0.993 in HEK293, and 0.968 in THP1. On held-out tuples within a library, building-block models substantially outperform centroid baselines. The strongest case is ZEL024 in HEK293, where a building-block plus pair model reaches median cosine 0.674 against a 0.397 centroid baseline (33.2% reduction in mean squared error). ZEL031 in THP1 follows at cosine 0.833 vs 0.783 (21.3% error reduction). Chemistry-only descriptors carry predictive signal as well: in ZEL024 / HEK293, a combined building-block plus chemistry-embedding model reaches cosine 0.621 with a 29.3% error reduction, the highest gain among any chemistry-only configuration evaluated.

Full-structure descriptors add signal where the system is best sampled, but not uniformly. In ZEL024 / HEK293, a combined building-block plus chemistry-embedding model reaches cosine 0.621 with a 29.3% error reduction, exceeding either representation alone. In weaker or sparser settings, full-structure augmentation is neutral or harmful, which makes ZEL024 / HEK293 the current positive case rather than proof of a universal structure model. The same calibration applies to reference-state recovery. ZEL024 / HEK293 contains a library tuple near the internal reference ZF-104 at cosine 0.901 against a HEK293 cross-control background of median 0.200 and 99th percentile 0.931. ZEL031 / THP1 contains high-cosine neighborhoods near sorafenib, MZ1, and STC-15, but THP1 also has a higher cross-control background (median 0.589, 99th percentile 0.972), so these are candidate neighborhoods, not mechanism calls. Building-block effects recur across cell lines at chemistry-resolved positions, and imaging adds a smaller secondary benefit in additive benchmarks.

Together, the results support a constrained but useful claim: in the best-sampled Z-Screen libraries,

explicit chemistry provenance plus low-pass RNA learns reusable chemistry-resolved biological programs. Imaging is a secondary extension in the present package. The core asset today is the RNA-derived transcriptomic design map and the provenance that makes each measured state reusable.

Significance

The present analysis tests whether Z-Screen can convert a combinatorial screen into reusable transcriptomic memory. Because every compound carries building-block provenance and a measured RNA response, the dataset can ask which chemical pieces move cell state, which combinations are predictable, which new tuples occupy neighborhoods of known references, and which programs survive a change in cell line. The result is not a mechanism assignment engine. It is a calibrated design map that can prioritize chemistry for follow-up.

Introduction

Early-stage discovery needs two kinds of information at once. It needs scale, because most chemistry is inactive or uninteresting in a given context. It also needs enough biological resolution to make a hit useful after the primary screen. Imaging platforms provide scale and rich morphology, but morphology alone is often difficult to translate into a chemical design decision. Transcriptomic profiling is closer to pathway state, but it is usually too expensive to apply deeply across combinatorial chemistry. Public transcriptomic atlases such as LINCS L1000 and Tahoe-100M established the value of perturbational RNA at large scale [2,3], while public multimodal efforts such as scGeneScope show the value of aligning imaging and RNA across matched treatments [4].

The gap addressed here is chemistry provenance. A conventional compound atlas can tell a discovery team that one molecule resembles another in expression space. It usually cannot say which building block, linker, or combinatorial position contributed to that state, because the chemistry was not generated as a measured design grammar. Z-Screen starts from the opposite end. Each compound is a tuple from a one-bead-one-compound library, and each measured RNA response can be traced back to building-block coordinates and full compound structure. That makes the primary question practical: does the RNA readout contain enough reproducible, chemistry-resolved signal to predict held-out programs and guide the next synthesis decision?

The present manuscript treats imaging as a current extension rather than the main claim. Same-well morphology is valuable, and companion paper1 focuses on that readout. In this paper, the evidence chain is RNA-first: reproducibility establishes the assay floor, building-block provenance tests predictability, calibrated reference-state neighborhoods test discovery-style triage, cross-cell comparisons ask whether programs can be reused, and the final multimodal benchmark asks whether imaging adds signal on top of the RNA map.

The analyses below characterize what the canonical dataset, itself assembled from a small number of pilot experiments, currently supports. Six concrete questions are addressed:

1. Are control compounds reproducible enough to support modeling?

2. Does building-block composition predict held-out transcriptomic states?
3. Where does full chemical structure add predictive signal beyond building-block identity?
4. Do library tuples recover recognizable reference-state neighborhoods once cross-control similarity backgrounds are accounted for?
5. Do building-block effects recur across cell lines?
6. Does imaging contribute secondary value in a restrained multimodal benchmark?

Each question is answerable on the present pilot. The aim is to define what the current data already supports, where it is strongest, and which next analyses are required before broader claims about out-of-family chemical generalization or mechanism can be made.

Results

Control profiles establish a stable assay floor across cell lines

The chemistry-prediction question is only meaningful if the RNA readout is stable enough to support it. Split-half reproducibility for named controls is high across the main systems. Median control cosine is 0.993 in A549, 0.991 in H1650, 0.993 in HEK293, and 0.968 in THP1. THP1 sits below the others because its compound-control library coverage is sparser, not because the readout is weaker on monocytes.

These numbers establish the floor for the rest of the paper. They also leave headroom between technical reproducibility and model performance: a model reaching cosine 0.674 is not capped by an unstable assay. With reproducibility established, the next question is whether the chemistry provenance attached to each RNA profile is predictive.

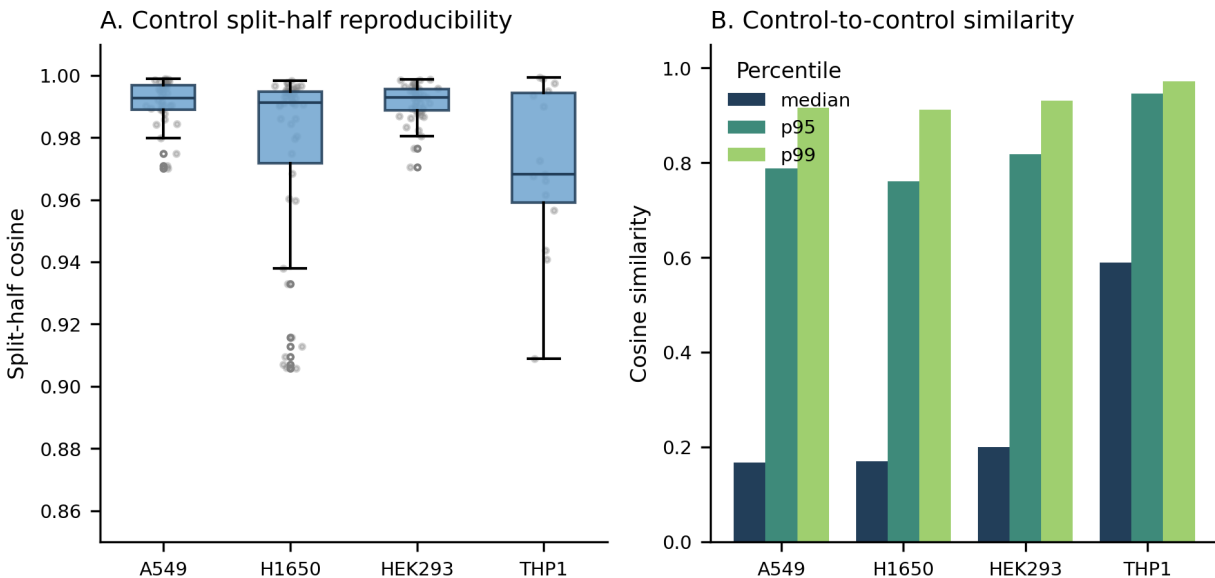


Figure 1: Figure 1. Named controls show high split-half reproducibility across all four cell lines, establishing a strong assay floor for downstream chemistry-to-phenotype modeling.

Building-block models predict held-out compound programs

The central modeling question is whether compositional chemistry carries enough signal to predict the transcriptomic state of a held-out library member. Across several library and cell-line systems, the answer is yes, with the strongest evidence coming from the best-sampled libraries.

ZEL024 in HEK293 is the strongest case. A building-block plus pair model reaches median cosine 0.674 against a 0.397 centroid baseline, a 33.2% reduction in mean squared error. ZEL031 in THP1 follows at cosine 0.833 vs 0.783 (21.3% error reduction). ZEL031 in A549 and ZEL024 in H1650 also showed positive gains, although less pronounced.

The model does more than memorize whole compounds within the tested split. It learns that particular chemical pieces and combinations of pieces carry recurring transcriptomic programs, which supports design-time reasoning about combinations not yet screened. The boundary is important: within the broader generalization-ladder framework laid out in companion paper³, this benchmark is interpolation within the shared building-block vocabulary of each library. Harder held-building-block and scaffold-family transfer regimes are evaluated separately there.

Once building-block provenance predicts RNA state, the next question is whether full chemical structure adds information beyond the library grammar.

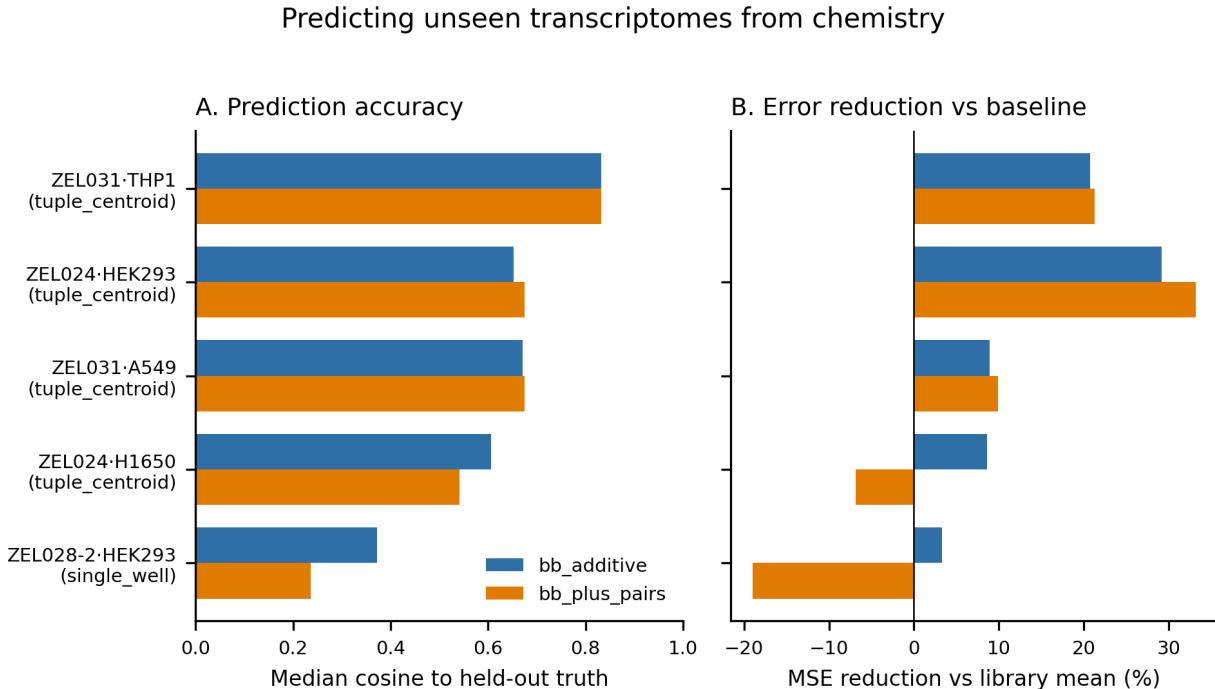


Figure 2: Building-block models improve prediction of held-out transcriptomic states across multiple library and cell-line systems, with the strongest gains in ZEL024 / HEK293 and ZEL031 / THP1.

Full chemical structure adds signal mainly in the strongest ZEL024 / HEK293 setting

Building-block identity is the strongest single signal in this dataset. Full structure adds further signal most clearly in ZEL024 / HEK293. In that system, a 256-dim chemistry embedding alone (a private random projection of ECFP4 fingerprints, see Methods) reached cosine 0.611 (27.8% error reduction), building-block identity alone reached 0.611 (27.5% error reduction), and combining the two reached 0.621 (29.3% error reduction).

That positive result should not be generalized to every current system. In ZEL024 / H1650, ZEL031 / A549, and ZEL031 / THP1, adding the full-structure embedding to building-block identity did not improve the mean error reduction over the building-block-only model. In sparse ZEL028-2 / HEK293, all chemistry-only variants performed below the mean-RNA baseline. The present conclusion is therefore narrow: full-structure descriptors can add design-time signal when coverage and assay context are favorable, with ZEL024 / HEK293 as the clearest example.

This hierarchy is useful rather than disappointing. Building-block provenance is the robust design coordinate today. Full structure is a candidate extension that becomes valuable when the measured library is dense enough for substructure information to be learned. Companion paper³ quantifies how much of the chemistry-only gain reflects interpolation within a library and how much survives stricter held-building-block evaluation. Having established predictability, the next question is whether the measured map can recover neighborhoods around known reference states.

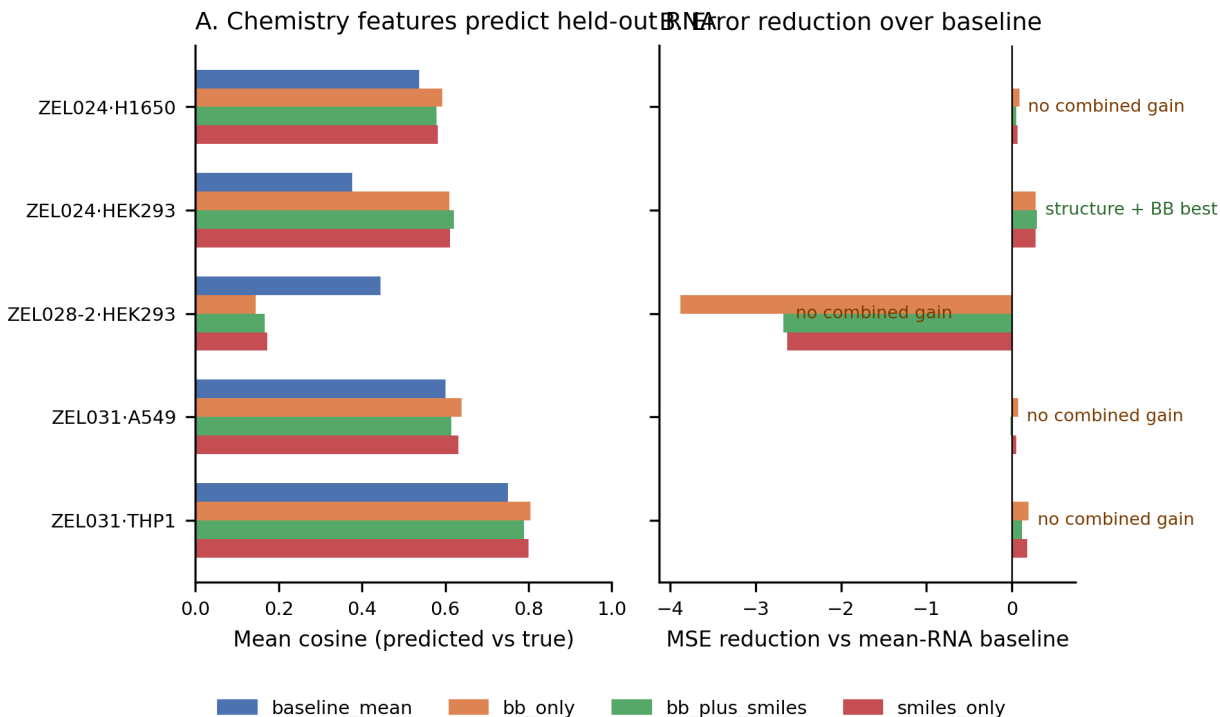


Figure 3: Chemistry-only models capture predictive signal unevenly across systems. Building-block provenance is the most reliable coordinate, and adding full-structure descriptors improves performance most clearly in ZEL024 / HEK293 while failing to improve or hurting in weaker and sparser settings.

Library tuples recover calibrated reference-state neighborhoods

The most direct discovery-style use of the map is not to prove mechanism, but to recover candidate neighborhoods around known reference compounds. A library tuple whose transcriptomic centroid lands near a reference centroid is worth follow-up only after calibration against the same-cell-line background of unrelated control-control similarities.

The cleanest neighborhood result comes from ZEL024 / HEK293, where a library tuple matched the internal reference ZF-104 at cosine 0.901. The HEK293 between-control similarity background sits at median 0.200 with a 99th percentile of 0.931. That places the 0.901 match far above the bulk of unrelated controls but still below the most extreme control-control tail. Several additional ZEL024 / HEK293 tuples reached cosine 0.85 to 0.89 against published references such as ZEL029-25 and ZEL029-28.

Candidate neighborhoods also appear in ZEL031 / THP1, with tuples reaching cosine 0.942 to sorafenib, 0.926 to MZ1, and 0.918 to STC-15. THP1 is useful for several pharmacology programs, but its cross-control similarity background is substantially higher (median 0.589, 99th percentile 0.972). The THP1 hits are biologically meaningful and populate the upper tail of plausible neighborhoods, but they should not be promoted as shared-mechanism evidence without orthogonal validation. The

THP1 background is reported explicitly so the calibration is visible to readers.

The defensible claim is calibrated candidate recovery. Z-Screen produces new chemistry that lands in recognizable RNA neighborhoods, and the strongest current case comes from a cell line where the cross-control background is low enough to make the neighborhood specific. These neighborhoods nominate follow-up experiments; they do not by themselves establish target engagement or mechanism. The next question is whether the chemistry programs behind these neighborhoods remain reusable when the cellular context changes.

Library tuples vs nearest reference compound

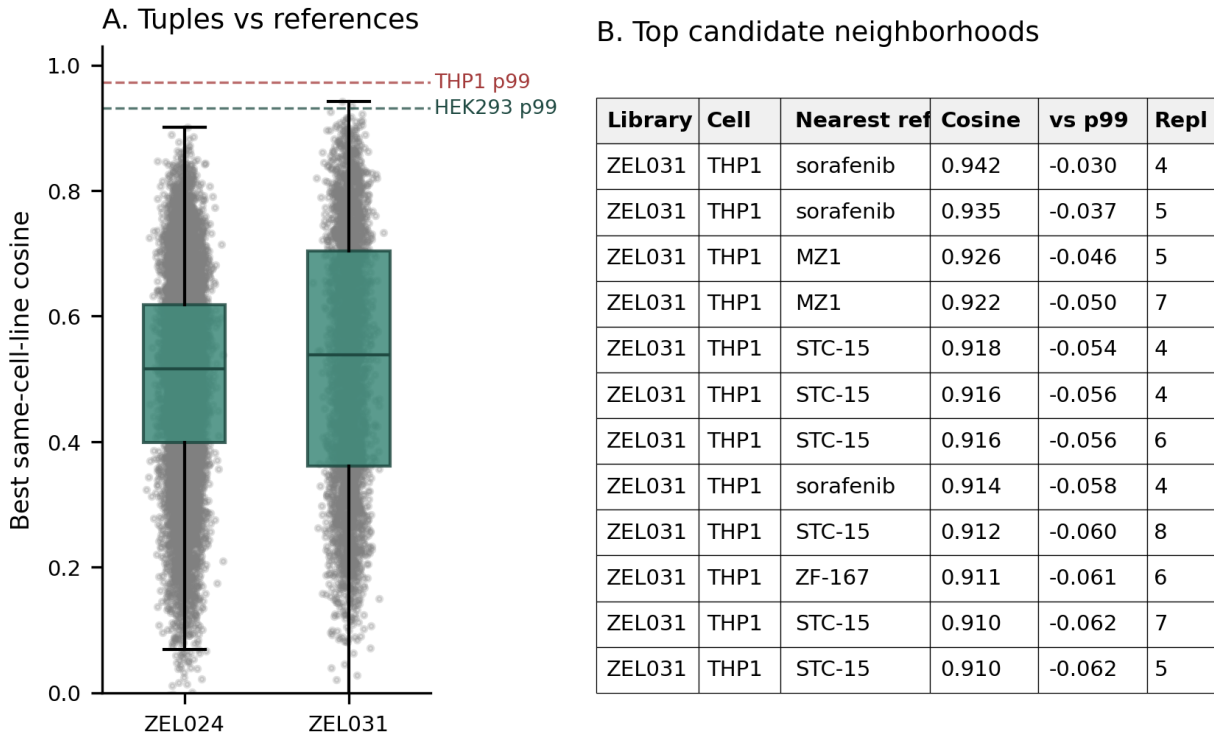


Figure 4: Figure 4. Multiple library tuples land near transcriptomic neighborhoods centered on known reference compounds. These are calibrated candidate neighborhoods, not mechanism assignments. The strongest current case is HEK293 (ZF-104 at cosine 0.901, against a HEK293 99th-percentile cross-control background of 0.931); THP1 hits to sorafenib, MZ1, and STC-15 occur in a system with a higher cross-control baseline.

Building-block programs recur across cell lines

A reusable chemistry model is more valuable when its building-block programs survive a change in cellular context. Per-building-block transcriptomic programs were compared in matched library systems across paired cell lines. Results are reported by building-block position because positions with very few unique values (e.g. ZEL024 bb0, bb1, bb2) collapse hundreds-to-thousands of distinct chemistries into a single average and therefore measure library-wide rather than chemistry-resolved

transfer. The chemistry-resolved positions are ZEL024 bb3 (83 building blocks, median 168 unique tuples averaged per signature) and ZEL031 bb0 / bb1 (a 2-position library, where each single-position signature still leaves the other position to vary across approximately 70 to 100 distinct partner chemistries).

Across these chemistry-resolved positions, building-block programs partially survive cell-context change. In ZEL024, the median cross-cell cosine at bb3 between HEK293 and H1650 was 0.602 ($n = 83$, IQR 0.51 to 0.69, strongest building blocks reaching 0.873). In ZEL031 between A549 and THP1, the median cross-cell cosine was 0.355 at bb0 ($n = 105$, IQR 0.23 to 0.48, strongest 0.747) and 0.526 at bb1 ($n = 84$, IQR 0.32 to 0.67, strongest 0.808).

These medians sit meaningfully below the within-cell reproducibility ceiling, so the result is not a claim of perfect transfer. The defensible interpretation is that many building blocks carry partially conserved biological directionality across cell types, which is the prerequisite for reusing chemistry across programs. The ZEL024 bb0 / bb1 / bb2 cross-cell results are not chemistry-resolved at this library architecture (each averaged signature reflects 900 to 7,000 distinct tuples) and are therefore reported as diagnostics in `paper2a/tables/bb_effect_consistency.csv` rather than as headline claims. Companion paper4 extends the cross-cell question to the molecule and grouped-program level, where the transfer story strengthens substantially in chemistry-resolved settings.

Cross-cell recurrence makes the RNA map more reusable, but it does not make imaging an equal pillar in this paper. The final Results analysis therefore asks a narrower question: whether the current image features add measurable signal on top of RNA in a simple additive benchmark.

Shared BB program consistency across cell lines

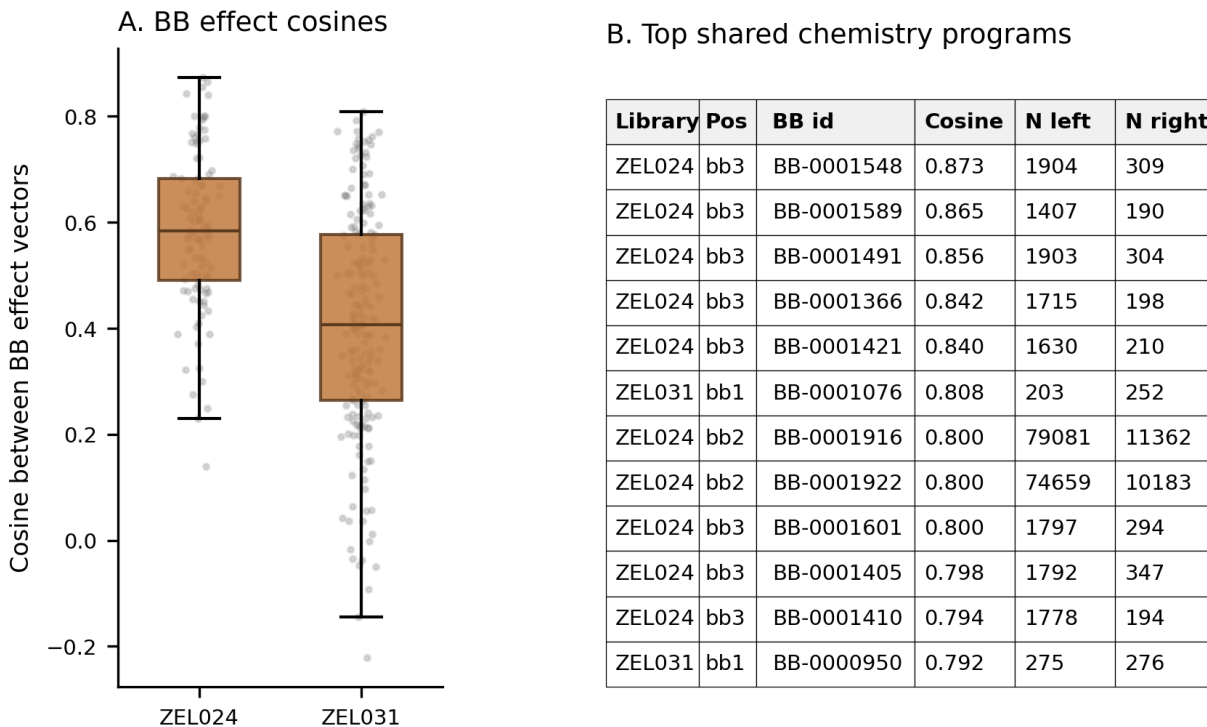


Figure 5: Building-block effects partially recur across cell lines at chemistry-resolved positions, with positive median transfer and a subset of building blocks showing strong cross-context conservation.

Imaging adds a modest secondary signal in a simple multimodal benchmark

The evidence in this paper is RNA-first, and the image branch is best read as a secondary extension in this analysis. In the strongest additive multimodal benchmark, based on ZEL024 controls, image-only classification reached 0.213 balanced accuracy, RNA-only reached 0.524, and image plus RNA reached 0.540. Adding chemistry to image plus RNA reached 0.611, consistent with the broader conclusion that provenance is the dominant non-RNA signal in this paper. The smaller ZEL031 recurrent-tuple benchmark followed the same ordering for image and RNA at lower absolute values: 0.127 for image, 0.146 for RNA, and 0.180 for image plus RNA. Chemistry-only and chemistry-containing models reach perfect accuracy in that small ZEL031 task, but that benchmark contains only 7 top-tuple classes with about 30 held-out examples and should be treated as a diagnostic rather than a broad multimodal claim.

The multimodal claim at the present pilot is measured. Imaging is not the lead modality today, but it adds directionally useful signal in an additive classification task. Companion paper1 shows a larger image contribution in the same-well ActiveSeq matched-readout setting after task-aware calibration of the per-well image latent, suggesting that the modest gain reported here reflects current image-pipeline maturity rather than a ceiling on same-well multimodality.

ActiveSeq-style additive multimodal probe

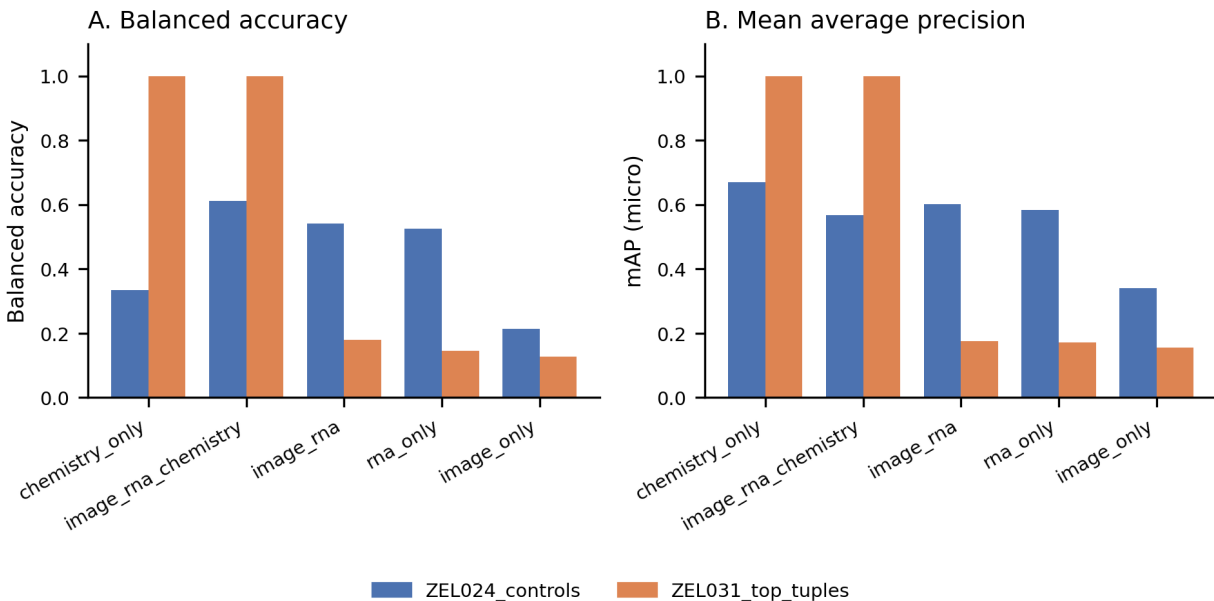


Figure 6: Figure 6. In a simple additive benchmark, image plus RNA modestly outperforms either single modality alone in ZEL024 controls, while RNA and chemistry provenance remain the dominant signals in the current dataset.

L1000 concordance provides an external validation check

The preceding analyses are internal to Z-Screen. A separate validation question is whether Z-Screen rank signatures resemble an independent transcriptomic platform where the same compounds and cell lines overlap. Of 47 named Z-Screen control compounds, 14 (29.8%) appear in the LINCS L1000 CMap LINCS 2020 build. Three of four Z-Screen cell lines (HEK293, A549, THP1) are present in LINCS L1000 with sample coverage; H1650 is absent.

For the concordance test, Z-Screen rank signatures were restricted to the 12,328-gene LINCS-visible universe so the two platforms were scored in the same gene space. Spearman rank correlation was then computed between Z-Screen up/down rank signatures and LINCS L1000 level-5 consensus signatures (NCBI GEO GSE70138 and GSE92742) for each adequately covered compound and cell-line pair. Across 11 such pairs (10 in A549, 1 in HEK293), all 11 produced positive cross-platform Spearman correlation. Median rho was 0.490, mean 0.393, and 8 of 11 pairs cleared $\rho = 0.2$. Five compounds reached empirical $p \leq 0.048$ against a same-cell-line unmatched-compound permutation null: ZM-336372 ($\rho = 0.625$, $p = 0.022$), rucaparib ($\rho = 0.530$, $p = 0.023$), sorafenib in A549 ($\rho = 0.505$, $p = 0.023$), veliparib ($\rho = 0.490$, $p = 0.046$), and crizotinib ($\rho = 0.679$, $p = 0.048$). Two compounds (palbociclib at $\rho = 0.182$, GSK126 at $\rho = 0.046$) fall in the bulk of the unmatched distribution; the GSK126 result is consistent with the documented slow transcriptional onset of EZH2 inhibition under LINCS’s 6 h treatment window.

This external comparison is not a benchmark victory claim. It is a sanity check that Z-Screen’s low-pass RNA signatures recover compound-specific transcriptional structure reported by an independent platform, using a gene-restricted comparison with no Z-Screen-specific calibration.

Discussion

Z-Screen behaves as a learnable chemistry-to-transcriptome system on data from a small number of pilot experiments. The evidence chain is intentionally sequential. Control reproducibility shows that low-pass RNA is stable enough for modeling. Building-block models show that explicit chemistry provenance predicts held-out transcriptomic programs. Reference-state analysis shows that library tuples can recover calibrated neighborhoods around known compounds. Cross-cell comparisons show that some building-block programs remain partially reusable across cellular contexts. Imaging then adds a modest secondary signal on top of the RNA-first map.

There is also a clear hierarchy in the current evidence. RNA carries the strongest biological signal today, and building-block provenance is the most reliable chemistry coordinate. Full-structure descriptors add signal in ZEL024 / HEK293, the best-sampled positive case, but they do not yet define a universal full-structure model across all libraries and cell lines. Imaging is directionally positive here and considerably stronger when calibrated for same-well matched readout in companion paper1. In this paper, the imaging branch is best framed as an additive layer on a working chemistry-to-RNA model.

The external position is therefore specific. Tahoe-100M is much larger on raw RNA scale, LINCS L1000 remains the deepest public small-molecule transcriptional catalog, scGeneScope is the closest public treatment-matched multimodal benchmark, and Cell Painting remains the standard high-throughput morphology assay [2-5]. Z-Screen is different because the chemistry is generated and measured as a compositional grammar. Every well carries building-block provenance, full compound identity, and an RNA response. The contribution is not raw atlas size; it is a design-relevant transcriptomic map in which a hit can be decomposed back into chemical choices.

The most informative next experiments follow directly. The first is greater depth in sparse libraries, especially systems that underperform when coverage per building block is low. The second is a prospective cross-library benchmark that asks whether a model trained on one scaffold family can predict transcriptomic programs in another; companion paper3 sets this up explicitly. The third is continued improvement in image consistency so that morphology can contribute more strongly to the readout. For a discovery team, the economic implication is that a screen run on this platform can leave behind a reusable model of chemistry-resolved RNA state rather than only a list of wells.

Methods

Dataset and preprocessing

The analysis uses the repaired canonical Z-Screen workspace, which contains RNA profiles, scVI latent coordinates, chemistry annotations, and selected image-derived features across 12 combinatorial libraries and 4 cell lines. Named controls were retained for reproducibility and reference-state neighbor-

hood analyses. Transcriptomic models were evaluated on library and cell-line subsets with sufficient compound coverage. The 72 ZIC004 / A549 rows without valid scVI latent coordinates were retained in the RNA count table but excluded from latent-space modeling; this is the source of the small mismatch between `canonical_obs.parquet` (615,793 rows) and `canonical_scvi_latents.parquet` (615,721 rows with valid coordinates) noted in the package README.

scVI latent space rather than gene-level expression

Most analyses operate on a 32-dimensional scVI latent space rather than full gene-level expression vectors, for two reasons. First, low-pass per-well mRNA-seq (each well is a pseudobulk of 5 to 10 cells) is dominated by sampling noise in any single gene; scVI’s variational latent absorbs that noise into shared structure across observations. Second, the latent is comparable across cell types to a degree that raw expression is not, which is the property the cross-cell transfer experiments in companion paper4 require. Where gene-level signal is the appropriate unit (reference-state neighborhood recovery, gene-program inspection in companion paper1) centered pseudobulk log-CPM is used directly.

Control reproducibility

Control split-half reproducibility was computed from repeated random partitions of wells belonging to the same reference compound. Similarity was summarized with cosine similarity between pseudo-replicate centroids in scVI latent space. Repeated partitioning rather than a single split was used to keep the variance of the metric low.

Building-block predictive modeling

Held-out tuple analyses fit additive and additive-plus-pair building-block models to transcriptomic centroids and evaluated them on compounds excluded from training by tuple identity. The choice of additive plus pair (rather than a high-order interaction model) reflects the present library coverage: pair-level interactions are well sampled in ZEL024, while higher-order terms run into combinatorial explosion at present scale. Performance was summarized with median cosine similarity to the measured transcriptomic target and mean squared error reduction relative to a centroid baseline. Companion paper3 extends this analysis to the harder building-block holdout and cross-library transfer regimes.

Chemistry-to-RNA prediction

Chemistry-only benchmarks compared models using a 256-dim per-compound chemistry embedding, building-block one-hot identities, or their concatenation. The chemistry embedding is a private fixed-seed Gaussian random projection from R^{2048} to R^{256} applied to per-compound 2048-bit ECFP4 (Morgan radius 2) fingerprints, then L2-normalized; this representation preserves Tanimoto-like geometry under cosine similarity (Johnson-Lindenstrauss) while being mathematically irreversible to substructure information without the projection matrix, which is held privately in the NDA bundle. A fixed deterministic projection was used in preference to a learned chemistry encoder

because the projection is reproducible across runs, requires no separate training step, and decouples the published chemistry features from the underlying SMILES strings. Outputs were scored against measured transcriptomic centroids using cosine similarity and mean squared error reduction relative to a centroid baseline. Full-structure claims were interpreted system by system rather than pooled across libraries.

Reference-state neighborhood recovery, cross-control normalization, and cross-cell building-block analysis

Reference-state neighborhood recovery identified library tuples whose transcriptomic centroids were most similar to named reference compounds within the same library and cell line. Hits were calibrated against the per-cell-line cross-control similarity background (median, 95th percentile, 99th percentile cosines among unrelated reference compounds in the same cell line), so that “high” similarity is interpreted relative to the cell-line baseline rather than against zero. The analysis nominates candidate neighborhoods for follow-up and was not used as proof of shared mechanism. Cross-cell building-block consistency compared the direction of per-building-block transcriptomic programs across paired cell lines within the same library.

Additive multimodal benchmark

The multimodal probe followed a conservative additive framing: image features, RNA features, and their concatenation were evaluated on held-out class prediction tasks constructed from the available image-linked data. Performance was summarized with balanced accuracy and mean average precision. The additive frame was chosen because it produces a result traceable to a specific feature axis; cross-attention or contrastive multimodal models can in principle do better, but the gain is harder to attribute to one modality.

External benchmark context

Tahoe-100M, scGeneScope, L1000, Cell Painting, and Z-Screen statistics were assembled into a comparison table covering RNA scale, image scale, cell-line coverage, perturbation diversity, pairing structure, and a representative result per dataset. The comparison was used only for Discussion-level positioning, not to claim parity on raw scale or multimodal balanced accuracy.

Limitations

The current analysis is based on a small number of pilot experiments rather than a fully saturated atlas. RNA carries the strongest signal today; the image branch is directionally positive and depends on derived feature artifacts rather than raw microscopy in this shareable package. Full-structure descriptors add clear signal mainly in ZEL024 / HEK293 and should not yet be treated as a universal structure-to-RNA model. Reference-state neighborhoods are calibrated against same-cell-line cross-control backgrounds, but even the strongest discovery-style hits require orthogonal experimental validation before they should be treated as confirmed mechanisms of action. Cross-library and

prospective scaffold-family transfer are set up by the present results but are not claimed as solved here.

Data availability

Data tables, derived image features, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper2a/`, `data/ZScreen_Canonical_Dataset/`, and `data/paper2_artifacts/`. Raw microscopy images are not included in the shareable bundle; the paper2a image-linked analyses use the derived parquet feature files documented in the package README. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper2a/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

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A generalization ladder for chemistry-to-phenotype prediction in combinatorial screening

Zafrens, Inc.

Abstract

Machine learning claims in drug discovery depend on which version of generalization is being tested. Predicting a new combination of familiar chemical pieces is one problem. Predicting a compound built from unseen pieces is harder. Predicting an entirely new scaffold family is harder still. The BELKA literature on DNA-encoded libraries made this hierarchy concrete for binary binder labels, and it gives useful vocabulary for asking the same question in dense phenotypic screening [1,2]. We use that vocabulary as motivation, not as a head-to-head superiority claim. The question here is narrower and more actionable: when each compound contributes a measured transcriptomic state, where does a chemistry-to-phenotype model interpolate, where does it extrapolate beyond retrieval, and where does it fail? We answer that question in Z-Screen, a 50,000-well one-bead-one-compound microchip assay that profiles each well by imaging and mRNA sequencing at 5 to 10 cells per well, using a five-rung ladder of progressively harder held-chemistry evaluations.

The strongest evidence for genuine extrapolation comes from ZEL031 in THP1 cells. Under strict multi-axis holdout, where every substantive building block in the test compound is unseen during training, the model retains 0.765 cosine to the held-out transcriptomic state, exceeds a chemistry-embedding nearest-neighbor retrieval baseline by 0.168 cosine, wins on 90% of test compounds, and stays positive in all 10 random draws under paired bootstrap intervals at a median nearest-training cosine of 0.642 in the public chem-embedding space (see Methods). Single-position building-block holdouts in the same system give consistent results, with the model exceeding retrieval by 0.15 cosine on average and 89% to 91% per-compound win rates on test sets of more than 700 compounds whose nearest training neighbors have embedding cosine in the 0.78 to 0.80 range.

ZEL024 in HEK293 produces high quality on a different problem. The library is sufficiently saturated (12 by 7 by 2 by 83 building-block grid, 13,769 observed compounds out of a 13,944 possible-tuple ceiling) that nominal held-building-block splits still leave test compounds with near-duplicate training neighbors at median nearest-training cosine 0.88 to 0.99. Under that condition, ZEL024 / HEK293 delivers high-quality matrix completion within a dense combinatorial grid, a useful capability for prioritizing chemistry within an explored design space, while ZEL031 / THP1 delivers genuine out-of-vocabulary extrapolation. The hardest test, cross-library scaffold-family transfer in the same cell line, remains the next decisive experiment rather than a solved problem on the current data: across three workable train-test pairs, the best absolute cosine to the held-out library

phenotype was 0.194 and the largest gain over nearest-neighbor retrieval was 0.096.

In aggregate, the ladder gives a bounded claim. Dense phenotypic readouts can support extrapolation beyond nearest-neighbor retrieval in a favorable system, and can support high-quality matrix completion in a saturated library. They do not yet show uniform scaffold-family transfer. The weak L5 result is therefore not hidden; it defines the prospective same-cell scaffold-family hop that should decide the hardest claim.

Significance

Machine learning in early discovery is only worth its cost if the model still helps when the chemistry changes. This paper separates the easy version of that claim from the hard one. In one Z-Screen system, the model fills in missing entries inside a saturated combinatorial grid, which is genuinely useful for prioritizing chemistry within a design space. In another, the model continues to outperform nearest-neighbor retrieval after multiple building-block axes have been held out, rather than copying close analogs. The contrast is the central result: the data show, with calibrated rigor, where this assay can interpolate, where it can extrapolate, and which experiment should come next.

Introduction

DNA-encoded library (DEL) platforms transformed the economics of binder discovery by reading very large chemical libraries against immobilized targets [3,4]. BELKA made a useful modeling point from that setting: a split that holds out recombinations of familiar chemistry is not the same as a split that withholds building blocks, and neither is the same as a scaffold-family shift [1,2]. Several follow-up studies reached compatible conclusions through different routes [5,6,7]. In this manuscript, BELKA is used for motivation and vocabulary. It is not treated as a direct comparator, because the readouts, targets, chemistry, and labels are different.

The open question for dense phenotypic screening is therefore not a contest with BELKA. It is whether a richer endpoint changes the generalization problem in useful ways. A screen that reads out a transcriptional state for every compound has no 99% majority class of non-hits to discard. Every training example contributes a structured endpoint. That structure could help a model move beyond nearest-neighbor lookup, but it could also simply make interpolation inside a dense combinatorial grid look better than it is. Testing the difference requires more than a single train/test split; it requires a ladder of held-chemistry evaluations, with explicit baselines at each rung.

The ladder is defined and run in Z-Screen.

- **L1: unseen tuple within a shared building-block vocabulary.** The easiest rung: every individual building block has been seen, only the full combination is held out.
- **L2: held-building-block at one position.** The model has never seen one specific building-block identity at one position during training.
- **L3: strict multi-axis holdout.** Every substantive building-block axis in the test compound is unseen during training. This is the hardest within-library rung.

- **L4: chemical-neighborhood holdout.** Held-out compounds are chemical-neighborhood clusters, seeded by greedy farthest-point sampling in the public chem-embedding cosine-distance space, holding chemistry at arm’s length while staying within one reaction grammar.
- **L5: cross-library scaffold-family transfer in the same cell line.** Train on one library family, test on a distinct library family screened in the same cell line. The closest analog to the BELKA hardest shift.

Each rung is evaluated against the same two baselines: a train-mean phenotype predictor and a chemistry-embedding nearest-neighbor retrieval baseline that copies the measured phenotype of the closest training compound by cosine similarity in the public 256-dim embedding space. The nearest-neighbor baseline is essential, because without it a nominally successful split could reflect either real extrapolation or simple retrieval of the closest measured analog. The ladder is designed to make that distinction visible at every rung.

Results

The five-rung ladder turns generalization into a decision tree

The first uncertainty is definitional. A model can look useful because it fills in missing tuples among familiar building blocks, because it learns something about unseen building blocks within one reaction grammar, or because it transfers to a new scaffold family. Those are different risks for a discovery program. We therefore organized the benchmark as a decision tree rather than a single OOD score.

The same modeling framework was applied at every rung: a 256-dim per-compound chemistry embedding (private Gaussian random projection of 2048-bit ECFP4 fingerprints, see Methods), training-split-refit building-block one-hot encodings where the vocabulary exists on both sides of the split, ridge regression with alpha equal to 10, and the two baselines above. Performance was scored with row-wise cosine similarity to the measured phenotype, mean squared error reduction relative to the train-mean baseline, delta cosine versus chemistry-embedding nearest-neighbor retrieval, and per-compound win rate against retrieval.

Ridge regression was used in preference to a neural network. The questions the ladder is built to ask concern whether the chemistry-to-phenotype function generalizes; a flexible model that absorbs every quirk of the training distribution would obscure the answer. A linear model on the chemistry embedding is a deliberately conservative test, and any gain it produces is easier to attribute to the structural axis being held out.

OOD ladder: Z-Screen within- and across-library prediction

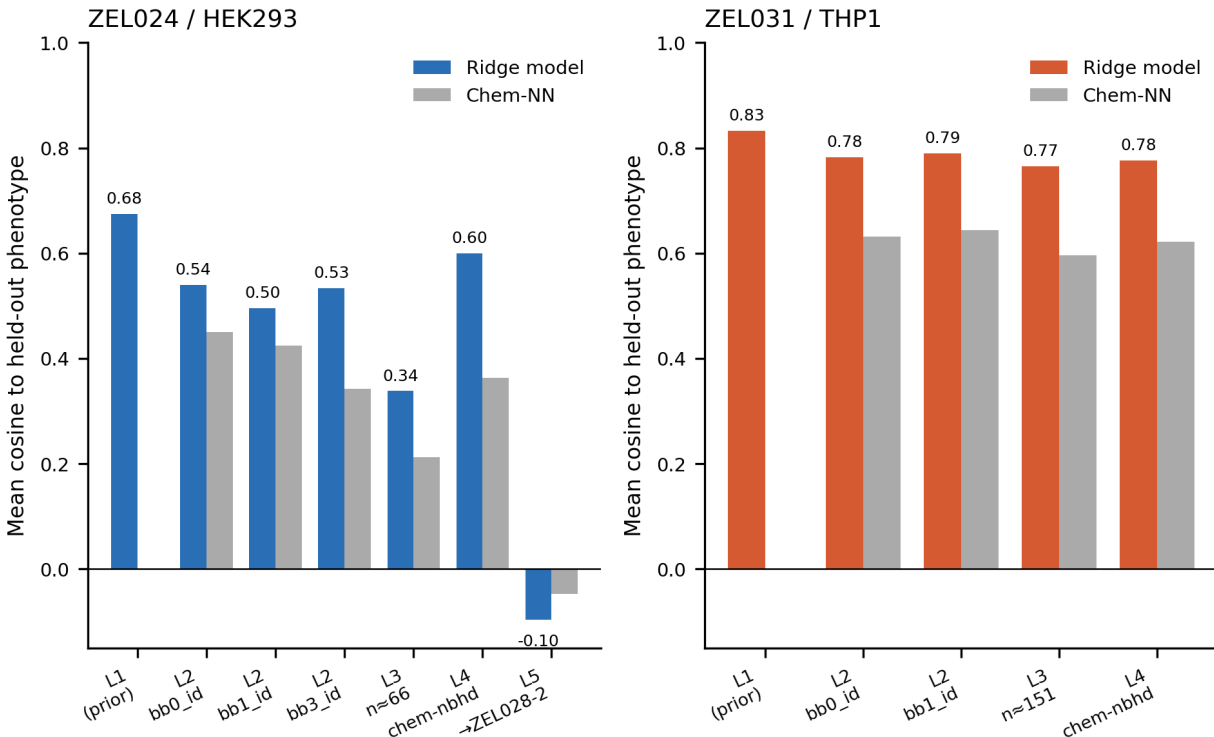


Figure 1: Figure 1. The generalization ladder is an experimental decision tree. L1 asks whether missing tuples can be filled in when all building blocks are familiar. L2 and L3 ask whether held building blocks can be predicted within a library. L4 asks whether gains survive chemical-neighborhood holdout within one reaction grammar. L5 asks whether a model transfers across scaffold families in the same cell line.

L2 to L3: ZEL031 / THP1 beats retrieval after held-building-block splits

After L1 establishes the easy interpolation case, the next uncertainty is whether any system still helps when the model cannot rely on already-seen building-block identities. The clearest answer comes from ZEL031 in THP1 cells. Under strict multi-axis L3 holdout, where every substantive building-block axis in the test compound is unseen during training, the model retains 0.765 cosine across 10 random draws of an average of 151 test compounds. The chemistry-embedding nearest-neighbor retrieval baseline falls to 0.597 on the same test sets, an average delta of 0.168. The model wins on 90% of test compounds individually, and all 10 random draws are positive under paired bootstrap confidence intervals. Median nearest-training cosine in the public chem-embedding space is 0.642 on the same compounds; this is random-projected ECFP4 cosine, not Tanimoto on raw fingerprints.

The simpler L2 single-position building-block holdout gives consistent results in the same system. Holding out bb0 identities, the model reaches 0.783 cosine versus 0.632 for nearest-neighbor retrieval,

a delta of 0.151 on an average of 728 test compounds. Holding out bb1 identities, the model reaches 0.790 versus 0.644, a delta of 0.146 on an average of 729 test compounds. Median nearest-training embedding cosine remains at 0.801 and 0.781 respectively, win rates are 89% and 91%, and all 10 random draws are positive under paired bootstrap intervals.

This is the beyond-retrieval held-chemistry result in the manuscript. It does not prove that every library will extrapolate, and it does not prove scaffold-family transfer. It shows that, in a favorable library and cell-line pair, held-building-block extrapolation survives a retrieval baseline and repeated random draws. That result creates the next question: whether the same apparent success in a denser system means the same thing.

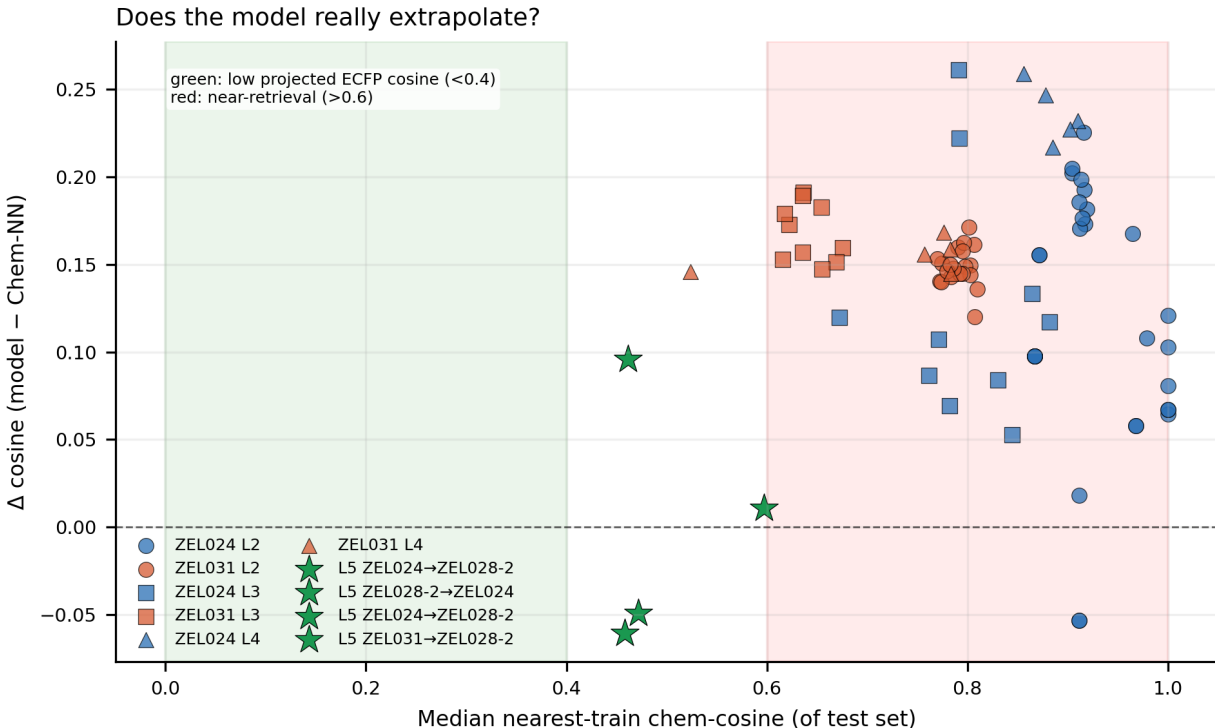


Figure 2: Figure 2. Retrieval-versus-extrapolation analysis plots model gain against median nearest-training similarity measured as cosine in the random-projected ECFP4 embedding, not raw-fingerprint Tanimoto. ZEL031 / THP1 sits in the lower-neighbor-similarity, high-model-gain regime, consistent with held-chemistry extrapolation rather than nearest-neighbor copying. ZEL024 / HEK293 sits in the high-neighbor-similarity regime, consistent with high-quality matrix completion within a saturated combinatorial grid.

L2 to L3: ZEL024 / HEK293 is positive matrix completion, not the same extrapolation claim

The ZEL031 result leaves an obvious risk: perhaps every positive held-building-block result should be called extrapolation. ZEL024 / HEK293 shows why that would be wrong. Single-position holdout in this library is positive and useful, but the library structure makes the task closer to matrix

completion. Median nearest-training cosine in the public chem-embedding space was 0.988 for bb0 holdout, 0.881 for bb1, and 0.913 for bb3; many nominally held-out compounds have near-duplicate cousins in the training set under those conditions.

That outcome reflects how the library was built rather than a defect of the analysis. High-quality matrix completion inside a dense 12 by 7 by 2 by 83 combinatorial grid (13,769 observed compounds out of a 13,944 possible-tuple ceiling) is valuable for prioritizing chemistry within an explored design space: the assay reliably fills gaps inside a saturated chemistry plate. It is a different deliverable from out-of-vocabulary extrapolation, and the strict L3 split clarifies the distinction. Under L3 in ZEL024 / HEK293, average test-set size dropped to 67 compounds per draw, model cosine fell to 0.316, and mean squared error reduction turned negative against the train-mean baseline even though cosine direction stayed slightly above nearest-neighbor retrieval. Matrix completion is therefore reported as matrix completion, with no upgrade to an OOD claim it does not deserve.

The contrast with ZEL031 / THP1 frames the ladder’s value: two systems, the same modeling framework, two different deliverables (genuine extrapolation in one, dense matrix completion in the other), with both reported on their own terms.

Different generalization regimes, not a head-to-head OOD comparison

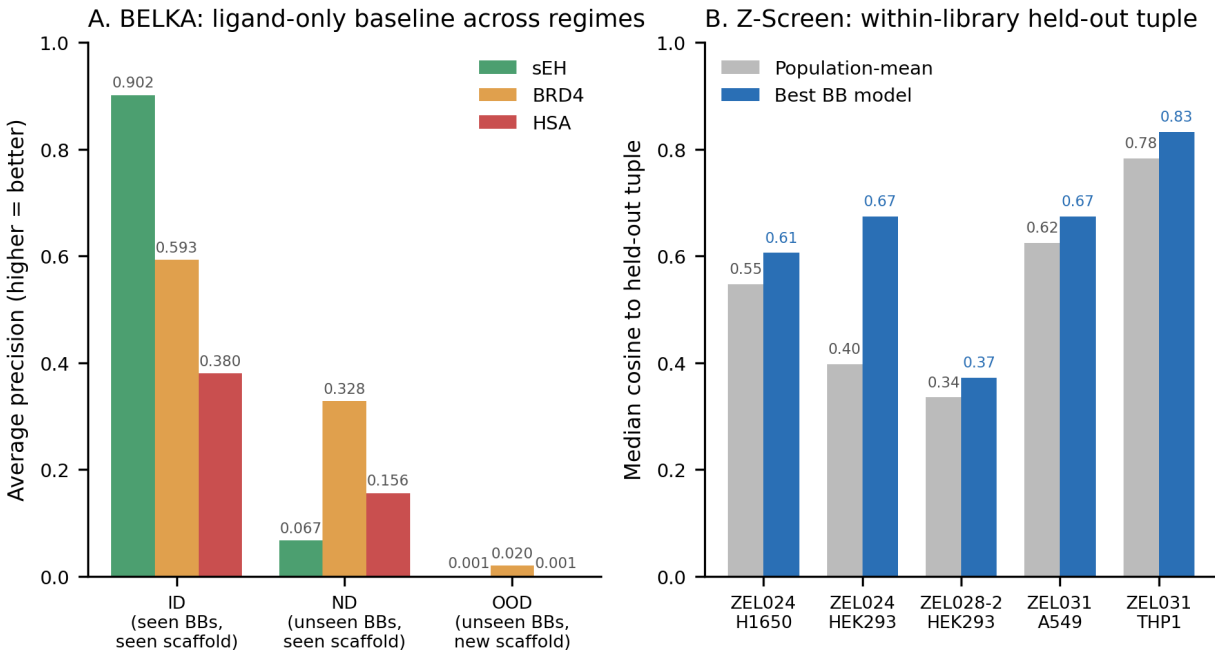


Figure 3: Figure 3. BELKA is used here as a vocabulary for generalization regimes, not as a head-to-head performance comparison. The Z-Screen panels separate matrix completion from harder held-chemistry regimes, with ZEL031 / THP1 carrying the strongest extrapolation signal in the present dataset.

L4: chemical-neighborhood holdout is supportive within-library evidence

After L2 and L3, the next uncertainty is whether the gains depend only on isolated building-block identities. L4 tests a different within-library split: chemical-neighborhood clusters are held out by farthest-point sampling in the projected ECFP cosine-distance space. This keeps the reaction grammar fixed while moving test compounds away from local training neighborhoods. Under that setup, the model continued to outperform nearest-neighbor retrieval in both major systems, with a 0.236 cosine gain in ZEL024 and a 0.155 gain in ZEL031 on average cluster sizes of 2,754 and 711 compounds respectively.

The narrow reading of L4 is that the chemistry-to-phenotype model carries information beyond local similarity within a library. That is supportive evidence for the ladder, but it is not scaffold-transfer proof. The scaffold-family question belongs to L5.

L5: retrospective scaffold-family transfer is weak and defines the next experiment

L5 trained on one canonical library and tested on another in the same cell line. Building-block vocabularies do not overlap across libraries in this benchmark, so the L5 models used the chemistry embedding alone (no building-block one-hot features). Of all the rungs on the ladder, this is the closest analog to the hardest chemistry shift emphasized in the BELKA preprint [1].

On current retrospective data, L5 was weak. Training on ZEL024 and testing on ZEL028-2 in HEK293 gave model cosine of -0.096, slightly below the nearest-neighbor baseline of -0.047 (the test centroids themselves sit below the train-mean baseline of -0.117 in this regime, so all approaches struggle). Reversing the direction was similarly weak. The best absolute performance came in H1650 for ZEL024 to ZEL028-2 (0.194 cosine versus 0.098 for nearest-neighbor retrieval, +0.096 gain on 765 test compounds). In A549, ZEL031 to ZEL028-2 reached 0.098 cosine versus 0.087 for nearest-neighbor retrieval. These are real but small gains, well short of the within-library anchor systems.

The weakness is the point of the rung. Cross-library transfer is where chemistry, assay, and latent-space drift compound, and it is the regime most affected by sparse coverage in ZEL028-2 (median 1 well per compound) and by cross-library scVI baseline drift. The result does not support a solved scaffold-transfer claim. It does define the next experiment precisely: a prospective same-cell scaffold-family hop with parity coverage and a pre-registered model.

OOD ladder performance matrix

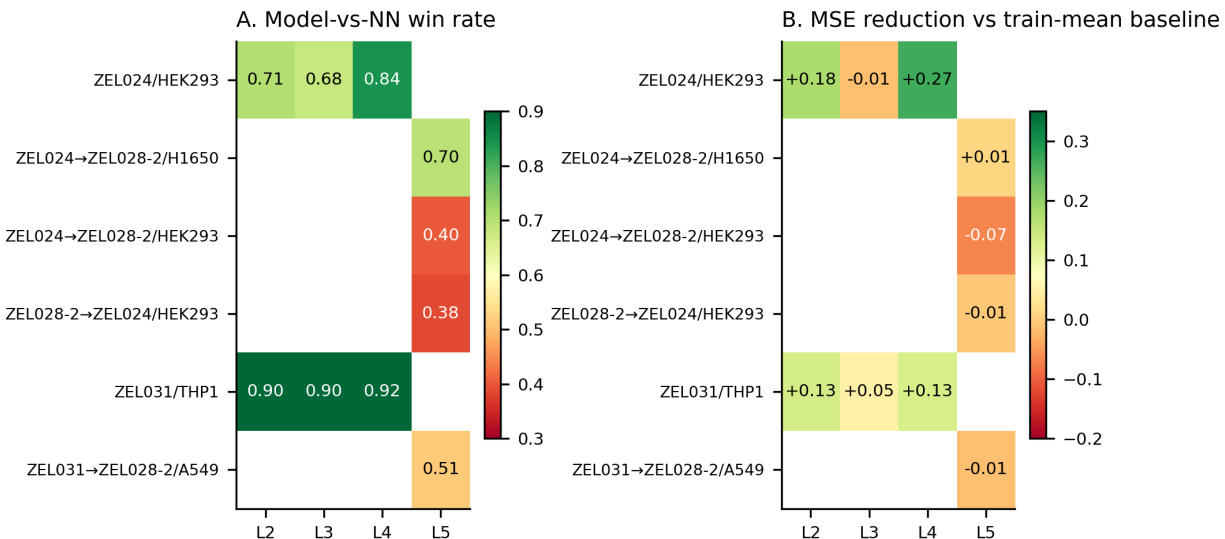


Figure 4: Figure 4. Per-system performance across the ladder shows strong within-library prediction in selected systems (ZEL031 / THP1 for genuine extrapolation, ZEL024 / HEK293 for matrix completion in a saturated grid), and a marked drop at retrospective cross-library scaffold-family transfer that defines the natural next experiment rather than a stable performance ceiling.

Table 1. Reader-facing summary of the ladder. Cosines are row-wise similarity to measured transcriptomic state; nearest-training similarity is cosine in the random-projected ECFP4 embedding.

Rung	System	Model	NN	Delta	Nearest train	Interpre- tation
L2	ZEL024 / HEK293	0.523	0.406	0.117	0.927	positive but high- neighbor matrix comple- tion
L2	ZEL031 / THP1	0.786	0.638	0.149	0.791	consistent held-BB extrapola- tion
L3	ZEL024 / HEK293	0.338	0.213	0.125	0.799	small-test matrix- completion boundary

Rung	System	Model	NN	Delta	Nearest train	Interpre- tation
L3	ZEL031 / THP1	0.765	0.597	0.168	0.642	strong held- chemistry extrapola- tion
L4	ZEL024 / HEK293	0.600	0.363	0.236	0.886	within- library signal beyond local retrieval
L4	ZEL031 / THP1	0.777	0.622	0.155	0.725	within- library signal beyond local retrieval
L5	ZEL024 -> ZEL028-2 / H1650	0.194	0.098	0.096	0.461	small positive retrospec- tive transfer
L5	ZEL024 -> ZEL028-2 / HEK293	-0.096	-0.047	-0.049	0.472	negative retrospec- tive transfer
L5	ZEL028-2 -> ZEL024 / HEK293	-0.095	-0.034	-0.061	0.458	negative retrospec- tive transfer
L5	ZEL031 -> ZEL028-2 / A549	0.098	0.087	0.011	0.597	near-flat retrospec- tive transfer

A prospective same-cell scaffold hop is now well defined

The ladder does more than score the present data. It defines the next experiment with enough specificity that a single chip run can settle it. A clean prospective test would hold cell line fixed,

choose two scaffold families with adequate coverage, train on family A and optionally family B, and nominate family C compounds for prospective measurement. Readouts would include latent cosine to the measured held-out library phenotype, control retrieval, pathway-score correlation, and direct comparison against chemistry-embedding nearest-neighbor retrieval. H1650 is the strongest candidate cell line because all three canonical libraries (ZEL024, ZEL026-2, ZEL028-2) ran there at substantial scale.

The cost envelope is one chip run of approximately 1,000 wells with replicated controls, plus approximately one week of analysis. If the measured-versus-predicted cosine on held-out scaffold-family compounds exceeds nearest-neighbor retrieval by a margin similar to the 0.15 to 0.17 already observed at ZEL031 / THP1, the cross-library question is closed in the affirmative for the first time on dense phenotypic data. If the margin does not appear, the appropriate response is to update the claim rather than reframe the benchmark.

Recommended next experiment: explicit scaffold-family transfer

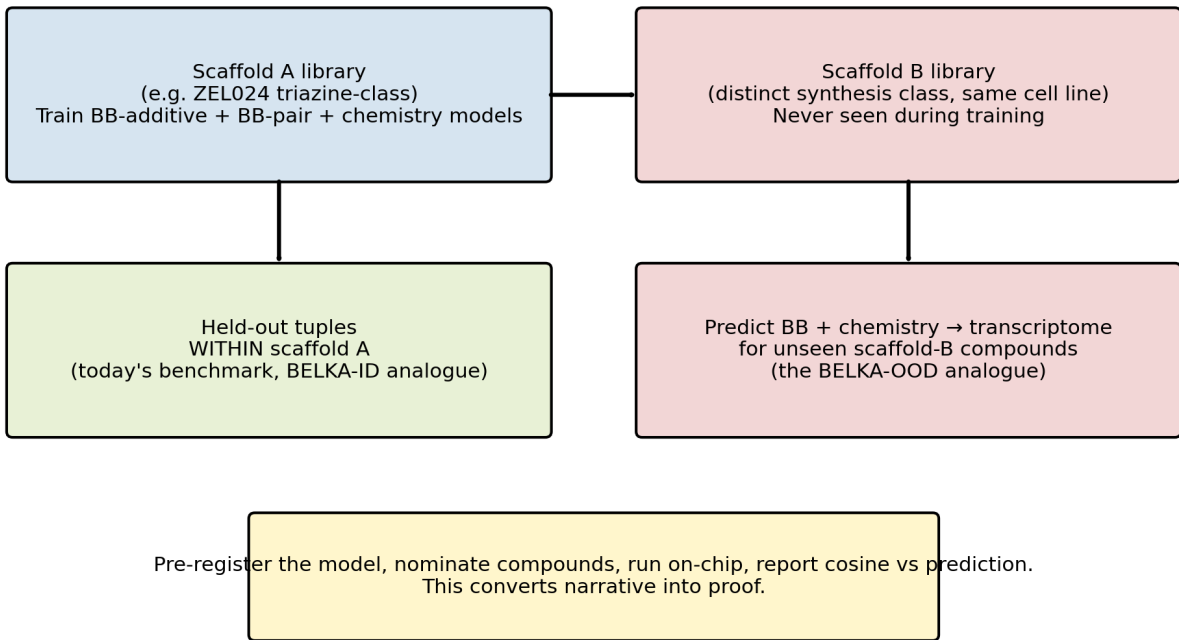


Figure 5: Figure 5. The next decisive experiment is a prospective same-cell scaffold-family hop with a pre-registered model, nominated compounds, and measured transcriptomic outcomes on the held-out family.

Discussion

Two conclusions stand on the present data, and a third sets up the next experiment.

First, dense phenotypic readouts can support genuine chemistry extrapolation in the right systems. ZEL031 / THP1 is the clearest case, with the model exceeding nearest-neighbor retrieval on held-

building-block and strict multi-axis benchmarks where the chemistry is not trivially close to training examples, and doing so consistently across 10 random draws under paired bootstrap intervals.

Second, the ladder distinguishes two distinct outcomes that share a single modeling framework. ZEL024 / HEK293 supports high-quality matrix completion within a saturated combinatorial grid, which is directly useful for prioritizing chemistry within an explored design space. ZEL031 / THP1 supports out-of-vocabulary extrapolation. Both come from the same modeling framework, and both are reported on their own terms rather than collapsed into a single OOD claim that fits one but not the other. Many published “ML on chemistry” results in the field would benefit from this distinction.

Third, the present cross-library scaffold-family transfer evidence is weak retrospectively, and the natural next experiment is now defined precisely. The same modeling framework that extrapolates inside ZEL031 / THP1 should be tested on a prospective same-cell scaffold hop in H1650. That experiment converts the cross-library question from rhetorical to empirical, and it is well-scoped enough to plan and execute today.

For the discovery community, the practical message is that “generalization” is not one thing. The current data support genuine held-building-block extrapolation in one well-characterized system and high-quality matrix completion in a saturated system. They do not yet support a broad cross-library scaffold-family claim. That is a stronger position scientifically because it tells the next chip what to test.

Methods

Benchmark ladder

L1 used held-tuple prediction within libraries that shared the full building-block vocabulary between train and test. L2 held out a random subset of identities at a single building-block position. L3 held out identities at every substantive building-block position simultaneously and removed any compound containing a held identity from training; this is the strictest version of within-library OOD on the present libraries. L4 held out chemical-neighborhood clusters within a library, with seeds chosen by greedy farthest-point sampling in the public chem-embedding cosine-distance space; this produces test sets that are diverse rather than locally similar. L5 trained on one library and tested on another in the same cell line, with no building-block-vocabulary overlap.

Features and models

Chemistry features were precomputed once as a 256-dim per-compound embedding shipped in `data/ZScreen_Canonical_Dataset/RNASeqAggregate/chem_embed.parquet`. The embedding is a fixed-seed Gaussian random projection from R^{2048} to R^{256} applied to per-compound 2048-bit ECFP4 fingerprints (Morgan radius 2), then L2-normalized. The projection matrix is held privately (NDA bundle) so that the embedding is not invertible to substructure information without it; under cosine similarity it preserves Tanimoto-like retrieval geometry approximately (Johnson-Lindenstrauss). Where building-block vocabularies overlapped between train and test, one-hot

encodings were fit on the training split only and combined with the chemistry embedding. Ridge regression with alpha equal to 10 was used as the primary prediction model. Ridge was chosen over a flexible nonlinear regressor because the goal of the ladder is to attribute generalization to the chemistry-to-phenotype function rather than to a model’s flexibility. Ridge alpha sweeps over four orders of magnitude were verified separately and do not change the qualitative ranking of rungs; the headline numbers are stable across the obvious hyperparameter envelope.

Baselines and metrics

Two baselines were used throughout. The first is a train-mean phenotype predictor (the constant prediction that minimizes squared error on the training set). The second is a chemistry-embedding nearest-neighbor retrieval baseline that copies the measured phenotype of the closest training compound by cosine similarity in the 256-dim embedding space. The nearest-neighbor baseline is the most consequential one: in any chemistry split where the test molecules are close to training molecules, retrieval will appear strong, and reported model gains have to be evaluated against retrieval rather than against a trivial constant. Performance was summarized with row-wise cosine to the measured phenotype, mean squared error reduction relative to the train-mean baseline, delta cosine versus nearest-neighbor retrieval, and per-compound win rate against retrieval.

Statistical reporting

Each random draw was evaluated with paired bootstrap resampling of test compounds. Confidence intervals were computed from 2,000 paired bootstrap samples, and summary tables report how many draws remained positive under the paired interval criterion. Ten random draws per rung were used rather than a single split because the size of test sets varies by holdout depth (some L3 splits drop to 67 compounds per draw in ZEL024) and a single random split would not give a stable estimate of model gain at small test-set sizes.

Cosine nearest-neighbor implementation

The chemistry-embedding nearest-neighbor baseline was implemented as an exact dot-product search on the L2-normalized 256-dim public embedding (cosine = dot product because the embedding is L2-normalized). Exact rather than approximate search was used because the same baseline appears in every cross-comparison in this manuscript and its public reproduction needs to be deterministic to the byte.

Limitations

The ladder is retrospective and strongest within selected library / cell-line systems rather than across every system in the canonical dataset. ZEL031 / THP1 supports the clearest extrapolation claim; ZEL024 / HEK293 is better interpreted as dense matrix completion. Cross-library scaffold-family transfer remains weak in the current data and should be treated as the next prospective experiment, not a solved capability. All transcriptomic targets are evaluated in the shared scVI latent space, so

full transcript-level validation and prospective measurement are needed before using the ladder as a final model-selection benchmark.

Data availability

Data tables, manuscript figures, and analysis inputs needed to reproduce this working draft are organized in this repository under `paper3a/`, `paper2/tables/`, and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper3a/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

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Cross-cell-type transcriptomic transfer of combinatorial chemistry programs in Z-Screen

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Abstract

A scalable answer to “how will this compound behave in a cell type that was not directly assayed?” would change the economics of preclinical drug discovery. The current answer requires running every program in every cell type of interest, which scales poorly. Cell-intrinsic pharmacology is rarely identical across tissues, so a single reference cell line is often insufficient; running ten cell lines for every campaign is what teams do when they have to.

The canonical Z-Screen combinatorial-library transcriptomic dataset supports a useful first version of cross-cell-type projection, but only for comparisons with enough paired chemistry to evaluate honestly. We first define that eligible set, then benchmark held-out transfer at the level of individual molecules and grouped building-block programs in a 32-dimensional scVI latent space.

At the molecule level, direct reuse of the source-cell signature was inadequate, while learned transfer maps recovered target-cell information. In ZEL024 (HEK293 $\leftrightarrow$ H1650), centered-response cosine for HEK293 $\rightarrow$ H1650 improved from 0.146 with direct reuse to 0.279 with ridge transfer; the reverse direction reached 0.371. In ZEL031 (A549 $\leftrightarrow$ THP1), A549 $\rightarrow$ THP1 improved from 0.094 to 0.242. These molecule-level gains were real but modest, which motivated a chemistry-program view. Aggregating by chemistry-resolved building-block programs produced stronger transfer: in ZEL024, bb0+bb1 pair programs reached support-weighted cosine 0.863 for H1650 $\rightarrow$ HEK293 and 0.808 for HEK293 $\rightarrow$ H1650; in the two-position ZEL031 library, bb0 programs reached 0.714 for A549 $\rightarrow$ THP1 and 0.664 for THP1 $\rightarrow$ A549.

The strongest current claim is therefore not universal single-molecule prediction. It is that paired landmark data can translate chemistry-linked response programs across cell types when the program is defined at an aggregation level that still preserves chemical specificity. We deliberately demote attractive single-building-block results from ZEL028-2 because each centroid averages hundreds of distinct tuples, making the signal too broad to treat as a chemistry-resolved program. Cross-library projection was suggestive in a few directions but asymmetric and underpowered, and gene-level decoding should be read as hypothesis generation. The practical implication is operational: profile a new compound or chemical series once in an accessible source cell type and use a learned transfer map to prioritize target-cell follow-up, with the strongest current confidence at the level of grouped chemical programs rather than isolated molecules.

Significance

Every added cell type increases screening cost, and skipping the wrong cell type is how programs miss toxicity, selectivity, or therapeutically relevant biology. This paper presents a first practical route through that tradeoff. When the same chemistry is observed across paired cellular contexts, Z-Screen can learn transfer maps that recover target-cell signal better than direct reuse of the source profile. The strongest current unit is the grouped chemical program rather than the single molecule, which matches how early-stage discovery decisions are usually made: at the level of chemical series, not isolated compounds. For a discovery organization, this changes the budget calculus. Instead of running every program through every cell type of clinical interest, a smaller paired landmark atlas plus a single source measurement begins to do useful work for the rest.

Introduction

A substantial share of preclinical cost goes to running the same compound, panel, or library in every cell type of practical interest. The economics of early-stage discovery would change if a sufficiently broad paired landmark atlas allowed teams to:

- profile a new compound in one accessible source cell type,
- project its likely transcriptional response in unmeasured target cell types,
- and use those projections to prioritize efficacy work, counter-screen liabilities, and decide where wet-lab follow-up is worth the cost.

The right form of that claim is narrow but still important. Cross-cell-type transcriptomic transfer is a model for cell-intrinsic transcriptional response, and it does not substitute for pharmacokinetics, metabolism, tissue architecture, or organism-level toxicology. It is a practical decision-support primitive for the upstream half of discovery, where even partial accuracy is valuable.

The most directly comparable public reference points are L1000 [2], which provides a chemistry-side compound atlas in a small number of cell lines (mostly MCF7, PC3, A549, A375), and Tahoe-100M [3], which scales perturbational scRNA-seq across 50 cancer cell lines for approximately a thousand compounds. Both are valuable for cross-cell signature comparisons of the same compound, and neither provides combinatorial-chemistry provenance: a hit in L1000 or Tahoe-100M is a single profiled compound with no internal structure to its identity. scGeneScope [4] adds matched imaging but uses one cell line. Z-Screen, by contrast, uses one-bead-one-compound combinatorial libraries profiled in 50,000-well microchips by paired imaging and mRNA sequencing at 5 to 10 cells per well. That design is well suited to the cross-cell question because every well in the canonical dataset carries full chemistry provenance (building blocks plus full SMILES), the same combinatorial libraries are run across multiple cell lines, and the readout is a low-pass transcriptomic state for every compound rather than a sparse hit or non-hit label.

Compound-level overlap with these public atlases is also relevant for cross-cell positioning. Of 47 named Z-Screen control compounds, 14 (29.8%) are present in LINCS L1000 (palbociclib, sorafenib, crizotinib, ceritinib, rucaparib, veliparib, lapatinib, cobimetinib, GSK126, CP-673451, and others), with 3 of 4 Z-Screen cell lines (HEK293, A549, THP1) covered by L1000 sample

metadata. Tahoe-100M’s drug metadata covers 4 of the 47 named compounds (palbociclib, crizotinib, cobimetinib, capmatinib), with A549 as the only matched cell line. A direct cross-platform Spearman concordance test against LINCS L1000 level-5 consensus signatures (NCBI GEO GSE70138 and GSE92742), restricted to the 12,328-gene LINCS-visible universe, recovers positive correlation for all 11 (compound, cell_line) pairs with adequate coverage on both platforms. Median $\rho = 0.490$, mean = 0.393, with 8 of 11 above $\rho = 0.2$ and 5 reaching empirical $p = 0.048$ against an unmatched-compound permutation null (crizotinib, ZM-336372, sorafenib, rucaparib, veliparib). The HEK293 cell-line-pooled comparison separates matched from unmatched compound pairings at one-sided Mann-Whitney $p = 0.009$. Full overlap and concordance tables are in **Benchmarking/** in this repository.

With those ingredients, the cross-cell question can be posed in a controlled way: how much of a source-cell transcriptomic response can be projected into a target cell type using only a learned transfer map fit on a paired landmark subset. This paper characterizes how well that question is answered today, on data from a small number of pilot experiments, and defines what additional data would close the remaining gaps.

Results

Dataset coverage defines the eligible comparisons

The first question was not which model performs best. It was which cell-pair comparisons have enough overlapping chemistry to support a transfer claim at all. The canonical dataset is a pilot corpus with uneven depth across libraries and cell lines. ZEL024 has approximately 10,000 to 14,000 unique SMILES per cell line, including the densest system in the workspace: HEK293 at a median of 11 wells per compound. ZEL031 has 8,000 to 9,000 unique SMILES in A549 and THP1 at a median of 2 wells per compound. ZEL028-2 is broad but shallow, with 40,000 to 61,000 unique SMILES in A549 and HEK293 at a median of 1 well per compound.

That difference matters because a cross-cell benchmark needs the same chemistry in both source and target cells. At a minimum support filter of two cells, ZEL024 H1650 $\leftrightarrow$ HEK293 contributed 6,299 shared SMILES and ZEL031 A549 $\leftrightarrow$ THP1 contributed 3,561 shared SMILES. By contrast, ZEL028-2 cell-pair overlaps fell to 9 to 19 shared SMILES at the same filter, and ZEL031 comparisons involving H1650 were similarly too small. At a five-cell filter, the eligible set narrowed further: ZEL024 H1650 $\leftrightarrow$ HEK293 retained 353 shared SMILES, and ZEL031 A549 $\leftrightarrow$ THP1 retained 140.

These coverage facts set the order of the evidence. Molecule-level transfer can be tested most cleanly in ZEL024 and ZEL031. ZEL028-2 can still contribute to grouped and cross-library exploratory analyses, but its singleton-heavy structure prevents it from carrying the main per-molecule claim.

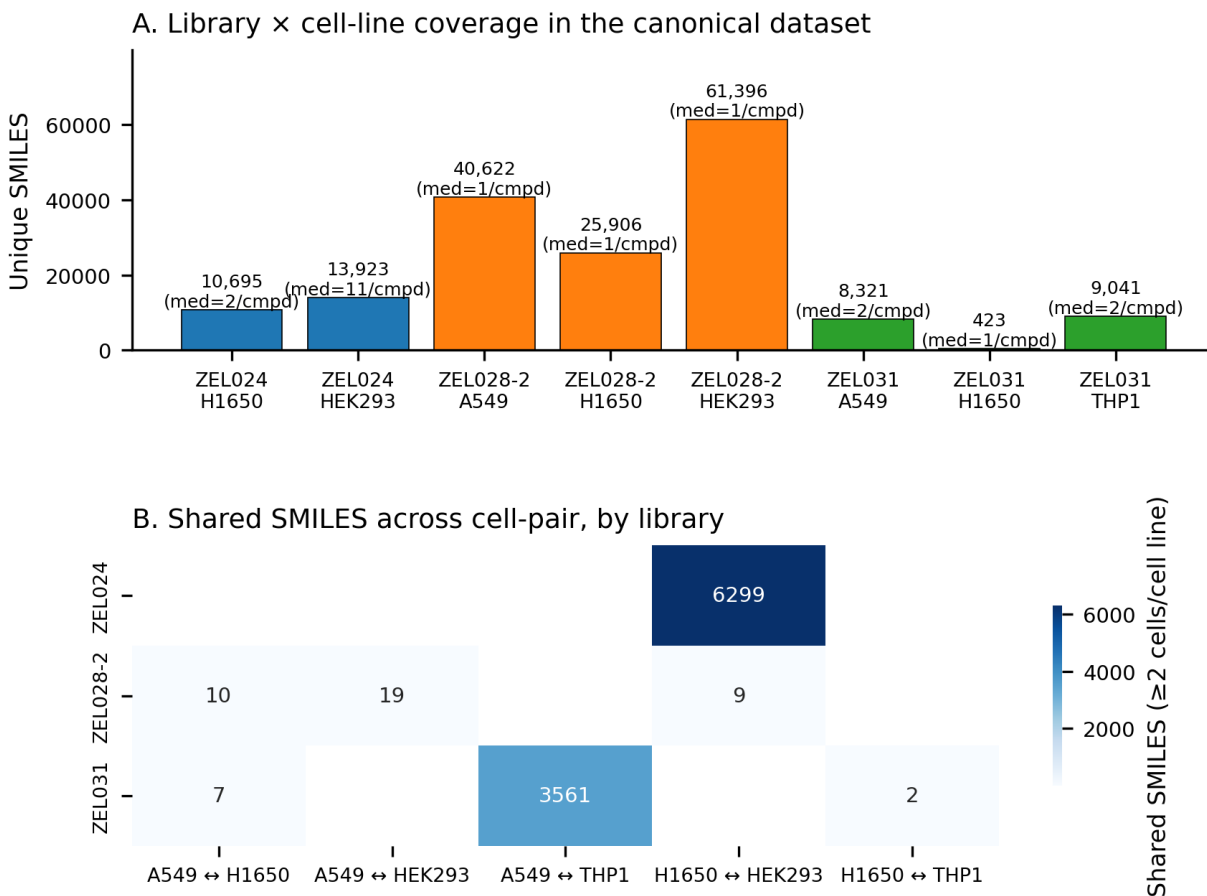


Figure 1: Figure 1. Library-by-cell-line coverage in the canonical Z-Screen dataset. Top: unique SMILES and median wells per compound per system. Bottom: shared-SMILES counts across cell-pair directions, by library. The two dense paired systems used in the molecule-level transfer benchmark are ZEL024 (H1650 $\leftrightarrow$ HEK293, 6,299 shared) and ZEL031 (A549 $\leftrightarrow$ THP1, 3,561 shared).

Molecule-level transfer is real in dense paired libraries

The molecule-level benchmark used held-out individual molecules in the two cell-pair systems with sufficient shared SMILES coverage:

- ZEL024, H1650 $\leftrightarrow$ HEK293, 6,299 shared molecules,
- ZEL031, A549 $\leftrightarrow$ THP1, 3,561 shared molecules.

Each molecule was represented by the mean of its 32-dimensional scVI latent coordinates within a given library and cell line. Direct reuse of the source-cell signature (identity) was compared against learned source-to-target transfer models. The centered-response benchmark is the primary readout because matched vehicle controls are unavailable in the canonical dataset, and the centered formulation is the closest available proxy for perturbational response. Under that benchmark, learned transfer consistently outperformed direct reuse at the largest training sets:

- ZEL024, HEK293 -> H1650, train 5,000: identity 0.146, ridge 0.279, PLS 0.275.
- ZEL024, H1650 -> HEK293, train 5,000: identity 0.146, ridge 0.371, PLS 0.358.
- ZEL031, A549 -> THP1, train 3,000: identity 0.094, ridge 0.242, PLS 0.242.
- ZEL031, THP1 -> A549, train 3,000: identity 0.094, ridge 0.212, PLS 0.214.

Two readings follow. Direct source-signature reuse alone is inadequate, so cross-cell projection is a real prediction problem rather than an artifact of cell-state offset. A simple regularized global linear or low-rank map already captures useful molecule-specific information once enough paired examples are available. The gains are also modest in absolute terms, especially in centered response space. This is the key reason to ask whether chemistry-resolved groups, rather than isolated molecules, are the more stable unit of transfer in the current dataset.

The raw-profile benchmark is reported in the supplementary material as a diagnostic. Performance is much higher in raw space (for example, ZEL024 HEK293 -> H1650 reaches 0.546 with ridge transfer at train 5,000, versus 0.063 with direct reuse), and a substantial portion of that gain reflects baseline cell-state translation rather than perturbational response. The centered-response benchmark above is the cleaner main readout for a discovery use case, because it removes the constant offset that exists between two cell lines regardless of any particular perturbation.

Molecule-level cross-cell transfer in centered response space

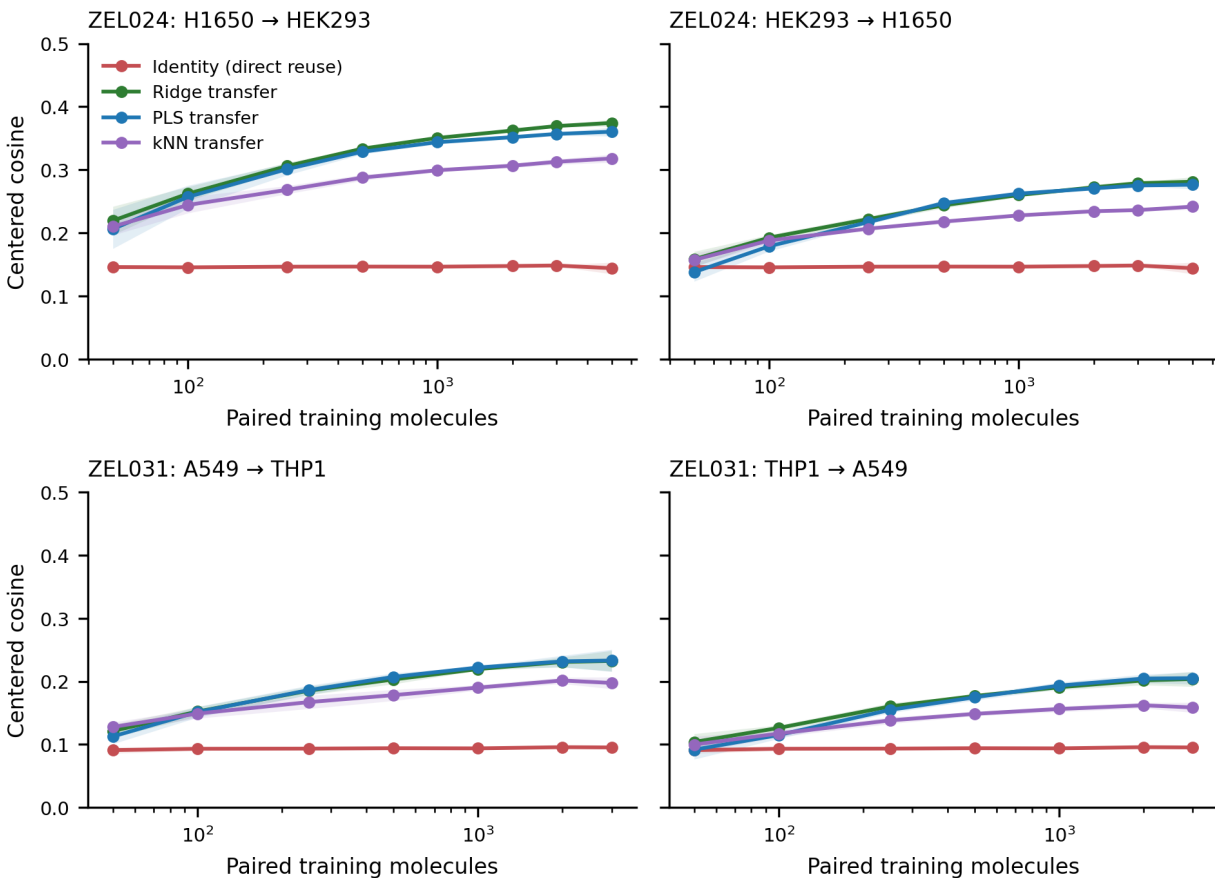


Figure 2: Figure 2. Molecule-level cross-cell transfer in centered response space across four well-supported cell-pair directions in ZEL024 and ZEL031. Direct reuse of the source signature is consistently weaker than learned transfer; ridge and PLS transfer are nearly tied. The absolute centered-response cosines remain modest, motivating the grouped-program analysis that follows.

Chemistry-resolved grouping exposes stronger transferable programs

Single-molecule signatures are noisy because each compound is often represented by a small number of wells and by a low-pass transcriptomic readout. The next test therefore asked whether source-cell and target-cell responses align more cleanly after averaging molecules that share a defined combinatorial chemistry program.

Reporting is restricted to aggregation levels that remain chemically interpretable. For ZEL024, which has four substantive building-block positions, the reported program is a BB-pair (bb0+bb1); each pair-level signature averages roughly 129 to 166 distinct tuples rather than collapsing the whole library. For ZEL031, which has only two substantive positions, a single bb0 group is the closest analog because the other position remains the varying partner chemistry.

Under those chemistry-resolved aggregations, grouped transfer was substantially stronger than the

within-cell direct-reuse baseline:

- ZEL024, H1650 $\rightarrow$ HEK293, grouped by bb0+bb1: 84 shared pairs, median support 240.5 wells, identity 0.471, ridge transfer **0.863**.
- ZEL024, HEK293 $\rightarrow$ H1650, grouped by bb0+bb1: 84 shared pairs, median support 240.5 wells, identity 0.471, ridge transfer **0.808**.
- ZEL031, A549 $\rightarrow$ THP1, grouped by bb0 in a two-position library: 129 shared groups, median support 157 wells, identity 0.419, ridge transfer **0.714**.
- ZEL031, THP1 $\rightarrow$ A549, grouped by bb0 in a two-position library: 129 shared groups, median support 157 wells, identity 0.419, ridge transfer **0.664**.

This is the strongest current result because it answers the limitation exposed by the molecule-level benchmark. The claim is not that arbitrary unseen molecules are solved. It is that chemistry-resolved building-block programs define reusable response units that transfer across cell types more cleanly than sparse individual molecules do. That is a practical unit for early discovery, where teams often advance or deprioritize chemical series before they have enough evidence to treat every analog as an independent biological object.

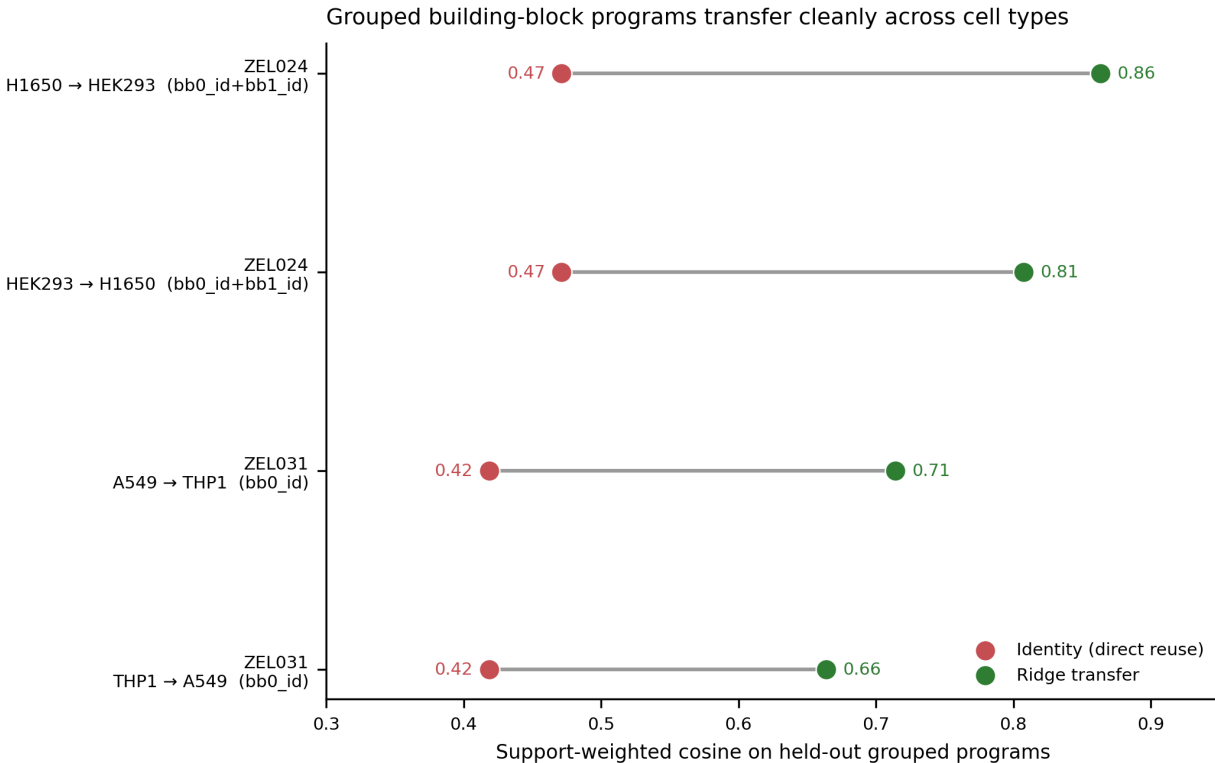


Figure 3: Figure 3. Grouped building-block program transfer at chemistry-resolved aggregation levels (BB-pair for ZEL024, single position bb0 for the two-position ZEL031 library). Identity (direct reuse of the source-cell signature) sits in the 0.42 to 0.52 range, while a regularized learned transfer map reaches support-weighted cosine 0.66 to 0.86 across four chemistry-resolved program-and-direction combinations.

The same point is visible when the molecule-level and grouped-program results are placed side by side. At the molecule level, learned transfer roughly doubles the centered cosine over identity in ZEL031 (0.094 to 0.21 to 0.24) and lifts it by 1.9 to 2.5 times in ZEL024. At the chemistry-resolved grouped-program level (ZEL024 BB-pair, ZEL031 bb0), learned transfer increases support-weighted cosine by 0.25 to 0.39 absolute over identity, with all four chemistry-resolved directions ending in the 0.66 to 0.86 cosine range. The grouped-program unit is the stronger current asset and is the unit recommended for early use in discovery decisions.

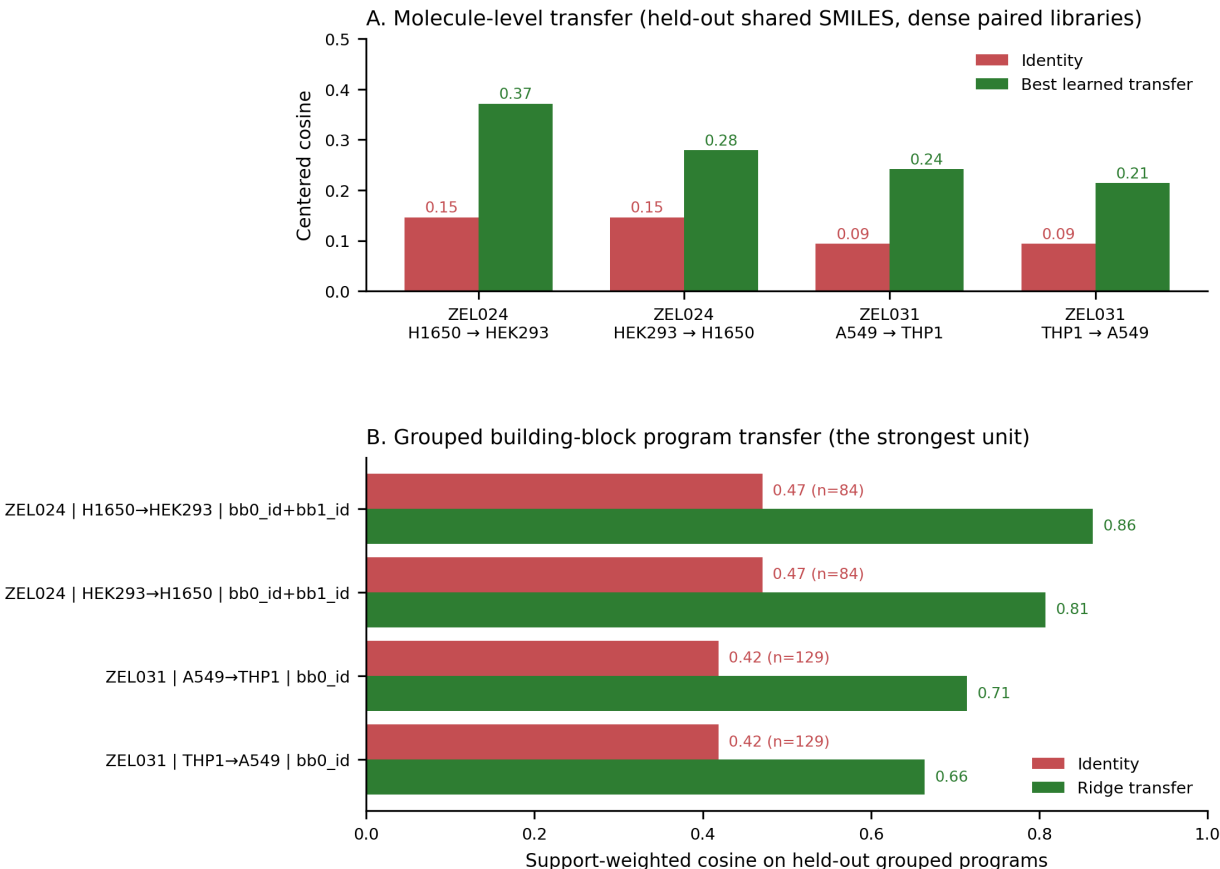


Figure 4: Figure 4. Best learned transfer versus identity at both molecule level (top) and grouped building-block program level (bottom). Learned transfer wins at every cell-pair direction tested, with the largest gains at the grouped-program level.

Guardrail: over-aggregated single-BB results should not headline

Some apparently attractive grouped results fail the chemical-specificity test. In ZEL028-2, single-building-block transfer reached high weighted cosines in the full grouped-analysis table: for example, bb1 grouping reached 0.837 for H1650 → HEK293 and 0.850 for A549 → HEK293. Those numbers are not reported as main results because ZEL028-2 has four substantive building-block positions. A single-position centroid therefore averages over broad chemistry, while the more specific BB-pair rows are much shallower per group.

That aggregation is useful as a guardrail, not as a headline. It says that broad library-level and cell-pair structure can be transferred, but it no longer isolates a chemistry-resolved program in the way a ZEL024 BB-pair or ZEL031 bb0 group does. Reporting those ZEL028-2 values beside the chemistry-resolved results would make the model look stronger while making the biological claim less precise.

Table 1. ZEL028-2 grouped-transfer guardrail. Median support is the minimum matched-cell support across source and target for each grouped centroid.

Direction	Group	Class	Groups	Median support	Transfer	Gain	Interpretation
A549 -> HEK293	bb1_id	single-BB aggregate	86	526.5	0.850	0.332	high transfer but over-aggregated
H1650 -> HEK293	bb1_id	single-BB aggregate	86	333.5	0.837	0.366	high transfer but over-aggregated
HEK293 -> A549	bb1_id	single-BB aggregate	86	526.5	0.762	0.244	high transfer but over-aggregated
A549 -> HEK293	bb1_id+bb1_id pair group	BB pair group	6,690	5.0	0.227	0.100	more specific but shallow per group
A549 -> HEK293	bb3_id+bb3_id pair group	BB pair group	7,158	5.0	0.215	0.089	more specific but shallow per group
HEK293 -> A549	bb1_id+bb1_id pair group	BB pair group	6,690	5.0	0.213	0.086	more specific but shallow per group

Cross-library projection is suggestive, asymmetric, and underpowered

A further question is whether a centered source-to-target transfer matrix learned in one library can project molecules from a different library measured in the same cell-pair direction. The answer is mixed.

Several projections were directionally positive:

- ZEL024 -> ZEL028-2, HEK293 -> H1650: weighted cosine 0.033 -> 0.059.
- ZEL024 -> ZEL028-2, H1650 -> HEK293: 0.033 -> 0.055.
- ZEL028-2 -> ZEL031, A549 -> H1650: 0.120 -> 0.216.

Other directions were weak or negative:

- ZEL028-2 -> ZEL024, HEK293 -> H1650: 0.119 -> 0.067.
- ZEL031 -> ZEL028-2, A549 -> H1650: 0.103 -> -0.009.

The pattern is not strong enough to support a clean dense-to-sparse rule. ZEL024 trained maps did improve transfer into ZEL028-2 in both HEK293/H1650 directions, but ZEL028-2 also improved one projection into ZEL031 while failing in the reverse ZEL024 direction. The safer interpretation is that a source-to-target cell transformation can sometimes survive a library change, but the present cross-library tests are dominated by small overlaps, singleton-heavy target sets, and disjoint building-block vocabularies. Cross-library projection is therefore the next prospective dataset to design, not a settled generalization result.

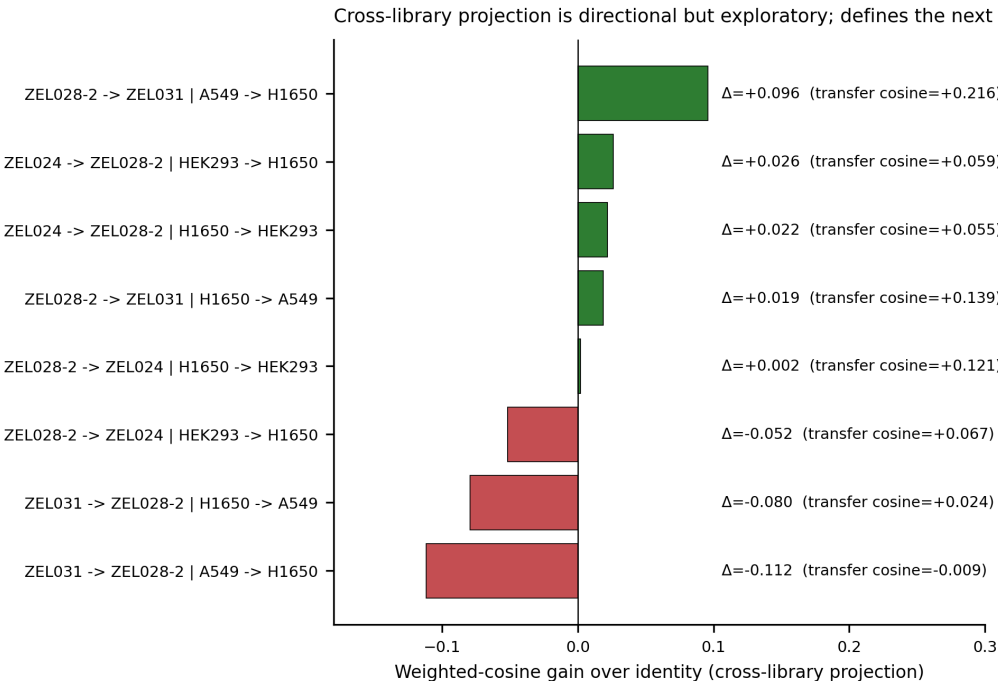


Figure 5: Figure 5. Cross-library projection results. Some cross-library projections are positive, but performance remains small, asymmetric, and sensitive to which library acts as teacher. The result is best read as a design brief for a prospective cross-library dataset.

Exploratory gene-level decoding partially recovers held-out target signatures

A separate experiment tested whether the learned cell-transfer map can be lifted back from the 32-dimensional latent space into ranked gene-level hypotheses. This was treated as an exploratory decoder experiment rather than a main claim. For each dense paired direction, a source-to-target latent transfer map was fit on shared training compounds, a target-cell ridge decoder was fit from target latents to 800 training-selected variable genes, and held-out target-cell gene deltas were compared against measured pseudobulk deltas. The strongest direction was ZEL024 H1650 $\rightarrow$ HEK293: transferred latents reached mean per-compound gene-rank Spearman 0.182 and top-50 absolute-delta overlap 0.139, compared with 0.021 and 0.083 when the source latent was passed directly through the target decoder. The reverse ZEL024 direction and both ZEL031 directions were weaker, with transfer Spearman values of 0.073 to 0.098.

The result is useful as hypothesis generation. A transferred latent can nominate genes or pathways to inspect in the target cell type, but the decoder still depends on observed target-cell training profiles and the correlations are not high enough to claim full transcriptome reconstruction.

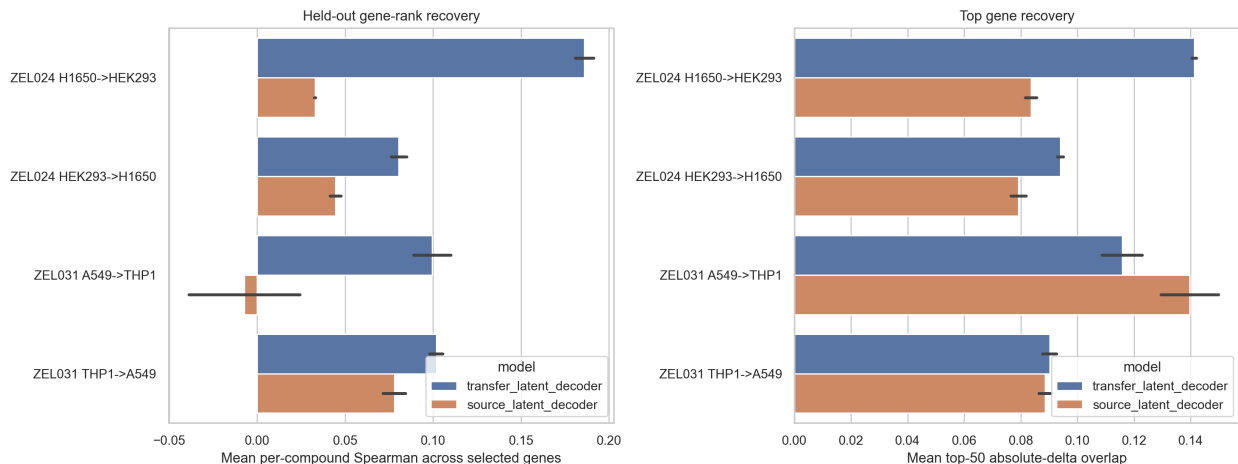


Figure 6: Supplementary Figure 2. Exploratory gene-level transfer. A target-cell decoder composed with the learned source-to-target latent map partially recovers held-out gene-rank structure, with the clearest gain in ZEL024 H1650 $\rightarrow$ HEK293 and weaker gains in the remaining directions.

Discussion

The current data support a two-level view of cross-cell transfer. At the molecule level, paired transfer is real in the eligible dense libraries: even after centering within cell line, learned transfer maps recover target-cell information that direct source reuse misses. The absolute molecule-level signal is still modest. At the chemistry-resolved grouped-program level (BB-pair in dense ZEL024; single-position bb0 in the two-position ZEL031 library), transfer becomes substantially stronger, with weighted cosine in the 0.66 to 0.86 range across four cell-pair directions. From a discovery perspective, the encouraging reading is that the first economically useful layer of cross-cell prediction is chemistry-program transfer rather than universal single-molecule prediction.

The current results also argue against an overly strong “one universal matrix solves everything” framing. ZEL028-2 shows why aggregation discipline matters: broad single-building-block centroids can produce appealing values while no longer representing a chemistry-resolved program. Cross-library projection remains asymmetric and fragile in sparse settings, and a more durable long-term architecture is likely a shared perturbational latent space with cell-type-specific decoders or adapters, coupled to an uncertainty model that rejects out-of-domain projections. That architecture fits naturally on top of the present transfer maps without replacing them, and it is consistent with the direction the public field is taking on cross-cell perturbation modeling.

In practical terms, a paired landmark atlas plus one source-cell measurement is enough to start prioritizing target-cell transcriptomic follow-up for the same chemistry, with the strongest current confidence at the level of grouped chemical programs rather than individual molecules. The immediate use cases are counter-screen prioritization, cross-cell-line selectivity ranking, and decision support for which downstream cell types deserve direct profiling. The next dataset should be designed around prospective cross-library scaffold-family transfer with enough repeated chemistry to distinguish true transfer from sparse-overlap variance.

Methods

Data inputs

All analyses use the canonical Z-Screen RNA aggregate dataset, specifically:

- `canonical_obs.parquet`,
- `canonical_scvi_latents.parquet`,
- `canonical_chemistry.parquet` for grouped building-block benchmarks.

Only rows with a non-null `smiles_hash` (the public-bundle stand-in for a chemistry-resolved compound identity) and `chemistry_grain == "building_block_annotation"` were retained. Compound-control-only rows were not used for the transfer benchmark because their annotations are mechanism-of-action labels rather than chemistry coordinates.

scVI latent space rather than full gene expression

The transfer benchmarks are built on 32-dimensional scVI latent coordinates rather than full gene-level vectors, for two reasons. First, low-pass per-well RNA depth in the present dataset (each well averages 5 to 10 cells, sequenced as one pseudobulk row) means individual genes are dominated by sampling noise; scVI absorbs that noise into shared structure across observations. Second, the latent is comparable across cell types to a degree that raw expression is not, which is the property the cross-cell transfer experiments require. The exploratory gene-level decoder experiment is included as a separate analysis specifically to assess how much of the latent gain survives when the result is lifted back to gene level.

Centered-response as the primary metric

Two cell lines have different baseline transcriptional states regardless of any perturbation. A cross-cell predictor that “knows” the constant baseline shift will show high raw-space cosine even if it has learned nothing about perturbational response. Centering removes that confound. For each train/test split, source and target training means were subtracted before fitting and evaluation. This is the closest available proxy for a vehicle-control normalization in a dataset that does not have matched vehicle controls.

Molecule-level transfer benchmark

For each (library, cell line, SMILES), scVI latent coordinates were averaged across wells. Shared molecules between a source and target cell line were then benchmarked under two evaluation spaces, raw_profile and centered_response. The models compared were identity, global_shift (raw space only), ridge_transfer, pls_transfer, and knn_transfer. The global_shift baseline was included so that the bulk-offset component of cross-cell variance is visible separately from chemistry-specific signal. Performance was measured by row-wise cosine similarity on held-out molecules and aggregated as mean cosine across multiple train-test splits.

Grouped building-block transfer

Within each library and cell-pair direction, molecules were grouped by candidate building-block keys (bb0_id, bb1_id, or multi-column combinations such as bb0_id+bb1_id). For each shared grouped key, source and target centered latent signatures were averaged and benchmarked using a centered ridge transfer model across repeated train-test splits. Performance was summarized as support-weighted cosine similarity across grouped programs. Support weighting prevents a few groups with thousands of supporting wells from being drowned out by many groups with five.

Cross-library transfer

For a fixed cell-pair direction, a centered ridge transfer matrix was trained in one library and applied to molecules from another library. Because the sparser libraries are highly singleton-heavy, adaptive minimum-well thresholds were used so that denser libraries retained stricter filters while still allowing exploratory sparse projections. Directional asymmetry is reported directly rather than corrected away, because teacher-library depth and target-library support are part of the question.

Exploratory gene-level transfer decoder

For the four dense paired directions, held-out shared SMILES were evaluated with a composed latent-to-gene workflow. A centered ridge model first mapped source-cell scVI centroids into target-cell scVI space. Separately, a target-cell ridge decoder was trained on the training compounds to map target-cell scVI centroids to log-CPM pseudobulk deltas over 800 variable genes selected from target training cells. Held-out predictions were scored by per-compound Pearson and Spearman correlation over gene deltas and by top-50 / top-100 absolute-delta gene overlap.

Limitations

- Matched vehicle controls are unavailable in the canonical dataset, so the centered-response benchmark is the closest available proxy for perturbational response rather than a fully DMSO-normalized effect.
- The strongest molecule-level benchmark exists only for the denser paired libraries ZEL024 and ZEL031; ZEL028-2 becomes too sparse under a strict per-molecule support filter.
- Cross-library transfer is limited by sparse sampling and by disjoint building-block vocabularies across libraries, which prevents a clean cross-library tuple-transfer formulation today.
- All analyses are based on averaged 32-dimensional scVI latent signatures rather than full transcript-level models or time- and dose-aware transfer.
- The gene-level decoder experiment is exploratory and uses a target-cell decoder trained on observed target profiles; it should be read as a way to generate ranked gene hypotheses from transferred latents, not as evidence that full transcriptomes can already be prospectively reconstructed from an unmeasured target cell state.
- The current dataset reflects a small number of pilot experiments; a prospective cross-library transfer study should be designed with repeated chemistry and predefined eligibility thresholds.

Data availability

Data tables, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper4a/` and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper4a/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

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Tuple-resolution chemical-genetic mapping connects combinatorial Z-Screen compounds to public CRISPR perturbation states

Zafrens, Inc.

Abstract

Public CRISPR atlases describe the transcriptional states produced by genetic perturbation, but they do not say which combinatorial chemistries can reproduce those states. This preprint asks whether Z-Screen compound tuples, measured with ordered building-block provenance, can be placed in the same rank-signature frame as public single-cell CRISPR perturbations.

In the best-covered system, 8,599 ZEL024 / HEK293 four-position tuples were compared with 8,603 Replogle K562 knockout signatures and calibrated with a size-matched permutation null. The calibrated map linked 1,276 distinct chemical tuples to 1,714 CRISPR programs at empirical $q \leq 0.05$, and remained broad after recurrent CRISPR hubs were removed. Resolution was essential: collapsing the same wells to single-building-block or pair averages gave almost disjoint top CRISPR neighborhoods, showing that the actionable unit is the full tuple. Same-cell THP1 controls recovered expected biology for MZ1/BRD4 and STC-15/interferon signaling. Documented-target recovery was enriched but imperfect in cross-cell scans. Norman CRISPR-activation doubles showed that tuple-level chemical scores preserve genetic sum-of-singles logic.

BAY-293 in A549 provided a discovery bridge: beyond its SOS1 annotation, it produced a coherent V-ATPase and vesicular-trafficking CRISPR neighborhood that included ATP6V1A. That target re-emerged independently in ZEL028-2, a shallower three-variable-position library, where noisy full-tuple calls were filtered through variable-pair subset support. Across ZEL028-2, the hierarchy separated 61,396 / 40,622 / 25,906 observed full tuples in HEK293 / A549 / H1650, 4,041 / 1,728 / 765 tuple signatures at ≥ 2 wells, and 2,151 deduplicated calibrated full-tuple calls used for mechanism support. Two non-hub calls had FDR-controlled subset corroboration: HEK293 ATP6V1A and A549 FKBP9. Together, the data support a conservative chemical-genetic state map: not target identification by itself, but a ranked, calibrated set of compound-level hypotheses with explicit chemistry coordinates for follow-up.

Significance

Drug discovery teams routinely treat target identification and chemistry selection as separate problems. Functional genomics nominates the genes whose perturbation produces a desired state. Medicinal chemistry then searches a separate chemical space for molecules that can create or exploit that state. The bridge between the two problems is profile matching: if the transcriptional

consequence of a molecule resembles the transcriptional consequence of perturbing a gene, the molecule is a starting point worth testing.

Z-Screen makes that bridge concrete at chemistry resolution. The output is not a finished target-deconvolution method; it is a calibrated hypothesis table. Thousands of combinatorial tuples can each be annotated with the CRISPR perturbation state they most resemble, with provenance back to exact per-position chemistry (the 4-position bb0 / bb1 / bb2 / bb3 tuple in the ZEL024 headline system), and with empirical false-discovery control. That is the layer required to move from “which gene matters” to “which molecule should we make or vary next.”

Introduction

Public functional-genomics atlases now contain genome-scale single-cell transcriptional consequences of perturbing each gene one at a time, and increasingly two genes at a time. Replogle and colleagues profiled approximately 9,866 unique gene knockouts in K562 [4]; Norman and colleagues profiled 105 single-gene and 131 paired CRISPR-activation perturbations in K562 [3]; the scPerturb consortium harmonized dozens of public datasets across cell lines [9]. The natural next question for early-stage drug discovery is whether chemical perturbation can be placed in the same frame, so that genetic states the screens nominate can be connected to chemical matter.

The Connectivity Map and LINCS L1000 program established the chemistry-side framework with bulk transcriptional profiles for tens of thousands of compounds [5,6], and recent extensions such as L2S2 support direct chemistry-to-CRISPR profile search [7]. The remaining gap is chemistry resolution. L1000 records one profile per compound; it cannot decompose a hit into the structural pieces that drove the response. Combinatorial Z-Screen libraries are different because each full tuple is a molecule with ordered building-block provenance. The main question here is whether those full-tuple states can be compared with public CRISPR atlases, while BB-pair and single-position averages are used only as aggregation diagnostics.

Z-Screen uses one-bead-one-compound combinatorial libraries, profiled in 50,000-well microchips by paired imaging and mRNA sequencing of the 5 to 10 cell colony in each micro-well, so each readout is a per-well pseudobulk that averages a small amount of cellular heterogeneity. Each library populates a different subset of building-block positions: ZEL024 uses four positions (bb0, bb1, bb2, bb3); ZEL028-2 nominally uses four (bb1, bb2, bb3, bb4) but only three are chemically variable in the canonical aggregate, since bb2 is fixed at a single identity (BB-0000312-002) across all wells, leaving bb1, bb3, and bb4 as the variable chemistry positions; ZEL031 uses two (bb0, bb1). Every well therefore links one compound (one bead, one tuple) to a per-well pseudobulk transcriptional readout, and the per-well pseudobulk is the unit of measurement throughout this preprint. Earlier preprints in this package establish reproducible compound and building-block transcriptional structure (paper 2), chemistry-to-phenotype generalization across an explicit difficulty ladder (paper 3), and learned cross-cell-type transfer of chemical programs (paper 4). The present preprint addresses the complementary question: whether chemical states measured at the level of individual combinatorial compounds can be placed in a genetic-perturbation frame, and whether the resulting map produces calibrated, interpretable, and discovery-relevant matches across many

distinct chemistries.

Terminology is important because two different forms of combinatorial biology are being compared. A Z-Screen tuple is a single combinatorial compound, defined by the full set of populated building-block identities, and measured as one molecule-level well state. A Norman CRISPR double is different: two independent genetic perturbations are applied simultaneously to the same cell, and the resulting state is compared with the states from each constituent single-gene perturbation. In this manuscript, tuple-level chemistry means full-molecule resolution; single-BB or BB-pair averages are aggregation diagnostics, not parallel definitions of a molecule.

The analysis is conservative. Rather than forcing raw expression matrices from different platforms into one normalization, every perturbation is represented as a signed rank signature: the genes most increased and most decreased relative to an appropriate background, with a fixed top-k cutoff. Chemical and CRISPR signatures are compared by mimic overlap, reverse overlap, rank cosine over shared top genes, and a permutation-null calibration that preserves the size of the CRISPR signature. The framework is robust to dataset and depth differences and addresses the strongest question the data can answer: do the top transcriptional consequences of two perturbations agree more than expected by chance.

Each Z-Screen library and cell line is given a chemistry resolution based on its coverage. ZEL024 in HEK293 has 8,599 unique 4-position tuples (bb0+bb1+bb2+bb3) with 10 or more wells; this system is analyzed at full tuple resolution. ZEL028-2 is treated as a 3-variable-position library (bb1, bb3, bb4) and analyzed at three complementary resolutions in parallel. The observed full-tuple universe is much larger than the usable signature universe: 61,396 observed full tuples in HEK293, 40,622 in A549, and 25,906 in H1650, but only 4,041, 1,728, and 765 respectively qualify at ≥ 2 wells. Those min2 full-tuple signatures have a median of 2 wells per tuple, so they are the most chemistry-specific but also the noisiest. Variable-pair subsets (bb1+bb3, bb1+bb4, bb3+bb4) at ≥ 10 wells per pair signature give 3,051 to 3,103 chemistry-pair signatures per pair in HEK293, 909 to 1,260 in A549, and 141 to 257 in H1650; this resolution sacrifices chemistry specificity but gains per-signature support. The variable-triple bb1+bb3+bb4 at ≥ 2 wells is equivalent to the full tuple in this library because bb0 is always missing and bb2 is constant. The hierarchical mechanism-support analysis then works on a third universe: 2,151 deduplicated calibrated full-tuple calls, one best CRISPR call per represented tuple query from the calibrated min2 scans. The 5- to 6-fold lower per-tuple well support relative to ZEL024 is recorded in the metadata for every ZEL028-2 signature, and the ZEL028-2 full-tuple results are read as a leaderboard of chemistry-resolved hypotheses rather than as confirmed per-tuple calls. ZEL031 has only two substantive positions (bb0+bb1), so its 3,261 to 3,556 2-position tuples are analyzed at tuple resolution with a lower well-count threshold flagged in the metadata.

Results

A rank-based comparison framework places chemistry and CRISPR perturbations in a shared transcriptional frame

Every headline Z-Screen perturbation was reduced to a signed rank signature against an appropriate background. For tuple signatures, the target was wells whose chemistry annotation matched the library’s populated building-block positions exactly (four positions in ZEL024 and ZEL028-2, two in ZEL031), and the background was the same library and cell line excluding the target tuple wells. BB-pair and single-position signatures were built as diagnostic aggregations for Figure 1 and metadata transparency; their targets pool wells matching a shared position or position pair across many distinct full tuples. Each signature is annotated with the number of wells aggregated and the number of distinct tuples averaged into it, so the level of chemical aggregation is explicit in the output tables.

Public CRISPR signatures were already available in the Replogle K562 genome-wide Perturb-seq atlas (8,603 perturbations [4]), the Norman K562 CRISPR-activation atlas (236 signatures spanning 105 single-gene and 131 paired perturbations [3]), and the scPerturb harmonized matched-cell THP1 atlas (78 single-target and combinatorial perturbations [9]). Each CRISPR perturbation was reduced to the same up/down rank representation as the Z-Screen signatures. Chemistry and CRISPR signatures were then compared by:

- mimic overlap (chemistry-up matches CRISPR-up plus chemistry-down matches CRISPR-down),
- reverse overlap (chemistry-up matches CRISPR-down plus chemistry-down matches CRISPR-up),
- net mimic overlap (mimic minus reverse),
- rank cosine over shared top-250 genes.

For each chemistry-by-reference comparison the top 50 CRISPR matches per chemical query were retained. The top 10,000 rows by net mimic overlap and total mimic overlap were then calibrated by drawing 1,000 random gene sets matched in size to each CRISPR signature from the dataset-specific gene universe and recomputing the score under each random draw. Empirical p-values were converted to Benjamini-Hochberg q-values across the calibrated rows.

Rank signatures were used in preference to raw expression vectors. Replogle K562 was generated on a different sequencing platform than Z-Screen and at much higher per-profile depth: Replogle profiles individual cells deeply, while Z-Screen reads each 50,000-well chip as a per-well pseudobulk over the 5 to 10 cell colony in each one-bead-one-compound micro-well. Norman is CRISPR-activation in K562; scPerturb is a harmonized merger of multiple ECCITE-seq panels in THP1. Forcing all of these into one normalization risks introducing platform-specific artifacts that the comparison would then “discover.” Rank signatures discard calibration differences and preserve the strongest property the data shares: the ordering of which genes go up and which go down. The full reproduction order is in `paper5a/README.md`.

Single-building-block aggregation is not a chemistry unit (Figure 1)

Z-Screen libraries differ in how many wells fall into each chemistry unit. ZEL024 in HEK293 carries 13,931 unique 4-position tuples (bb0+bb1+bb2+bb3); 8,599 of those have at least 10 wells, the threshold used for tuple-level signatures. For diagnostic aggregation, the same wells can also be collapsed into 84 bb0+bb1 pair signatures or 83 single-bb3 signatures. These are not parallel chemistry units. A bb0+bb1 pair signature averages a median of 165 distinct tuples, and a single bb3 signature averages a median of 168 distinct tuples; other single-BB positions in ZEL024 / HEK293 can average a median as high as 6,965 distinct tuples per signature. The point of Figure 1 is to test what is lost when molecule-level chemistry is over-collapsed.

Single-bb3 signatures were built from the same cells, and each tuple’s top-5 Replogle CRISPR matches were compared with the matches that the containing bb0+bb1 pair and the containing bb3 single-position diagnostic would produce. For 99.686% of tuples, the top-5 CRISPR matches at tuple resolution shared zero genes with the top-5 at bb3 single-BB aggregation; the median Jaccard overlap between the tuple top-5 and the bb3 top-5 was 0.000. The same pattern appeared for tuple-versus-pair: 99.721% zero overlap, median Jaccard 0.000.

This finding is a warning against chemical over-collapse. Coarser averages can produce almost independent CRISPR neighborhoods because they mix hundreds to thousands of distinct molecules. The practical implication is direct: a discovery team that wants to act on a CRISPR-like match needs that match attached to the specific chemistry responsible for it. Tuple-level signatures are therefore used for the headline ZEL024 / HEK293 result, and all coarser outputs are treated as diagnostics with explicit n-wells and n-tuples-averaged annotations.

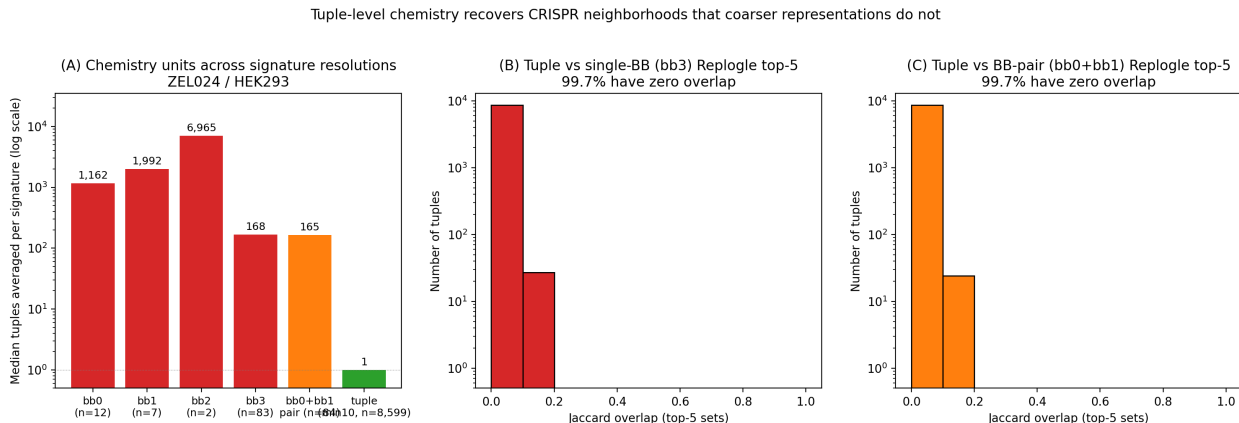


Figure 1. Aggregation diagnostics for ZEL024 / HEK293. (A) Median number of distinct chemical tuples averaged into each diagnostic signature, log-scale. Single-BB signatures average over many distinct compounds sharing one building-block position; tuple signatures average over one full molecule-level tuple. (B) Top-5 Replogle CRISPR matches at tuple resolution share zero genes with the containing single-BB bb3 diagnostic for 99.686 percent of 8,599 tuples. (C) The same comparison at tuple-versus-bb0+bb1 pair diagnostic resolution, with 99.721 percent zero overlap. The finding argues against chemical over-collapse, not for treating single-BB averages as chemistry units.

Tuple-level chemistry maps onto 1,714 distinct CRISPR programs in ZEL024 / HEK293 (Figure 2)

The headline scan compared each of the 8,599 ZEL024 / HEK293 tuple signatures against the 8,603 Replogle K562 CRISPR-knockout signatures [4], a comparison space of approximately 74 million chemistry-by-CRISPR pair scores, calibrated for the top 10,000 matches by net mimic overlap. The calibrated rows were dominated by hits at $q \leq 0.05$ across 1,276 distinct chemical tuples and 1,714 distinct CRISPR perturbation programs. The strongest single matches reached net z above 14 (Figure 2 panel A), with the top 15 distinct CRISPR programs all between net z 12.8 and 14.8. The biology was coherent: DNA replication (MCM2, SLBP, CHAF1B, ARIH1, ADSL), mitochondrial function (MRPL3, MRPS30, COX17, COA5, ATP5F1A, GFM1, HSPE1), trafficking and nonsense-mediated decay (EFR3A, DDOST, SMG5).

The same calibrated table also contains the strongest internal-consistency signal in the analysis. Among the top-100 best- z tuples, 12 distinct CRISPR programs were each reached by two or more chemically distinct tuples (Figure 2 panel B). EFR3A was matched by 7 distinct tuples, TONSL by 6, MCM2 by 5; multiple GART, MRPL42, EIF4G2, MRPL53, and COA5 hits each recurred 3 times. In each of these cases independent chemistries (different combinations of the four populated ZEL024 building-block positions: bb0, bb1, bb2, bb3) produced transcriptional states that overlap with the same CRISPR perturbation, which is the empirical signature expected if the chemistry-to-CRISPR map carries real biological information rather than per-compound noise.

A small number of CRISPR perturbations recurred as top neighbors of many otherwise unrelated chemistries. Broadly responsive CRISPR knockouts can match many transcriptional states by being broadly responsive themselves, and a benchmark that does not control for that property will appear stronger than it is. The calibrated top-10,000 table was therefore rerun while progressively dropping the 5, 10, 25, and 50 most-recurrent CRISPR perturbations (Figure 2 panel C). After dropping the 50 most-recurrent hubs, 1,236 distinct chemical tuples and 1,664 distinct CRISPR programs still passed $q \leq 0.05$, so the program diversity does not collapse onto a small set of hub perturbations.

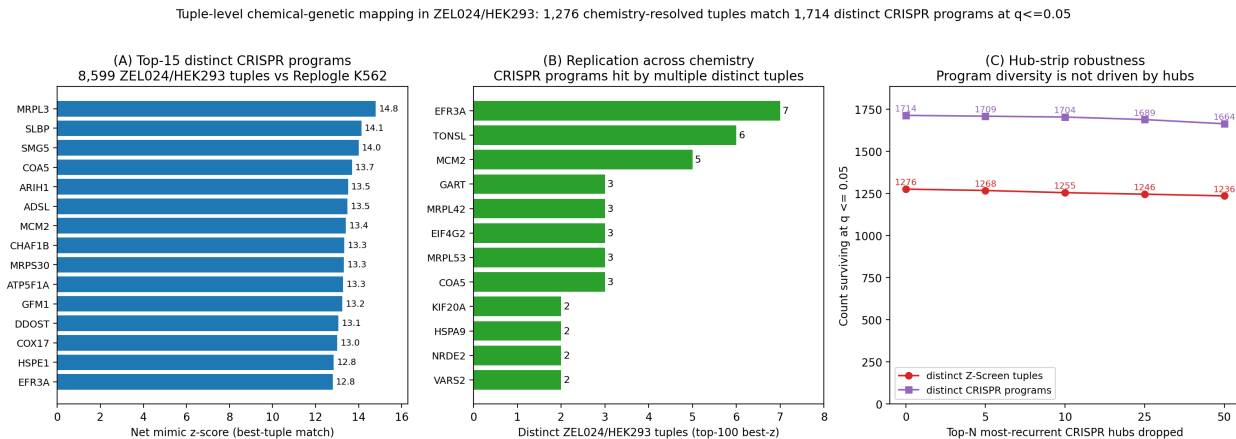


Figure 2. Tuple-level chemistry maps onto 1,714 distinct CRISPR programs in ZEL024 / HEK293. (A) Top-15 distinct CRISPR programs by best Z-Screen tuple z-score, deduplicated to one tuple per

CRISPR program. (B) Replication-across-chemistry: among the top-100 best-z tuples, CRISPR programs reached by multiple chemically distinct tuples. (C) Hub-strip robustness: the calibrated top-10,000 table after dropping the top-N most-recurrent CRISPR perturbations. Tuple and program diversity at $q \leq 0.05$ are essentially unchanged at $N = 50$.

Same-cell positive controls anchor the framework (Figure 3)

The ZEL024 map is cross-cell: HEK293 chemistry is being compared with K562 CRISPR knockouts. Before interpreting those matches, the method needs a same-cell sanity check. Two THP1 controls provide it. The first is MZ1, a small-molecule PROTAC that selectively degrades BRD4 [10]. MZ1 lives in a sparse THP1 control library that lacks adequate same-library background, so its signature was built against a same-cell compound-pool fallback background (all THP1 compound-treated cells excluding MZ1). Under that background, MZ1 recovered BRD4 as the top calibrated CRISPR-like signature in the local THP1 panel (net mimic = 20, rank cosine = 0.539, calibrated net $z = 6.84$, empirical $q = 0.015$).

The second control is STC-15, a METTL3 inhibitor. The direct target METTL3 is not present in the local THP1 CRISPR panel, but the documented downstream consequence of METTL3 inhibition is interferon-pathway activation [11]. STC-15 recovered IRF1 as the second calibrated CRISPR-like match ($z = 3.06$, $q = 0.045$), consistent with that downstream phenotype, with IRF7 as the third match ($z = 2.70$, $q = 0.096$).

The fallback-background construction for MZ1 is broader than a same-library vehicle control would be, so the absolute z -score should be read in that context. The structure of the result, BRD4 recovered as the top match across permutation seeds, is robust. Together, MZ1 and STC-15 anchor two cases this type of analysis must distinguish: the direct genetic analog is in the panel and is recovered as the top hit, or the direct analog is absent and the top hit is a downstream pathway rather than a generic stress signature.

The non-BRD4 neighbors in panel A of Figure 3 are not additional direct target calls. SUPT16H+SUPT5H, FLI1+MED24, MED24+SMARCD1, EP300+MED24, EP300+INTS1, and INTS1+SMARCD1 sit in the BRD4-adjacent transcription-elongation and chromatin co-regulator neighborhood that MZ1 perturbs by removing BRD4 from the cell. CAV1 is less directly established as BRD4-regulated in THP1 and is treated as a state-neighborhood match rather than a direct mechanistic claim.

Same-cell positive controls. MZ1 recovers BRD4 ($z=6.84$, $q=0.015$); STC-15 recovers IRF1 in top-2 as a downstream interferon-pathway proxy ($z=3.06$, $q=0.045$).

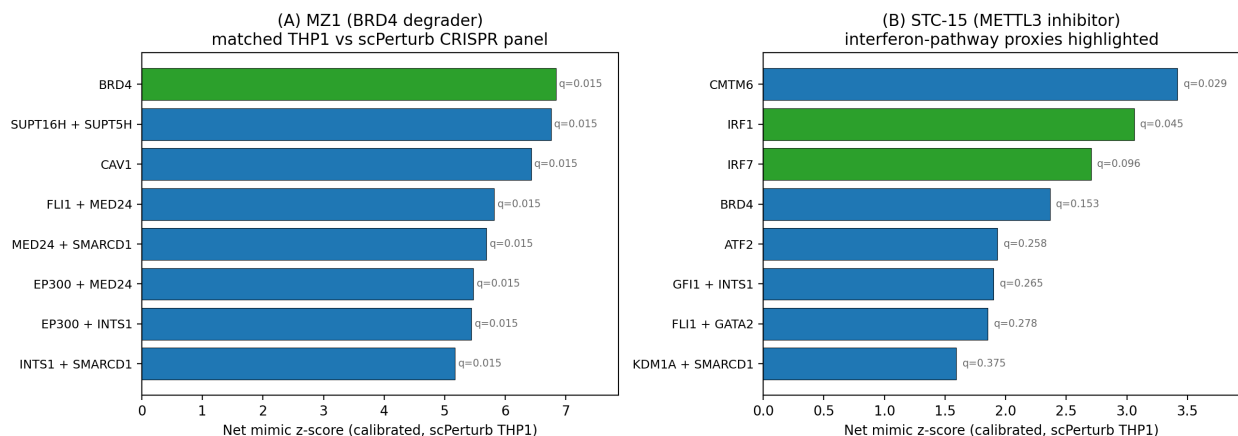


Figure 3. Same-cell positive controls in matched THP1 with calibrated empirical q-values from a permutation-null gene-set calibration. (A) MZ1 (BRD4-selective PROTAC degrader): BRD4 is the top calibrated CRISPR-like signature ($z = 6.84$, $q = 0.015$), with subsequent neighbors falling in BRD4-adjacent transcriptional and chromatin programs. (B) STC-15 (METTL3 inhibitor): METTL3 itself is absent from the local CRISPR panel; the downstream interferon-pathway proxy IRF1 is recovered in the top two calibrated matches ($z = 3.06$, $q = 0.045$).

Documented-target recovery is enriched but conditional (Figure 5B)

The same-cell controls ask whether the machinery works in clean examples. A broader documented-target scan asks a harder question: when a compound has a known primary target, does a cross-cell CRISPR rank list place that target unusually high?

In `known_biology_recovery.csv`, 34 compound-cell-line comparisons cover 21 Z-Screen compounds. Restricting to the 27 comparisons evaluated against the 8,839-perturbation Replogle K562 ranking (across 17 unique compounds with a documented primary target represented in Replogle), seven placed the documented target in the top 15 percent: FGFR1 for derazantinib in HEK293 (rank 145, about 1.6 percent), SMARCA4 for SMARCA ligand 1 in A549 (rank 531), MAPK14 for AZD-7624 in A549 (rank 621), METTL3 for STC-15 in A549 (rank 784, about 8.9 percent), FGFR1 for fexagratinib in A549 (rank 1,168), PARP2 for veliparib in A549 (rank 1,206), and KRAS for MRTX-1719 in A549 (rank 1,210). One additional entry, STC-15 in HEK293, was scored against a 17,678-item merged Replogle+Norman ranking (METTL3 rank 1,102, about 6.2 percent of the merged universe) and is reported separately because the reference universe is not directly comparable.

The failures are part of the result. Cobimetinib’s MAP2K2 ranked 4,694 in A549 and 6,685 in HEK293, consistent with allosteric MEK inhibition not always producing a knockout-like transcriptional consequence. GSK126’s EZH2 ranked 8,719 of 8,839 in A549, well in the anti-correlated tail, consistent with the long protein half-life of EZH2 under enzymatic inhibition. Cell-line dependence is also visible: BAY-293 recovers SOS1 at rank 2,106 in A549 but rank 7,802 in HEK293, and AZD-7624 recovers MAPK14 at rank 621 in A549 but rank 3,712 in HEK293. The cross-cell

scan is not target deconvolution. It is enriched for documented mechanisms when the molecular pharmacology produces a knockout-like transcriptional state, and it is informative when it does not.

Tuple scores preserve Norman combinatorial logic (Figure 5A)

The target-recovery scan tests single-gene plausibility. Norman tests a different property: whether chemical tuple scores follow the combinatorial structure of paired genetic perturbations. A Z-Screen tuple is one assembled molecule, measured as the per-well pseudobulk over the 5 to 10 cells in that one-bead-one-compound well; a Norman double is two CRISPR-activation perturbations applied together to one cell. The analysis therefore does not claim that a molecule is a two-gene perturbation. It asks whether the rank-signature scoring behaves consistently when the genetic reference itself has a known sum-of-singles structure.

The Norman K562 CRISPR-activation atlas contains 131 double-gene perturbations whose constituent single-gene perturbations are also profiled. For each Norman double, the Spearman correlation across the 8,599 ZEL024 / HEK293 tuples was computed between the tuple-vs-double match score and the sum of tuple-vs-single-A plus tuple-vs-single-B match scores. All 131 doubles produced positive correlations with $p < 0.05$ (median Spearman $\rho = 0.353$). The strongest pairs (ETS2+MAPK1, MAP2K3+ELMSAN1, CEBPB+MAPK1, MAP2K3+MAP2K6, FOSB+OSR2) reached ρ between 0.60 and 0.69. For 98.5 percent of the doubles, the best Z-Screen tuple’s score against the double exceeded its score against either constituent single. Full-tuple chemical signatures therefore track the paired-perturbation logic the Norman atlas was built to express.

Two separate checks on the chemistry-to-CRISPR rank-signature map

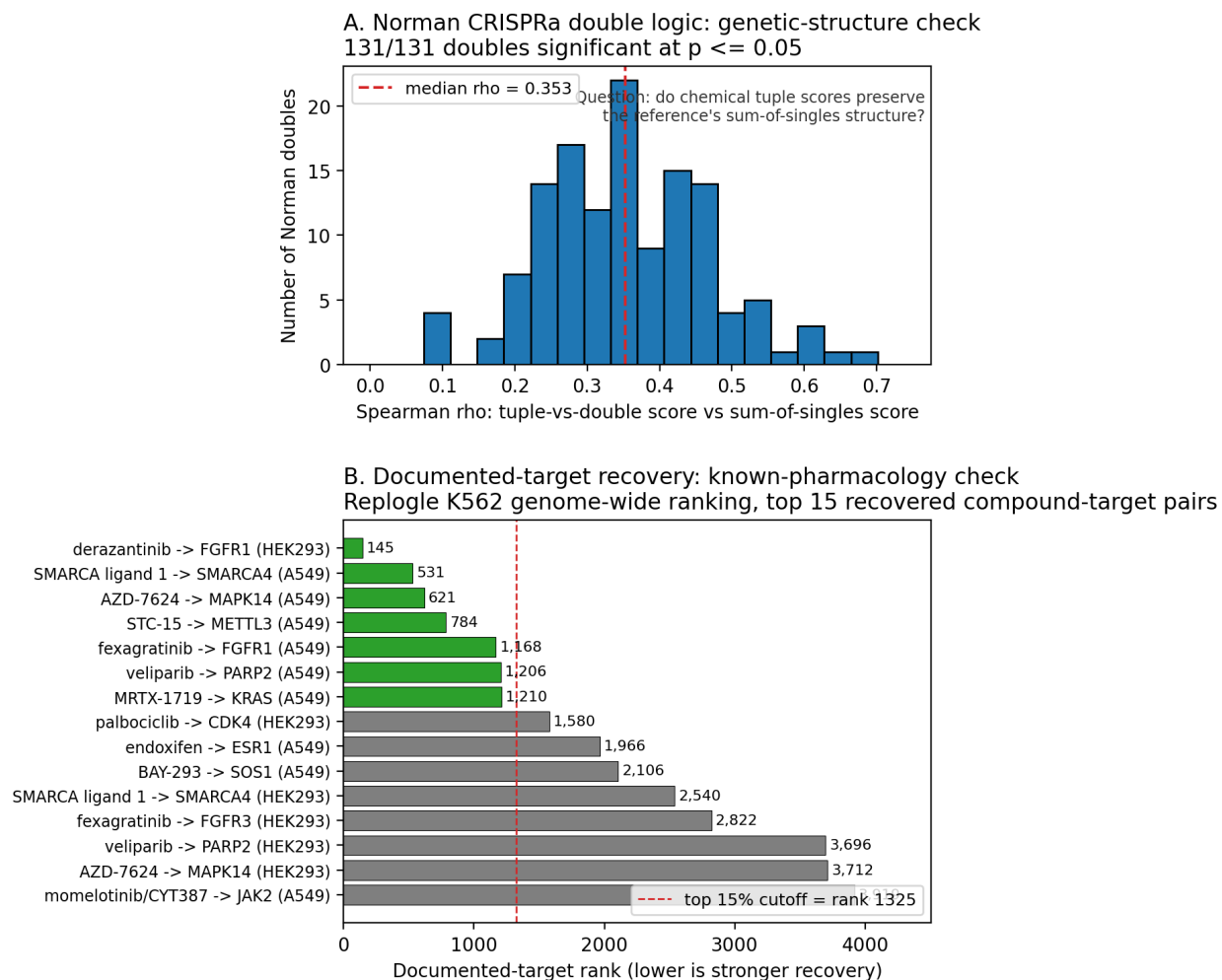


Figure 5. Two separate validation checks. (A) Norman combinatorial logic: distribution of Spearman correlations across 131 Norman K562 doubles between tuple-vs-double match scores and sum-of-singles match scores; all 131 doubles significant at $p \leq 0.05$; median rho = 0.353. (B) Documented-target recovery: ranks within the Replegle K562 cross-cell ranking for the top-15 best-recovered compound-target pairs; the dashed line marks the top 15 percent of the 8,839-perturbation reference.

BAY-293 bridges primary pharmacology to a V-ATPase hypothesis (Figure 4)

The control and enrichment analyses establish that expected biology can be recovered, but the useful next step is a case where the top CRISPR neighborhood is not just the annotated primary target. BAY-293 is an SOS1 inhibitor profiled in A549 in a non-combinatorial control library. In the documented-target scan above, SOS1 is visible but not dominant in A549 (rank 2,106). The calibrated Replegle neighborhood instead points to a coherent V-ATPase and vesicular-trafficking state.

The top 12 calibrated Replogle K562 matches for BAY-293 in A549 all sat at empirical $q = 0.003$, with net z between 9.5 and 11.9. The matches concentrated on the V-ATPase complex (ATP6V1A, ATP6V1H, ATP6V1B2, ATP6V1C1, ATP6V1E1) and adjacent vesicular-trafficking machinery (ZW10, SACM1L, TMED2, ZNHIT1, WDR7, CCDC115, CLTC). RAS signaling intersects endosomal traffic in multiple documented places, so a transcriptional state that overlaps both V-ATPase and trafficking knockdown signatures is biologically interpretable. It is also a useful bridge into the ZEL028-2 hierarchy below, where ATP6V1A re-emerges from an unrelated combinatorial tuple rather than from BAY-293's known SOS1 pharmacology.

BAY-293 transcriptional state matches V-ATPase + vesicular trafficking CRISPR programs (top-CRISPR $z=11.86$, $q=0.003$)

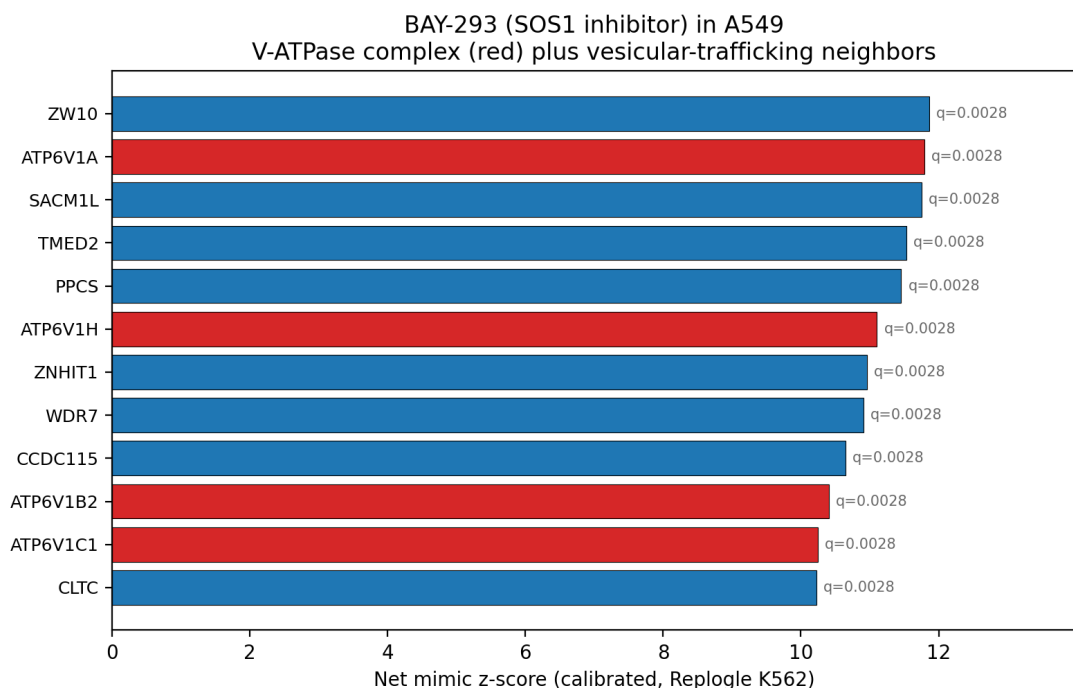


Figure 4. BAY-293 bridge example. The SOS1 inhibitor BAY-293 in A549 matches the V-ATPase complex (red) and vesicular-trafficking CRISPR knockouts (blue) at calibrated empirical $q = 0.003$. Top hit ZW10 reaches net $z = 11.86$; nine V-ATPase or trafficking components reach net z above 10.

ZEL028-2 uses hierarchy to rescue signal from shallow tuple coverage (Supplementary Figures 2 and 3)

ZEL028-2 is the stress test for the framework. It has many observed chemistries but shallow per-tuple support: 61,396 unique full tuples in HEK293, 40,622 in A549, and 25,906 in H1650, but only 4,041, 1,728, and 765 full-tuple signatures qualify at the relaxed ≥ 2 -well threshold. The median qualifying tuple still has only 2 wells. The full-tuple scan is therefore useful for hypothesis generation, but it cannot be read like the ZEL024 / HEK293 min10 tuple scan.

The calibrated min2 full-tuple scans nevertheless show real structure. The HEK293 deduplicated table covers 943 distinct chemical tuples and 285 distinct CRISPR programs; the strongest full-tuple

matches reach net $z = 24.7$ in HEK293, 20.5 in A549, and 18.6 in H1650. Top distinct CRISPR programs concentrate on RNA processing (DICER1, DROSHA, CNOT3, SMG5, and the EXOSC2 / EXOSC4 / EXOSC6 / EXOSC8 exosome subunits), mitochondrial translation (MRPL3, MRPL10, MRPL18, MRPL19, MRPL35, MRPL43, IARS2, VARS2, GFM1), and helicase or translation-initiation programs (DHX36, EIF3H, RPS4X, RPS6, RACK1). Fifteen CRISPR programs appear in the top-50 deduplicated hits in all three ZEL028-2 cell lines. At the same time, exact-tuple cross-cell replication is sparse: only 36 qualifying min2 full tuples appear in at least two cell lines, and only one appears in all three.

The solution was to treat ZEL028-2 as a three-resolution evidence problem. First, full tuples at ≥ 2 wells provide chemistry-specific hypotheses. Second, variable-pair subset signatures across bb1+bb3, bb1+bb4, and bb3+bb4 at ≥ 10 wells provide better-supported partial-chemistry evidence: 3,051 to 3,103 signatures per pair in HEK293, 909 to 1,260 in A549, and 141 to 257 in H1650. Third, the variable-triple bb1+bb3+bb4 at ≥ 2 wells is reported alongside the full tuple because bb0 is absent and bb2 is fixed. The ZEL028-2 variable-pair panels also preserve Norman logic: across all 12 panels (4 panels per cell line), 90.8 to 100 percent of the 131 Norman doubles were significant at $p \leq 0.05$; median Spearman ranged 0.325 to 0.371; and the best Z-Screen signature score against the double exceeded both single scores in 86.3 to 98.5 percent of doubles.

The mechanism-support table starts from a narrower row universe than the raw signature counts. The raw observed universe is 61,396 / 40,622 / 25,906 full tuples. The min2 signature universe is 4,041 / 1,728 / 765 full-tuple signatures. The hierarchical support universe is 2,151 deduplicated calibrated full-tuple calls, one best CRISPR call per represented tuple query from the full-tuple min2 calibrated scans: 943 in HEK293, 760 in A549, and 448 in H1650. Each row is annotated for whether the same CRISPR perturbation appears in same-cell variable-pair top-50 lists, same-cell variable-pair calibrated $q \leq 0.05$ rows, cross-cell variable-pair lists, cross-cell calibrated rows, the variable-triple scan, and whether the CRISPR program is a top-50 recurrent hub.

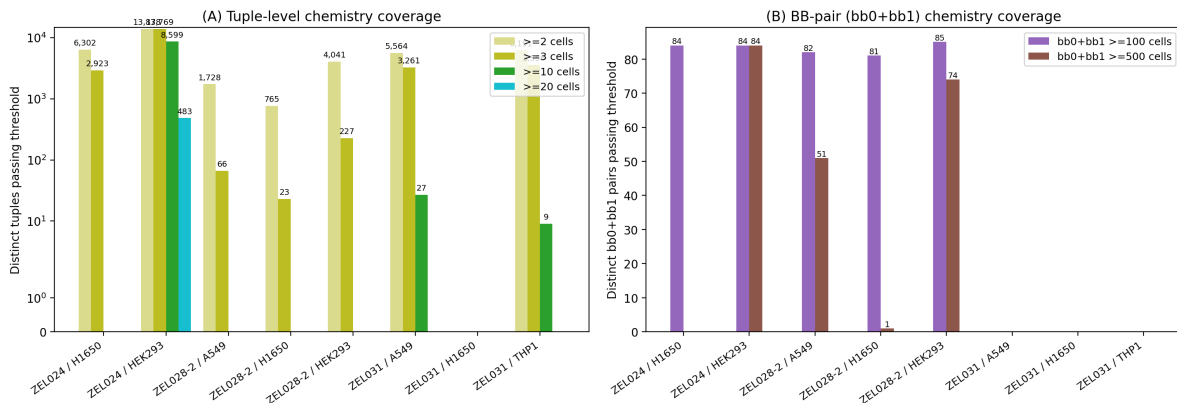
Of those 2,151 deduplicated calls, 1,967 (91.4 percent) target a top-50 recurrent hub program. The remaining non-hub set contains 171 candidates. Within that non-hub group, 54 have the same CRISPR perturbation in at least one same-cell variable-pair top-50 list, and 25 of those have it at calibrated $q \leq 0.05$ in at least one same-cell pair scan. Forty-five have the same CRISPR perturbation in at least one cross-cell variable-pair top-50 list, and 13 of those have it at calibrated $q \leq 0.05$ in at least one cross-cell pair scan. Two non-hub candidates pass the strictest FDR-controlled subset-support tier: HEK293 ATP6V1A (full-tuple net $z = 15.47$, $q = 0.001$; bb3+bb4 variable-pair calibrated subset $z = 7.94$, $q = 0.001$) and A549 FKBP9 (full-tuple net $z = 11.87$, $q = 0.001$; bb1+bb4 variable-pair calibrated subset $z = 8.96$, $q = 0.001$).

ATP6V1A is the key reason BAY-293 was treated as a bridge rather than a side anecdote. BAY-293 independently produced a calibrated V-ATPase / trafficking neighborhood in A549, including ATP6V1A. ZEL028-2 then surfaced ATP6V1A from an unrelated HEK293 combinatorial tuple, with same-cell variable-pair calibrated support. That recurrence does not prove target engagement, but it converts an isolated compound-control observation into a concrete chemistry-resolved follow-up: a specific ZEL028-2 full tuple, its bb3+bb4 support, and a V-ATPase hypothesis that can be tested

biochemically and transcriptionally.

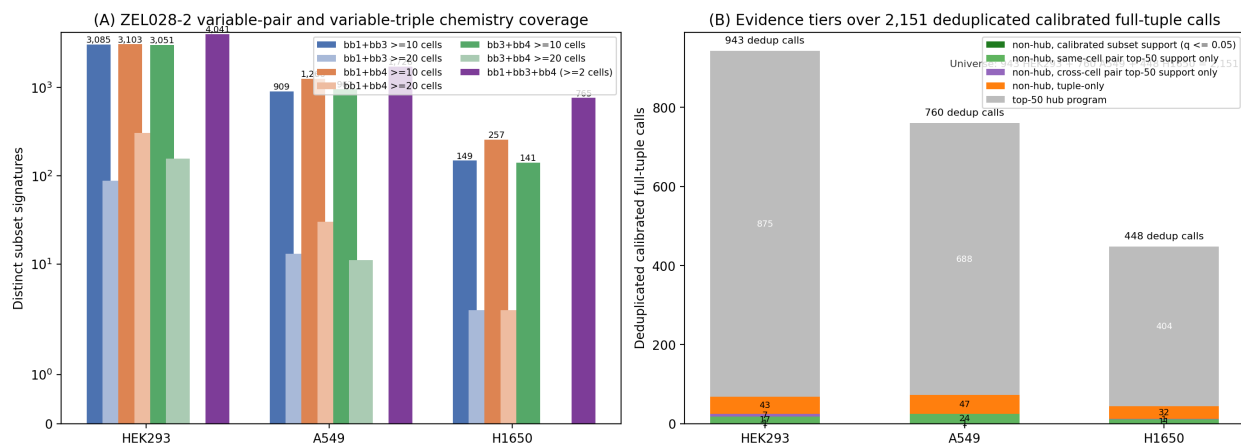
The broader top-50-supported tier, which lacks FDR-controlled subset support, contains 11 additional non-hub candidates beyond the two FDR-controlled ones. These include MRPL42, EXOC4, and DONSON in HEK293; PSMG1, MRPL51, POLD3, and a second MRPL51-supported A549 tuple; ZCRB1 and ARIH1 in H1650; plus H1650 FKBP9 and H1650 NUP54 supported by A549 pair panels. The boundary between calibrated subset support and top-50-only support is deliberately kept visible. The full tuple-by-tuple table is `paper5a/tables/ZEL028-2_hierarchical_mechanism_support.csv`; the per-cell-line summary is `ZEL028-2_hierarchical_mechanism_support_summary.csv`; the FDR-controlled and broader subset-supported shortlists are in `ZEL028-2_hierarchical_mechanism_support_top_candidates.csv`.

Chemistry coverage across Z-Screen systems sets the analytical resolution per system: ZEL024 / HEK293 supports tuples at ≥ 10 cells, ZEL031 systems at ≥ 3 cells, and ZEL028-2 systems at ≥ 2 cells.



Supplementary Figure 2. Z-Screen chemistry coverage across systems. (A) Distinct tuples passing ≥ 2 , ≥ 3 , ≥ 10 , and ≥ 20 well thresholds in each system. ZEL024 / HEK293 supports tuple-level signatures at ≥ 10 wells, ZEL031 systems at ≥ 3 wells, and ZEL028-2 systems at ≥ 2 wells. (B) Distinct bb0+bb1 pairs passing 100 and 500 well thresholds in libraries that populate bb0; ZEL028-2 has no substantive bb0 position, so its bb0+bb1 pair counts collapse to the same single-bb1 aggregation regardless of bb2/bb3/bb4 and are not used as a chemistry unit anywhere in this manuscript. The chemistry-resolved ZEL028-2 analysis uses variable-pair (bb1+bb3, bb1+bb4, bb3+bb4) and variable-triple (bb1+bb3+bb4) signatures (Supplementary Figure 3). ZEL024 supports both tuple and bb0+bb1 pair resolution; ZEL031 has too many distinct pairs for any single pair to carry meaningful well counts.

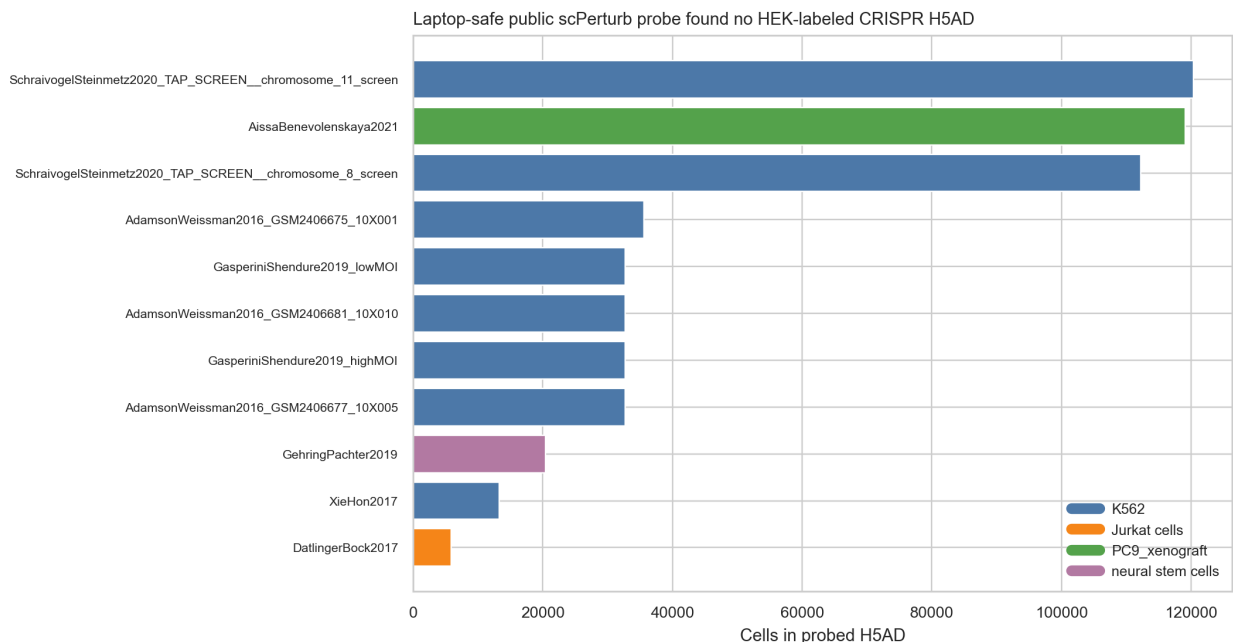
ZEL028-2 coverage and support tiers for 2,151 deduplicated calibrated full-tuple calls.



Supplementary Figure 3. ZEL028-2 multi-resolution chemistry coverage and hierarchical mechanism support. (A) Variable-pair (bb1+bb3, bb1+bb4, bb3+bb4) subset signatures qualifying at ≥ 10 wells per signature and at ≥ 20 wells per signature, plus the variable-triple bb1+bb3+bb4 at ≥ 2 wells, across HEK293, A549, and H1650. (B) Hierarchical mechanism-support breakdown of the 2,151 deduplicated calibrated full-tuple calls. Bars are stacked from bottom to top: non-hub with calibrated subset support at $q \leq 0.05$ (FDR-controlled tier; 2 candidates total: HEK293 ATP6V1A and A549 FKBP9), non-hub with same-cell variable-pair top-50 support only, non-hub with cross-cell variable-pair top-50 support only, non-hub tuple-only, and top-50 hub programs. Total counts above each bar give the number of deduplicated calibrated full-tuple calls per cell line.

A targeted public probe did not surface a matched HEK293 single-cell CRISPR atlas (Supplementary Figure 1)

Eleven scPerturb H5AD files under 500 MB each were probed by reading observation metadata only, looking for a HEK293 or HEK293T single-cell CRISPR dataset that would allow direct same-cell evaluation against the HEK293-rich Z-Screen libraries. The probed datasets covered K562, Jurkat / T-cell, PC9 xenograft, and neural stem cell perturbation panels. None were HEK-labeled CRISPR datasets, so the present manuscript treats HEK293-to-K562 Replogle comparisons as cross-cell rank-signature state hypotheses rather than matched-cell HEK293 target calls. Paper 4 separately reports learned cross-cell transfer of chemical programs. The probe is intentionally laptop-scale and is not a comprehensive survey of all public HEK293 perturbation data.



Supplementary Figure 1. Targeted probe of 11 small (<500 MB) scPerturb H5AD files for matched HEK293 single-cell CRISPR data. Each row reports cell-line label, perturbation type, perturbation count, and dataset dimensions; no HEK-labeled CRISPR dataset was identified locally.

Discussion

The result that frames the rest of the paper is the headline ZEL024 / HEK293 scan: 1,276 distinct chemical tuples each carry a calibrated CRISPR-like hypothesis at empirical $q \leq 0.05$, and the resulting hits cover 1,714 distinct CRISPR perturbation programs. The numbers themselves are not the central result; the central result is that each one of those tuples is a specific 4-position (bb0 / bb1 / bb2 / bb3) chemistry that a discovery team can synthesize, vary, and test, with the underlying CRISPR program available for orthogonal validation. The framework therefore turns Z-Screen output from a catalog of measurements into a structured search engine over genetic perturbation space.

The matched-cell controls validate the framework’s calibration. MZ1 recovers BRD4 with a clean separation against the same-cell THP1 CRISPR panel; STC-15 recovers IRF1 within the top two calibrated matches as a downstream proxy of METTL3 inhibition. Where the genetic analog is present, the framework recovers it; where the genetic analog is absent, the top hit is a documented downstream pathway rather than a generic stress signature. BAY-293 in A549 demonstrates the discovery layer: the framework surfaces a coherent V-ATPase / vesicular trafficking hypothesis ranked alongside, but not predicted by, primary pharmacology.

The combinatorial-logic test is the strongest evidence that the framework is reading real combinatorial chemistry rather than per-tuple noise. All 131 Norman doubles tested produce positive Spearman correlations between tuple-vs-double scores and tuple-vs-sum-of-singles scores in the ZEL024 / HEK293 headline. The same combinatorial logic reproduces independently in ZEL028-2 across

HEK293, A549, and H1650 and at three different chemistry resolutions (variable-pair, variable-triple, full-tuple), with 90.8 to 100 percent of doubles significant at $p \leq 0.05$ across all 12 ZEL028-2 panels. The chemistry follows the genetics’ combinatorial structure at chemistry-resolved level in three different cell lines and across multiple library architectures, which is the same structure the Norman atlas was built to express. That is not a result a one-profile-per-compound platform such as L1000 can produce, because there is no internal chemistry structure inside a single L1000 entry to test combinatorial logic against.

The ZEL028-2 hierarchical mechanism-support model also matters in its own right. ZEL028-2 has shallow per-tuple coverage (median 2 wells per qualifying full tuple), so individual full-tuple calls are noisy. A naive use of the calibrated full-tuple leaderboard would over-claim. The revised count universe is explicit: 61,396 / 40,622 / 25,906 observed full tuples, 4,041 / 1,728 / 765 min2 tuple signatures, and 2,151 deduplicated best full-tuple CRISPR calls used for mechanism support. Building variable-pair subset signatures at ≥ 10 wells, scanning each pair panel against the same CRISPR atlases, permutation-calibrating those subset scans with the same size-matched null used for the full-tuple headline, and then asking whether each deduplicated full-tuple call also appears at calibrated $q \leq 0.05$ in independent pair subsets converts a noisy signal into an FDR-controlled shortlist. Two non-hub candidates pass the strictest tier: HEK293 ATP6V1A (full-tuple net $z = 15.47$; bb3+bb4 pair calibrated $z = 7.94$) and A549 FKBP9 (full-tuple net $z = 11.87$; bb1+bb4 pair calibrated $z = 8.96$). ATP6V1A’s re-emergence is the clearest cross-system biological coherence in the analysis: the same V-ATPase target is the calibrated top-12 Replogle hit for BAY-293 in A549 (Figure 4), and the HEK293 ZEL028-2 chemistry that surfaces ATP6V1A independently from a four-position tuple at calibrated subset support is precisely the kind of entry that justifies a specific medicinal-chemistry follow-up. Eleven additional non-hub candidates carry uncalibrated subset top-50 support and form a broader investigation queue; the boundary between FDR-controlled and uncalibrated support is reported transparently rather than collapsed into a single shortlist.

L1000 remains a useful external reference for data-quality positioning, and a separate cross-platform concordance analysis comparing Z-Screen rank signatures with LINCS L1000 level-5 consensus signatures lives in the repository root at **Benchmarking/**. The quantitative claims in this manuscript are restricted to outputs in **paper5a/tables/** (rank-signature comparison, permutation calibration, recurrence diagnostics, controls, resolution diagnostics, and Norman combinatorial logic) so that every reported number is reproducible from the bundled reproduction package alone.

CRISPR similarity is not target identification. A small molecule may resemble a knockout because it directly engages the encoded protein, because it perturbs an upstream regulator, because it triggers a compensatory state, or because both perturbations converge on a broad stress program. The framework therefore treats every match as a state-level hypothesis until orthogonal validation upgrades it. The recurrent-neighbor diagnostic and the hub-strip analysis are reported alongside the calibrated headline numbers because broadly responsive CRISPR perturbations need to be flagged rather than silently lost in the leaderboard.

The framework is naturally tied to follow-up. The top combinatorial chemistries pointing toward a desired CRISPR state are concrete tuples with explicit per-position chemistry: bb0 / bb1 / bb2 /

bb3 in ZEL024, bb1 / bb2 / bb3 / bb4 in ZEL028-2, bb0 / bb1 in ZEL031. A follow-up library can enrich the partner chemistry around those tuples, profile the new compounds, and update the model. Paper 4 separately reports cross-cell transfer at building-block-position centroid resolution, complementary to but not tuple-resolved in paper 5.

A compact internal validation panel falls out of the analysis. BRD4 as a positive control; MED24, INTS1, EP300, FLI1, HDAC3, MYC, and SMARCD1 to test the recurrent transcription-and-chromatin programs; PDCD1LG2 for the strongest checkpoint-like hypothesis from the HEK293 diagnostic BB-pair signatures; SMAD4 as a stress / state comparator; JAK2, IFNGR1, IFNGR2, and STAT1 to test the interferon-pathway proxies that emerged for STC-15 and several non-annotated discovery candidates. Running that panel in matched THP1 and HEK293 would directly test the two main claims: that known chemical-genetic analogs are recovered when present, and that the chemistry-resolved hypotheses replicate in a matched CRISPR context.

For drug discovery teams, the practical implication concerns the front end of a campaign. Instead of starting from a target and searching a separate compound database, a team can start from a desired CRISPR state and pull a ranked, calibrated list of combinatorial chemistries that move cells toward that state, with the chemistry decomposable into building-block coordinates that a medicinal chemistry team can act on.

Methods

Z-Screen data inputs

The analysis used the repaired canonical Z-Screen RNA-seq aggregate at `data/ZScreen_Canonical_Dataset/RNASEq`. Primary files: `Full_ANNDData_object_compressed.h5ad` (615,793 wells x 36,601 genes; rows are well-level pseudobulks of 5 to 10 cells per well, stored under the AnnData `obs` convention), `canonical_obs.parquet`, `canonical_chemistry.parquet`, and `canonical_scvi_latents.parquet`. The H5AD repair removed 140 wells affected by corrupted sparse-matrix chunks; the linked parquet files were aligned to the same `obs_index` after repair. See `DATA_REPAIR_REPORT.md`.

Rank signatures rather than expression vectors

The three CRISPR references and the Z-Screen RNA aggregate were generated on different platforms with different normalization conventions and at different sequencing depths per profiled unit. The CRISPR references are per-cell measurements; Z-Screen rows are per-well pseudobulks over the 5 to 10 cell colony in each one-bead-one-compound well. Forcing all of them into a common normalization risks introducing platform-specific artifacts that would then appear as similarity. Rank signatures discard calibration differences and preserve the strongest property the data shares: the ordering of genes most up- and most down-regulated relative to background. The choice is conservative. It costs some statistical power compared with full expression-vector regressions, and provides the ability to compare across heterogeneous public datasets without assuming they were normalized the same way.

Z-Screen tuple signatures

For each library and cell line, building-block-annotation wells (`chemistry_grain == "building_block_annotation"`) were grouped by the concatenated tuple identifier `bb0_id|bb1_id|bb2_id|bb3_id` with unused positions filled as the literal string "NA" (so a ZEL024 tuple has the form `bb0|bb1|bb2|bb3|NA`, a ZEL028-2 tuple has `NA|bb1|bb2|bb3|bb4`, and a ZEL031 tuple has `bb0|bb1|NA|NA|NA`). Tuples with at least 10 wells were retained for the headline ZEL024 / HEK293 analysis (8,599 tuples, median 13 wells per tuple). A supplementary ≥ 20 -well threshold yielded 483 tuples and is reported alongside the headline as a stricter sensitivity check. ZEL031 systems used a ≥ 3 -well threshold (3,261 in A549, 3,556 in THP1) because the two-position library is too diverse for higher thresholds. ZEL028-2 systems used a ≥ 2 -well threshold because per-tuple coverage in that library does not support the ≥ 10 -well or ≥ 3 -well thresholds; under ≥ 2 wells, 4,041 (HEK293), 1,728 (A549), and 765 (H1650) tuples qualify, with a median of 2 wells per qualifying tuple. The shallower per-tuple support in ZEL028-2 is recorded in every signature's `n_cells` field (the field name follows the AnnData convention; the values count well-level pseudobulks), and is the reason the ZEL028-2 tuple analysis is interpreted as a chemistry-resolved leaderboard rather than as per-tuple definitive calls. For each tuple, target-well counts were aggregated by direct sparse-matrix slicing of the H5AD; the background was the same library and cell line excluding target-tuple wells. The effect vector was $\log\text{CPM}(\text{target}) - \log\text{CPM}(\text{background})$. Genes were ranked separately in the up and down directions; the top 250 per direction were retained.

The same-library, same-cell-line background was used in preference to a global baseline so that bias from chemistry-independent differences across libraries (e.g. the synthesis chemistry of one library affecting baseline cell state) does not appear as signal. Built by `paper5a/scripts/run_tuple_signatures.py`.

Diagnostic aggregation: BB-pair and single-position signatures

BB-pair and single-position signatures were built as aggregation diagnostics and metadata transparency checks, not as molecule-level chemistry units. They use the same target-vs-same-library-background logic as tuple signatures, but the target is a collection of many distinct molecules sharing one position or one position pair. ZEL024 / HEK293 carries a diagnostic `bb0+bb1` pair anchor (84 pair signatures, median 165 distinct tuples per signature) and diagnostic `bb3` single-position signatures used in the Figure 1 resolution comparison. The metadata for every signature records `pair_value_a`, `pair_value_b`, and the number of distinct tuples averaged in (`n_tuples_averaged`), so the level of chemical aggregation is explicit on every row. Built by `paper5a/scripts/run_bbpair_signatures.py`.

ZEL028-2 variable-position subset signatures

ZEL028-2 has only three variable chemistry positions (`bb1`, `bb3`, `bb4`); `bb2` is fixed at BB-0000312-002 across all wells, and `bb0` is always missing. A `bb0+bb1` "pair" signature in ZEL028-2 therefore degenerates to a single-`bb1` aggregation regardless of `bb2`, `bb3`, `bb4` and is not a chemistry unit. Variable-position subset signatures (`bb1+bb3`, `bb1+bb4`, `bb3+bb4` at ≥ 10 and ≥ 20

wells, and the variable-triple bb1+bb3+bb4 at ≥ 2 wells) were therefore built directly with `paper5a/scripts/run_bbsubset_signatures.py`. The script uses the same target-vs-same-library-background construction as the tuple builder, with the target defined by the requested subset of populated bb positions. Variable-pair panels at ≥ 10 wells per signature were used as the primary supplementary chemistry-pair grain (3,051 to 3,103 signatures per pair in HEK293; 909 to 1,260 in A549; 141 to 257 in H1650). The variable-triple at ≥ 2 wells is exactly equivalent to the full tuple in this library because bb0 is always missing and bb2 is constant. Each subset panel was scanned against the public CRISPR references with `paper5a/scripts/run_chemistry_vs_crispr.py`; the resulting top-50-per-query tables were used as input to the hierarchical mechanism-support score. The nine variable-pair Replogle scans (3 pair grains x 3 cell lines at ≥ 10 wells) were also permutation-calibrated with `paper5a/scripts/run_calibrate_matches.py` using the same size-matched null procedure as the headline ZEL024 / HEK293 scan (1,000 permutations per row, seed 13), so the subset-support layer in the hierarchical model is FDR-controlled rather than top-50 rank-overlap only.

Public CRISPR rank signatures

Public CRISPR rank signatures used in this manuscript are bundled in `paper5a/external_data/`:

- Replogle K562 genome-scale Perturb-seq, 8,603 perturbations [4].
- Norman K562 CRISPR-activation, 105 single + 131 paired perturbations [3].
- scPerturb harmonized THP1, 78 perturbations from Papalexi/Satija ECCITE-seq and Wesels/Satija combinatorial CRISPR-Cas13 [9].

The rank signatures were generated upstream from the source single-cell CRISPR data with `paper5a/scripts/build_replogle_rank_signatures.py`, `paper5a/scripts/build_norman_rank_signatures.py` and `paper5a/scripts/build_scperturb_crispr_signatures.py`. Each builder converts the public single-cell perturbation data into the same up/down rank format used for Z-Screen, with control cells from the same dataset as background.

Rank-based chemistry-to-CRISPR comparison

Each rank signature was converted into a signed rank vector. Top up-genes received positive scores from $+(\text{top_k} + 1 - \text{rank}) / \text{top_k}$; top down-genes received the negative analog. Each Z-Screen signature was compared with each CRISPR reference signature by:

- mimic up-overlap (Z-up genes hitting C-up genes),
- mimic down-overlap,
- reverse up-down overlap,
- reverse down-up overlap,
- mimic total = mimic up + mimic down,
- reverse total = reverse up-down + reverse down-up,
- net mimic overlap = mimic total minus reverse total,
- rank cosine over shared top genes.

Comparisons used `top_k = 250` and recorded shared-top-gene counts per pair. To keep file sizes bounded for the 8,599-tuple by 8,603-CRISPR scan (74 million pair scores), only the top 50 CRISPR matches per Z-Screen query were retained as raw output. Rank cosine was reported as NaN when fewer than 10 genes were shared between the two signatures. Net mimic overlap (rather than rank cosine alone) is the headline scoring metric because it penalizes anti-correlated matches, which would otherwise inflate cosine for compounds that produce broadly opposite-direction transcriptional states. Run by `paper5a/scripts/run_chemistry_vs_crispr.py`.

Permutation-null calibration

For each top-K-per-query table the top 10,000 rows by net mimic overlap were calibrated by drawing 1,000 random gene sets matched in size to each CRISPR signature’s up-set and down-set from the dataset-specific gene universe. For each random draw the same mimic / reverse / net-mimic statistics were recomputed against the fixed Z-Screen up/down set. Empirical p-values were computed as $(1 + \text{count of nulls} \geq \text{observed}) / (1 + \text{n permutations})$ and converted to Benjamini-Hochberg q-values across the calibrated rows. The headline ZEL024 / HEK293 scan used seed 13 and 1,000 permutations per row.

A size-matched permutation null was used in preference to a parametric test because parametric distributions for rank-overlap statistics depend on assumptions (gene universe, top-k cutoff, signature size) that vary by reference. A permutation null preserves the size of each CRISPR signature exactly, so the same null procedure is valid across heterogeneous references without requiring a separate parametric calibration for each one. Run by `paper5a/scripts/run_calibrate_matches.py`.

Recurrent-neighbor diagnostics and deduplicated headline tables

After calibration, per-CRISPR-signature recurrence tables were built that record how many distinct Z-Screen queries each CRISPR perturbation appeared as a top match for, both across the full top-K-per-query table and across the calibrated table at each q threshold. Two deduplicated views were also computed: one row per CRISPR program (best Z-Screen match by net z) and one row per Z-Screen query (best CRISPR match by net z). Finally, a hub-strip robustness summary records how many distinct programs and queries survive at $q \leq 0.05$ after dropping the top N most-recurrent CRISPR signatures (N in 0, 5, 10, 25, 50). The hub-strip analysis is reported alongside the headline rather than as a footnote because broadly responsive CRISPR perturbations are a known structural feature of large CRISPR atlases, and any chemistry-to-CRISPR framework that does not control for them will over-claim. Run by `paper5a/scripts/run_recurrent_neighbor_diagnostics.py`.

Resolution comparison

For each ZEL024 / HEK293 tuple at ≥ 10 wells, single-bb3 signatures were built from the same library at ≥ 500 wells per bb3, with explicit n-tuples-averaged metadata. Single-bb3 signatures were compared with the Replogle K562 reference and the top 50 matches per query were recorded. For each tuple the top 5 Replogle CRISPR matches at tuple resolution, the matches the containing bb0+bb1 pair would produce, and the matches the containing

bb3 single-position signature would produce were collected, and pairwise Jaccard overlaps were computed. The summary table reports the per-tuple overlaps and the aggregate distribution. Run by `paper5a/scripts/run_resolution_comparison.py` plus the figure builder `paper5a/scripts/make_resolution_figure.py`.

Norman combinatorial logic

For each Norman double-gene perturbation A+B (131 doubles), the constituent single-gene signatures A and B were located in the Norman metadata. Across the 8,599 ZEL024 / HEK293 tuples, the Spearman correlation was computed between the tuple-vs-double net mimic vector and the sum of the tuple-vs-single-A and tuple-vs-single-B net mimic vectors. The best Z-Screen tuple's individual scores against the double and each single were also recorded, along with whether the best double score exceeded both single scores. Run by `paper5a/scripts/run_norman_combinatorial_logic.py`.

The same combinatorial-logic test was rerun on each ZEL028-2 variable-pair panel (bb1+bb3, bb1+bb4, bb3+bb4 at ≥ 10 wells across HEK293, A549, H1650) and on each ZEL028-2 variable-triple panel (bb1+bb3+bb4 at ≥ 2 wells) using `paper5a/scripts/run_norman_subset_combinatorial_logic.py`. Per-panel summary outputs are at `paper5a/tables/ZEL028-2*_subset*_min*_vs_norman_norman_combinatorial`. The 12 ZEL028-2 panels yield median Spearman correlations between 0.325 and 0.371, with 90.8 to 100 percent of doubles significant at $p \leq 0.05$.

Hierarchical mechanism support for ZEL028-2

The script `paper5a/scripts/build_z028_hierarchical_support.py` starts from `ZEL028-2*_tuples_min2_vs_replogle_dedup_top_per_zscreen.csv`, not from every row in the 10,000-row calibrated tables. This defines the 2,151-row support universe as one best calibrated CRISPR call per represented full-tuple query: 943 HEK293, 760 A549, and 448 H1650 rows. For each row, the script annotates (a) whether the same CRISPR perturbation appears in the top 50 of any same-cell variable-pair top-K-per-query table (bb1+bb3, bb1+bb4, bb3+bb4 at ≥ 10 wells), (b) whether it appears at calibrated empirical $q \leq 0.05$ in any same-cell variable-pair calibrated table, (c) whether it appears in the top 50 of any cross-cell variable-pair top-K-per-query table, (d) whether it appears at calibrated $q \leq 0.05$ in any cross-cell variable-pair calibrated table, (e) whether it appears in the variable-triple top-50, and (f) whether the CRISPR program is among the top 50 most-recurrent CRISPR signatures of the calibrated full-tuple table. A composite support score sums the corroborating sources, with calibrated-tier support weighted above top-50 rank-overlap support. The strictest tier (`calibrated_subset_supported_nonhub`) requires a non-hub CRISPR call AND at least one calibrated $q \leq 0.05$ subset support; this tier identifies HEK293 ATP6V1A and A549 FKBP9. The broader tier (`subset_supported_nonhub_uncalibrated`) requires top-50 subset support without calibrated subset support and identifies 11 further non-hub candidates. Outputs: `ZEL028-2_hierarchical_mechanism_support.csv` (one row per deduplicated calibrated full-tuple call, with full set of support columns), `ZEL028-2_hierarchical_mechanism_support_summary.csv` (per-cell-line counts at every tier), `ZEL028-2_hierarchical_mechanism_support_top_candidates.csv` (the 13-row non-hub shortlist with

explicit tier annotation), and `ZEL028-2_hierarchical_mechanism_support_by_crispr.csv` (per-CRISPR-program aggregate).

Same-cell positive controls

The MZ1 and STC-15 positive controls used the matched-cell THP1 scPerturb reference. MZ1 lives in a sparse THP1 control library that lacks adequate same-library background, so its compound signature was built with `paper5a/scripts/build_compound_rank_signatures_with_fallback.py` using a same-cell compound-pool fallback background (all THP1 compound-treated cells excluding MZ1). The same fallback was used for STC-15, STM2457, sorafenib, Crizotinib, and capmatinib in their respective sparse libraries. The fallback background is broader than a same-library vehicle control would be; the absolute z-score should be read in that context. The structure of the MZ1 result (BRD4 as the top match across permutation seeds) is robust. The internal-control summary integrating direct-target and proxy-target recoveries is in `paper5a/tables/internal_control_summary_combined.csv`; the top-3 calibrated matches per same-cell THP1 compound are in `paper5a/tables/second_positive_control_thp1_compounds_top3.csv`.

Limitations

The framework compares transcriptional states, not biochemical binding. A chemistry-to-CRISPR match is a state-level hypothesis until validated by orthogonal target engagement.

The MZ1 fallback background for matched-cell THP1 is broader than a same-library vehicle control would be. The structure of the MZ1 to BRD4 result is robust across permutation seeds; the absolute z-score should be read in the context of that broader calibration.

Local same-cell CRISPR coverage is THP1-dominant, with no HEK293 or HEK293T single-cell CRISPR atlas surfaced by the local public probe. The HEK293 chemistry results therefore use cross-cell K562 Replogle as the genome-wide reference; the cross-cell setting means the matches are chemistry-resolved CRISPR-like state hypotheses rather than direct cell-of-origin target calls.

Tuple-level chemistry resolution is dataset-dependent. ZEL024 in HEK293 supports 8,599 tuples at ≥ 10 wells with median 13 wells per tuple, the cleanest tuple-level system in the canonical workspace. ZEL028-2 systems use a relaxed ≥ 2 -well threshold for the calibrated full-tuple scan because per-tuple coverage in that library does not support the ≥ 10 -well or even ≥ 3 -well threshold; the resulting tuple signatures aggregate a median of 2 wells, which makes individual tuple signatures substantially noisier than the ZEL024 / HEK293 tuple signatures. The ZEL028-2 tuple-level results are therefore interpreted as a chemistry-resolved leaderboard whose strongest entries reproduce the same biology themes ZEL024 / HEK293 surfaced (RNA processing and mitochondrial translation) rather than as confirmed per-tuple calls. The hierarchical mechanism-support model addresses this directly by re-running ZEL028-2 chemistry at three resolutions (full tuple, variable triple `bb1+bb3+bb4` at ≥ 2 wells, and variable pair subsets `bb1+bb3` / `bb1+bb4` / `bb3+bb4` at ≥ 10 wells per signature) and using the subset signatures as orthogonal support for each calibrated full-tuple call. The variable-pair Replogle scans are also permutation-calibrated

with the same size-matched null used for the headline ZEL024 / HEK293 scan, so the strictest tier of the hierarchical model (calibrated subset support at $q \leq 0.05$) is FDR-controlled and identifies a small high-confidence shortlist (HEK293 ATP6V1A; A549 FKBP9) rather than a hub-stripped count. ZEL031 systems are tuple-resolution at lower well counts. The reporting tables annotate each signature with n-wells and n-tuples-averaged so the reader can directly judge the chemistry resolution behind any specific match.

Cross-cell transfer of chemical programs is reported in paper 4 at building-block-position centroid resolution; tuple-resolution cross-cell transfer requires deeper per-tuple cell coverage than the current canonical workspace supports and is left as future work.

Recurrent CRISPR perturbations exist; the recurrence diagnostic and hub-strip summary report how the headline counts change when the top hubs are dropped from the calibrated table. After dropping the 50 most-recurrent CRISPR signatures, 1,236 chemistry-resolved tuples and 1,664 distinct CRISPR programs still passed $q \leq 0.05$, so the diversity is not driven by hubs alone.

The targeted public scPerturb probe is laptop-scale (11 H5AD files under 500 MB each) and is not a comprehensive survey of all public HEK293 perturbation data.

Data availability

The full reproducibility package, including the canonical Z-Screen RNA-seq dataset, bundled public CRISPR rank signatures, all derived match tables, calibrated headline outputs, recurrence diagnostics, and figure-building scripts, is contained in this repository at `paper5a/`. The repaired canonical RNA-seq aggregate is at `data/ZScreen_Canonical_Dataset/RNASeqAggregate/`. Public CRISPR rank signatures are bundled in `paper5a/external_data/`. Their upstream source datasets are referenced below.

Code availability

All analysis and figure-generation scripts are in `paper5a/scripts/`. The reproduction order, including dependencies on the bundled public CRISPR rank signatures, is in `paper5a/README.md`. Package-level dependencies are listed in the repository root `requirements.txt`. The codebase was tested with Python 3.14.3 on Windows.

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